



## **Home-Based Portable IoT ECG**

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Date

## **Declaration**

This is to declare that the graduation project entitled "Home-Based Portable IoT ECG", under the supervision of Dr. Sameer Hasan and Eng. Omar Yasin, with acknowledgments to Dr. Samer Al-Abed, is our own original work and does not contain any unacknowledged material from any other source.

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## **Acknowledgment**

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Thank you all for being part of this milestone.

## **Abstract**

Cardiovascular diseases (CVDs) represent the leading cause of mortality in Jordan and the broader MENA region, yet access to essential diagnostic tools like portable ECG devices remains critically limited due to high costs and lack of domestic production. This project addresses this healthcare and technological gap by designing a modular, scalable, low-cost, portable and web integrated ECG device tailored to the Jordanian healthcare environment. The device leverages a custom PCB for improved reliability and cost-efficiency and incorporates Centralized data storage and web-based connectivity to facilitate ECG data reviewing and follow-up between patients and physicians. By designing locally, the project supports national goals of economic independence in medical technology and aims to reduce reliance on imported equipment.

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## **Chapter 1: Introduction**

The electrocardiogram (ECG) is a powerful and indispensable tool in modern cardiology, playing a pivotal role in diagnosing and managing cardiovascular diseases, which remain the leading cause of mortality worldwide. According to the World Health Organization, cardiovascular diseases account for an estimated 17.9 million deaths each year, underscoring the urgent need for early detection and intervention [1]. An ECG records the heart's electrical activity over time, offering crucial insights into cardiac rhythm, conduction pathways, and overall structural integrity [2]. By detecting abnormalities such as arrhythmias, ischemic changes, and conduction defects, the ECG has saved countless lives by enabling timely medical intervention and guiding therapeutic strategies [3]. Its non-invasive nature and accessibility make it an essential component of both emergency and routine clinical settings, reinforcing its significance in reducing cardiovascular-related morbidity and mortality [4].

Despite its clinical importance, access to ECG monitoring remains limited by the nature of existing systems. Conventional ECG devices are typically hospital-based, requiring specialized equipment, trained personnel, and controlled clinical environments. While high-end standalone and wearable devices have emerged in recent years, many of these solutions are either costly, limited in functionality, or designed primarily for short-term or single-lead monitoring, [5] as well as being limitedly available in Jordan. As a result, continuous and comprehensive cardiac monitoring remains inaccessible to a large portion of patients, particularly in low-resource settings.

The need for accessible ECG monitoring is especially critical for patients requiring long-term observation outside hospital environments. Individuals with mild, moderate, or chronic cardiac conditions often benefit from extended monitoring to detect intermittent or early-stage abnormalities that may not appear during brief clinical examinations. Home-based monitoring

solutions can reduce hospital burden, improve patient comfort, and enable earlier medical intervention. However, such systems must be affordable, easy to use, and capable of providing meaningful data to be truly effective. [6]

In response to these challenges, this project presents the design and implementation of a low-cost, portable, and user-friendly multi-lead ECG monitoring system. The proposed system aims to bridge the gap between hospital-grade ECG devices and consumer-level solutions by offering reliable signal acquisition, wireless data transmission, and web-based visualization suitable for both patients and healthcare providers. The focus of this work is on affordability, accessibility, and practicality, making cardiac monitoring more widely available without compromising signal quality or clinical relevance.

This dissertation is structured as follows. Chapter 2 provides a background on cardiac electrophysiology and ECG fundamentals. Chapter 3 presents a review of related work and existing ECG monitoring systems. Chapter 4 details the system's methodology, including hardware design, signal acquisition, embedded firmware implementation, and software architecture. Chapter 5 discusses system testing and validation of results. Finally, Chapter 6 concludes the dissertation and outlines potential future improvements.

## **1.1 Anatomy and Physiology of the Heart**

The heart is a muscular pump whose structure and function are orchestrated through a symphony of electrical and mechanical processes [7]. Its structure includes four chambers, two atria and two ventricles—each with specialized roles in the process of blood circulation. The chambers are separated by robust fibrous tissues and valves that maintain unidirectional blood flow, as seen in Figure 1.

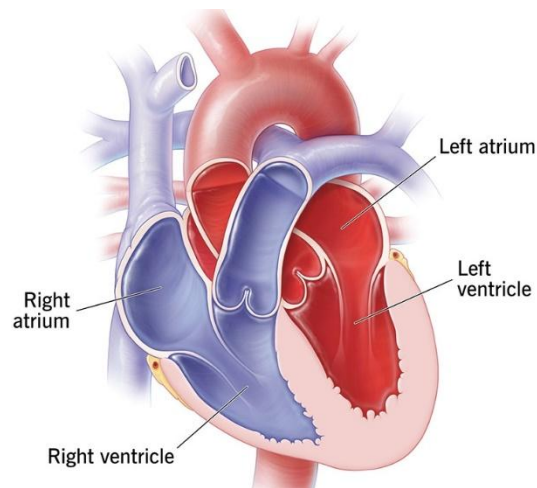


Figure 1. The gross anatomy of the heart [8].

The atria, ventricles, and valves are fundamental to cardiac function. The atria, with their thin walls, receive blood and initiate contraction via specialized conduction tissue. The ventricles, particularly the robust left ventricle, generate the force needed to propel blood throughout the body. The heart valves, including the atrioventricular (mitral and tricuspid) and semilunar (aortic and pulmonary) valves, ensure unidirectional blood flow by opening and closing in synchrony with the cardiac cycle, thereby preventing regurgitation and optimizing hemodynamics [7][9][10], ensuring efficient hemodynamics even under variable physiological demands [11][12].

At the microscopic level, the heart structure is equally complex. It comprises three distinct layers: the endocardium (inner lining), myocardium (thick muscular middle layer), and epicardium (outer layer) [9].

The electrical conduction system is the cornerstone of cardiac function. It is interplay between its various components - the SA node, AV node, Bundle of His, bundle branches, and Purkinje fibers and Bachmann's Bundle, seen in Figure 2, ensures that the heart beats with remarkable regularity and efficiency.

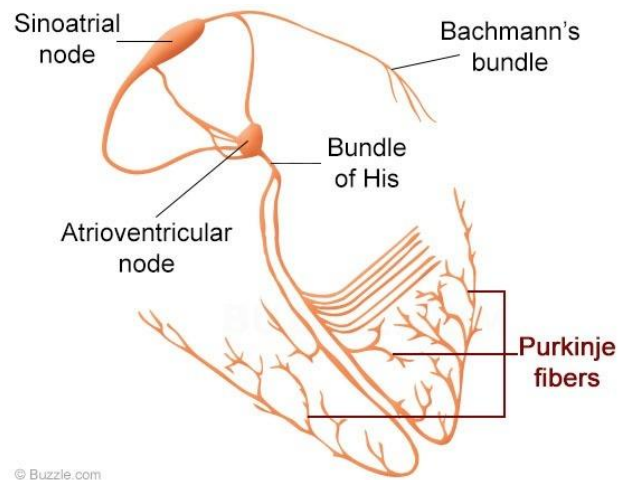


Figure 2. The electrical conduction system of the heart [11].

The heart's electrical activity represents a complex interplay of ionic exchanges and signal propagation culminating in mechanical contraction. Electrical signals in the heart directly correspond to the phases of depolarization and repolarization. When an action potential is generated in a cardiomyocyte, depolarization occurs first, initiated by the rapid influx of sodium ions ( $\text{Na}^+$ ). This sudden positive shift in membrane potential triggers the opening of voltage-gated calcium channels, allowing calcium ions ( $\text{Ca}^{2+}$ ) to enter the cell. This influx of  $\text{Ca}^{2+}$  not only sustains the plateau phase of the action potential but also plays a crucial role in excitation–contraction coupling, where it stimulates the sarcoplasmic reticulum to release additional  $\text{Ca}^{2+}$ , ultimately leading to myocardial contraction [13][14].

Repolarization follows as potassium ions ( $\text{K}^+$ ) exit the cell, restoring the resting membrane potential and allowing the cardiomyocyte to prepare for the next action potential. This repolarization phase ensures that the electrical cycle resets properly, preventing sustained depolarization and allowing coordinated contractions [10].

This entire process is regulated by the autonomic nervous system, hormones, and local paracrine factors, allowing the heart to adapt swiftly to changing metabolic demands [7].

Oxygen plays a crucial role in cardiac excitation–contraction coupling, which is the process that links the electrical signals in the heart to its physical contraction [15]. This happens by enabling the production of ATP (adenosine triphosphate), the energy currency of cells, through a process called oxidative phosphorylation, which takes place inside the mitochondria. ATP is vital because it powers the work of the ion pumps, including the sarcoplasmic reticulum  $\text{Ca}^{2+}$ -ATPase (SERCA), which helps store calcium inside a compartment of the cell. Another is the  $\text{Na}^+/\text{K}^+$ -ATPase, which keeps the balance of sodium and potassium ions in and out of the cell, creating the electrical environment needed for signals to move across heart cells [16].

When the body does not get enough oxygen, a condition known as hypoxia, less ATP is made. This weakens the activity of pumps like SERCA, causing too much calcium to stay in the fluid part of the cell instead of being stored [15][16]. As a result, the heart's action potential can last too long, the reset period between beats (the refractory period) may change, and dangerous irregular arrhythmia can occur [17]. Low oxygen also causes harmful molecules known as reactive oxygen species (ROS) to build up. These molecules can damage proteins and interfere with ion channels, which are gateways that control how ions flow in and out of the cell, affecting both the heartbeat and the strength of heart muscle contractions (myocardial contractility) [17].

Initiation of the heartbeat begins at the sinoatrial (SA) node, the heart's natural pacemaker located in the right atrium near the entrance of the superior vena cava. Histologically, the SA node comprises specialized cardiomyocytes with an intrinsic ability to depolarize spontaneously, a property enabled by a unique mix of ion channels [18][19]. The SA node generates electrical impulse at atrial depolarization. Its spontaneous depolarization is regulated by autonomic input, which is sympathetic stimulation that accelerates the heart rate, while parasympathetic stimulation decelerates it.

Following the SA node, the impulse reaches the atrioventricular (AV) node, positioned at the junction of the atria and ventricles. The AV node serves as a critical relay station that introduces a physiological delay of approximately 0.1 seconds, allowing for complete atrial contraction and optimal ventricular filling [10][19]. Histologically more compact than the SA node and rich in connexins (gap junction proteins), the AV node facilitates a slow yet coordinated propagation of electrical impulses. This delay is not only mechanically efficient but also protective, preventing excessively rapid atrial impulses from overloading the ventricles [11].

After the AV node, the electrical impulse travels through the Bundle of His, a slender fibrous structure that represents the only electrical connection between the atria and ventricles. The Bundle of His then branches into the right and left bundle branches, each rapidly transmitting the impulse to their respective ventricles [20]. These fiber tracts, insulated by connective tissue, ensure that the electrical impulse is delivered with speed and efficiency [13][18].

At the terminal ends of the conduction network lie the Purkinje fibers, which are large, fast-conducting cells responsible for rapidly spreading the electrical impulse throughout the ventricular myocardium. Their extensive branching and high conduction speed ensure that the ventricles contract almost synchronously, which is essential for optimal blood ejection during systole. Purkinje fibers possess a unique cellular structure characterized by fewer myofibrils and a high density of mitochondria, making them specialized for rapid impulse propagation rather than force generation [13][21].

As for the Bachmann's bundle, it is a crucial interatrial conduction pathway that facilitates the rapid transmission of electrical impulses from the right atrium to the left atrium, ensuring that both chambers contract in a coordinated fashion. This synchronization is vital for effective atrial emptying and optimal ventricular filling, which ultimately enhances the efficiency of the cardiac



cycle. Its function is integral to maintaining normal atrial rhythm, and disturbances in this pathway have been linked to the development of atrial arrhythmia [10][19].

At the molecular level, the heart's efficiency is further augmented by the precise regulation of ion channels, gap junctions, and intracellular signaling pathways. The connectivity provided by gap junctions allows for the rapid propagation of electrical impulses between adjacent myocytes, ensuring that contraction occurs in a unified and synchronized [19][21].

## 1.2 Pathology of the Heart

The pathology of the heart represents a variety of clinical conditions that manifest distinct electrocardiographic features, each reflecting specific underlying structural and functional abnormalities. At its core, cardiac pathology encompasses a wide array of disorders ranging from ischemic injuries and myocardial infarctions to various conduction disturbances and arrhythmia, each altering the balance of electrical and mechanical activity. The ECG waveform consists of several distinct components—the P wave, QRS complex, T wave, and sometimes a U wave—each corresponding to specific electrical and mechanical events in the heart, as seen in Figure 3. These alterations are often revealed through detailed analysis of the ECG, where deviations in the PQRST complex offer clues to the nature and extent of the disease process [9].

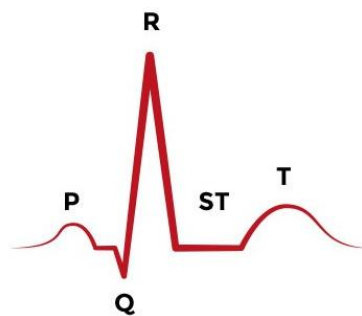


Figure 3. A PQRST wave.

The P wave represents atrial depolarization, indicating the initiation of the heartbeat at the SA node and subsequent atrial contraction. It is usually small and upright in most leads, with abnormalities signifying atrial enlargement or conduction delays [18].

The QRS complex corresponds to ventricular depolarization, marking the most critical phase of electrical activation. This complex includes the Q wave (initial downward deflection), the R wave (large upward deflection), and the S wave (subsequent downward deflection). The width and morphology of the QRS complex provides insight into conduction disorders such as bundle branch blocks or ventricular hypertrophy [7].

Following ventricular contraction, the T wave reflects ventricular repolarization, a crucial phase where the heart resets for the next beat. Flattened or inverted T waves may indicate ischemia, electrolyte imbalances, or ventricular strain [22].

In some cases, a U wave appears after the T wave, which is often linked to hypokalemia or abnormal repolarization processes [21].

Fortunately, the use of an ECG paved a path for detecting many dangerous, yet common, conditions.

#### a. Myocardial Infarction

Myocardial infarction (MI), commonly known as a heart attack, is perhaps the most extensively studied cardiac pathology, characterized by the abrupt occlusion of coronary arteries leading to ischemia and subsequent necrosis of the affected myocardial tissue. The ischemic cascade produces specific ECG changes, such as ST-segment elevation or depression, pathological Q Waves, and T-wave inversions, each correlating with the infarct size and location [23]. In particular, ST-segment elevation, when persistent as seen in Figure 4 is a hallmark for diagnosing

acute MI, while the subsequent evolution of Q waves signifies irreversible myocardial damage [14].

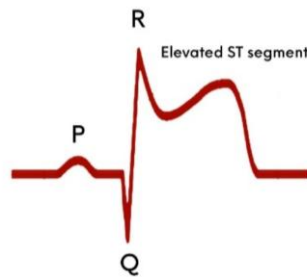


Figure 4. An MI ECG wave.

b. Cardiac arrhythmia – Atrial Fibrillation

Cardiac arrhythmias are a variety of critical pathologies that have a direct effect on the ECG profile. Atrial fibrillation, the most common sustained arrhythmia in clinical practice, is typified by an irregular rhythm and the absence of distinct P waves, reflecting the chaotic atrial electrical activity, as seen in Figure 5.

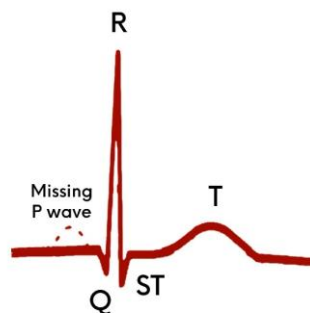


Figure 5. An atrial fibrillation wave.

This condition is often linked to atrial enlargement, fibrosis, or systemic conditions that predispose to electrical instability [19].

### c. Ventricular Tachycardia & Fibrillation

Ventricular tachycardia and ventricular fibrillation are life-threatening disturbances of ventricular rhythm that lead to rapid and unsynchronized contraction, dramatically altering the QRS complex morphology and, in the case of fibrillation, obliterating any discernible pattern [18]. Ventricular tachycardia and Ventricular Fibrillation are seen in Figures 6 and 7, respectively.



Figure 7. A ventricular fibrillation wave.

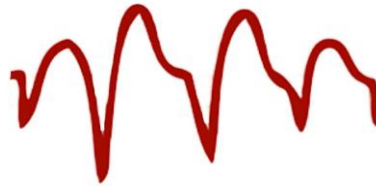


Figure 6. A ventricular tachycardia wave.

### d. Conduction abnormalities - Right and left bundle branch blocks

Conduction abnormalities further complicate the electrical narrative of the heart. A bundle branch block, whether right or left, results in delayed depolarization of one ventricle, thereby widening the QRS complex and often producing secondary ST-T changes. Right bundle branch block (RBBB) occurs due to impaired conduction in the right bundle branch, leading to sequential rather than simultaneous activation of the ventricles. This manifests in lead V1 as a characteristic RSR' pattern, commonly described as "rabbit ears," where the initial small R wave represents normal septal depolarization, the deep S wave reflects dominant left ventricular activation, and the prominent second R wave (R'), indicates delayed right ventricular depolarization, as shown in

Figure 8. Meanwhile, lateral leads (e.g., Lead V6) exhibit a broad S wave, signifying prolonged right ventricular activation [24][25].

Left bundle branch block (LBBB), in contrast, disrupts conduction through the left bundle branch,

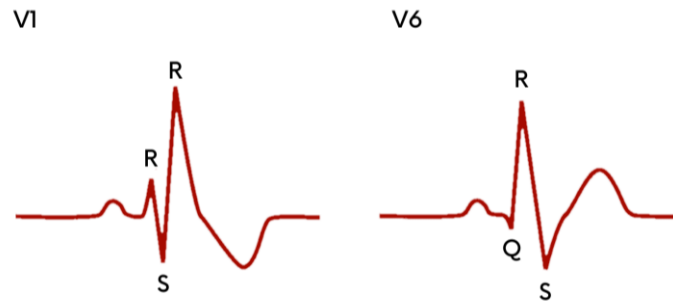


Figure 8. RBBB ECG in Leads V1 & Lead V6.

delaying left ventricular depolarization. This is seen in Figure 9, where in lateral leads (I, aVL, V5, V6) have a broad, notched, or slurred R wave in the shape of an 'M' and an absence of the normal septal Q wave. The right ventricle depolarizes first, causing a deep S wave in leads V1-V3. Additionally, LBBB produces a discordant ST-T pattern, where the ST segment and T wave deflect in the opposite direction of the QRS complex [26][27].

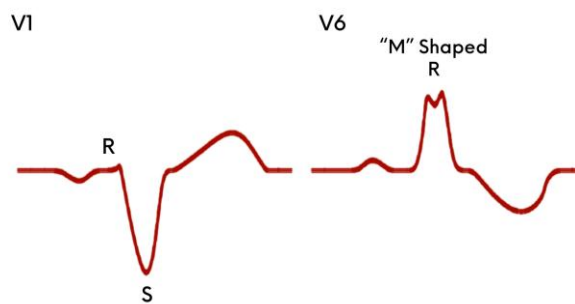


Figure 9. LBBB ECG in Leads V1 & Lead V6.

Both RBBB and LBBB can be benign or indicative of underlying pathology such as ischemic heart disease, cardiomyopathy, or structural heart disorders, so further clinical correlation and evaluation is needed.

e. Conduction abnormalities - atrioventricular (AV) block

AV block, categorized into first, second, and third degrees, manifests as prolonged PR intervals or intermittent non-conducted atrial impulses, leading up to a complete dissociation between atrial and ventricular activity in third-degree block. Each of these conduction defects imparts distinct ECG signatures, as seen in Figure 10, that not only aid in diagnosis but also provide prognostic information regarding the underlying structural integrity of the conduction system [13].

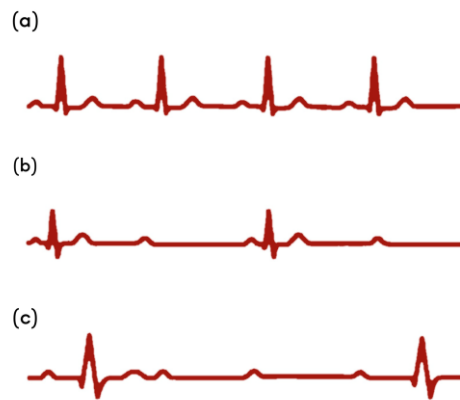


Figure 10. (a) First-degree AV block ECG, (b) Second-degree AV block ECG, (c) Third-degree AV block ECG.

f. Hypertrophic Cardiomyopathies

Cardiomyopathies, encompassing both hypertrophic and dilated forms, represent intrinsic disorders of the myocardium that often result in aberrant electrical conduction and mechanical dysfunction [9]. In hypertrophic cardiomyopathy, as shown in Figure 11, asymmetric septal thickening and myocardial disarray can lead to abnormal repolarization patterns, including deep, inverted T waves and potential ST-segment abnormalities [20][28].

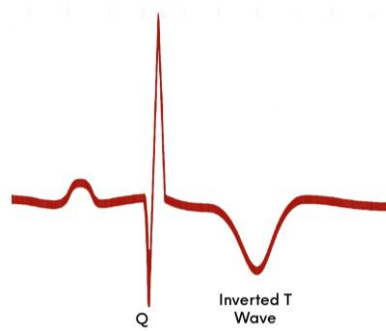


Figure 11. ECG of hypertrophic cardiomyopathy.

#### g. Dilated Cardiomyopathies

Dilated cardiomyopathy is more frequently associated with conduction delays and nonspecific ST-T changes, as shown through the leads in Figure 12 [9][20]. These conditions not only compromise cardiac output but also predispose patients to life-threatening arrhythmias, thereby underscoring the intricate interplay between structural pathology and electrical instability [14].



Figure 12. Multiple ECGs of dilated cardiomyopathy.

#### h. Hyperkalemia and hypokalemia

Hyperkalemia and hypokalemia are products of electrolyte disturbances, specifically changes in potassium intake. Hyperkalemia often produces peaked T waves, a shortened QT interval, and, in severe cases, widening of the QRS complex, as shown in Figure 13.



Figure 13. ECG of hyperkalemia.

While hypokalemia may lead to flattened T waves, the appearance of U waves, and a prolonged QT interval, as seen in Figure 14 [22].

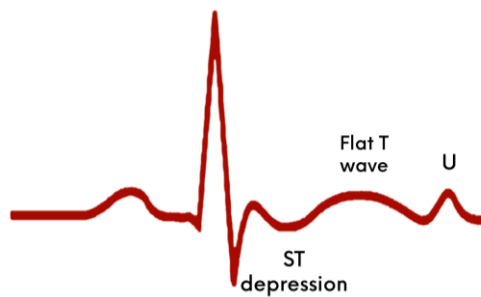


Figure 14. ECG of hypokalemia.

#### i. Pericarditis

Pericarditis is characterized by ST-segment elevations and PR depressions. Electrical activity often reflects the inflammatory process affecting the pericardium, and these changes tend to evolve over

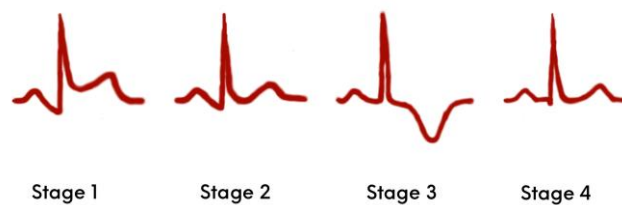


Figure 15. ECG wave across the 4 stages of pericarditis.



time, thereby requiring serial ECG recordings for accurate diagnosis in their different stages, as shown in Figure 15 [23].

Beyond these specific conditions, the ECG serves as a vital tool for detecting subtle and diffuse changes that may show the onset of cardiac pathology before overt clinical symptoms emerge. Early detection of conduction system disease can facilitate timely intervention and potentially prevent the progression to more severe forms like heart block or arrhythmia [21][19].

### **1.3 Problem Statement**

Cardiovascular diseases (CVDs) remain the leading cause of mortality in Jordan, with an age-standardized prevalence rate of 7,980 per 100,000 individuals as of 2019 [29]. In the Middle East and North Africa (MENA) region, ischemic heart disease alone affected approximately 19.98 million individuals in the same year, highlighting the growing burden of cardiac conditions that often require long-term monitoring and follow-up care [30]. Despite this prevalence, access to continuous and affordable cardiac monitoring tools remains limited, particularly outside hospital environments.

Conventional clinical ECG systems are designed for short-duration recordings conducted in hospitals or specialized clinics. While these systems provide high diagnostic accuracy, they are not suitable for frequent or continuous monitoring in everyday settings [31]. Patients with diagnosed or suspected cardiac conditions often experience intermittent symptoms that do not coincide with scheduled clinical visits, resulting in missed events and delayed clinical intervention [32]. Frequent hospital visits also impose logistical, financial, and accessibility burdens, especially for elderly patients and individuals living in rural or lower-income areas.

Home-based ECG monitoring offers a practical solution to these limitations by enabling patients to record cardiac signals in their natural environment and at the time symptoms occur. Portable

ECG devices combined with wireless data transmission allow physiological data to be captured continuously or on demand, reducing dependence on hospital-based testing and supporting post-diagnosis monitoring and follow-up care. However, in Jordan, affordable and locally developed home-based ECG solutions remain scarce, with most available systems being either cost-prohibitive or not tailored to regional healthcare needs.

This project addresses this gap by developing a low-cost, portable, home-based IoT ECG monitoring system that enables remote data acquisition, real-time visualization, and secure web-based access for clinicians. By focusing on accessibility, portability, and continuous monitoring outside the clinical setting, the proposed system aims to improve cardiac care delivery and support timely clinical decision-making within the local healthcare context.

#### **1.4 Project Goals and Objectives**

The primary goal of this project is to develop a user-friendly and affordable home-based IoT-enabled ECG monitoring system tailored to the healthcare and economic context of Jordan. Due to the high cost and limited availability of locally manufactured ECG devices, many patients lack access to frequent cardiac monitoring outside clinical settings. This project aims to address this gap by enabling accessible, at-home ECG acquisition with remote data access for clinicians.

##### **1.4.1 Main Goals**

1. Design and implement a portable home-based ECG device.

The project aimed to design a compact and lightweight ECG device suitable for home use, capable of acquiring and displaying reliable cardiac signals without requiring continuous clinical supervision.

2. Develop a custom printed circuit board (PCB)

A dedicated PCB was designed to integrate all essential analog, digital, and communication components, improving system reliability, signal integrity, and portability compared to general-purpose development boards.

### 3. Enable IoT-based Wireless data transmission and storage.

The system is designed to support wireless transmission of ECG data to a web-based backend platform, allowing recorded signals to be securely stored and accessed remotely by healthcare providers.

### 4. Support Local Design and Production

The project demonstrates the feasibility of locally designing and assembling ECG monitoring systems, contributing to reduced dependence on imported medical devices and supporting national innovation efforts.

## **1.4.2 Objectives**

- Enable home-based ECG monitoring for patients requiring frequent cardiac follow-up, particularly after hospital discharge.
- Reduce the need for repeated hospital visits by allowing ECG data to be recorded in real-world environments and shared remotely with clinicians.
- Improve access to cardiac monitoring tools in underserved and rural communities where healthcare facilities may be limited.
- Contribute to national healthcare development by aligning with Jordan's growing medical technology sector, which currently includes over 78 registered medical device companies [33].

- Support a bilingual web-based user interface with full Arabic language localization to improve usability and accessibility for home-based users in Jordan, where Arabic is the primary language.

By focusing on affordability and portability, this project aims to enhance continuity of cardiac care and provide a base for future advancements in home-based healthcare technologies in Jordan.

## **1.5 Scope and Limitations**

This graduation project focuses on the design and development of a locally assembled, affordable, and portable home-based ECG monitoring system aimed to contribute to the Jordanian healthcare system. The system aims to enable accessible cardiac monitoring outside clinical environments through IoT-based data transmission and web-based visualization, supporting remote clinician review and follow-up care.

### **1.5.1 Scope**

#### **1. Hardware development:**

A custom printed circuit board (PCB) was designed to replace general-purpose prototyping platforms. The hardware integrates ECG signal acquisition circuitry and wireless communication components, with emphasis on portability, compactness, and considerations for power efficiency for home use.

#### **2. Software System Integration:**

The project includes embedded firmware for ECG data acquisition and wireless transmission, as well as a web-based backend and frontend system for real-time visualization, session management, and ECG data storage. The software architecture enables recorded ECG sessions to be accessed remotely by authorized users through a browser-based interface.

### 3. Home-Based Use and Target Users

The system is designed for at-home use by patients requiring frequent cardiac monitoring, particularly during post-hospital recovery or long-term follow-up. The proposed solution reduces reliance on repeated hospital visits by allowing ECG data to be recorded in non-clinical environments and reviewed remotely.

### 4. Local Production and Innovation Focus

The design prioritizes local production and assembly processes where possible. This approach supports national efforts toward self-reliance in medical technology development and contributes to local technical capacity building.

#### **1.5.2 Limitations**

- Some specialized ECG components are not available locally and must be imported, increasing development costs and procurement time.
- The project was developed under limited time and resource constraints, restricting the depth of hardware optimization and large-scale validation.
- Human subject testing was not conducted due to the absence of ethical approval, preventing validation using real patient data. Full clinical validation and regulatory approval were outside the scope of this project and may require extended evaluation.
- A trusted clinical ECG simulator was unavailable, so system testing relied on a single pre-recorded ECG signal provided through university laboratory resources.

- Although the system supports multi-lead ECG visualization through mathematical derivation, clinical validation was limited and did not include comprehensive multi-lead physiological evaluation.
- The connected monitoring functionality depends on network availability, which may limit usability in areas with unstable or limited internet access.
- Commercial cloud services were not utilized due to subscription and operational costs; instead, network-based data handling was adopted.
- Jordan's medical device regulation is still in early development stages. Full clinical validation and approval may face delays due to evolving regulatory procedures under the Jordan Food and Drug Administration (JFDA) [34].

This project represents a foundational step toward accessible home-based cardiac monitoring in Jordan and provides a technical basis for future clinical validation, regulatory approval, and system expansion and adaptation.

## **Chapter 2: Theoretical Background**

The ECG machine functions by detecting electrical signals produced by cardiac muscle contractions, amplifying them, filtering out noise, and converting them into a readable waveform. The resulting tracings allow clinicians to assess heart conditions such as arrhythmias, ischemia, and infarctions [35].

### **2.1 The Historical Evolution of ECG Machines**

The development of ECG technology spans over a century, evolving from large and heavy early devices to highly sophisticated digital systems. The first practical ECG machine was introduced in

the early 1900s by Dutch physiologist Willem Einthoven, who developed the string galvanometer, a large, complex device capable of recording heart activity [36]. Despite its size and operational challenges, this invention laid the groundwork for modern electrocardiography.

By the mid-20th century, advancements in electronics enabled the transition from vacuum tube-based ECG machines to transistorized devices, significantly reducing their size and improving portability [37]. In 1957, Norman J. Holter pioneered ambulatory ECG monitoring, introducing the Holter monitor, which allowed continuous cardiac monitoring over extended periods. This breakthrough was instrumental in diagnosing intermittent arrhythmias [38][39].

The digital revolution in the 1980s and 1990s further transformed ECG technology, introducing computerized interpretation algorithms that enhanced diagnostic accuracy. Modern ECG machines integrate AI-driven analytics, wireless transmission, and cloud-based storage, facilitating remote monitoring and large-scale data analysis [40]. These innovations have expanded the role of an ECG in preventive cardiology, allowing for early detection of cardiac abnormalities [41].

## **2.2 Components of a Standard ECG**

An ECG machine consists of several key components, each playing a vital role in signal acquisition, processing, and interpretation.

### **2.2.1 Electrodes**

The electrodes are the first point of contact, placed on the skin to detect bioelectrical signals. These electrodes, usually made of silver/silver chloride due to its stable electrode-electrolyte interface, which minimizes polarization effects and enhances signal fidelity [42], they work by capturing potential differences generated by the depolarization and repolarization of cardiac muscle cells [43]. Electrodes are connected to the machine via leads, which provide different perspectives of cardiac electrical activity. The electrodes interface with the body through an electrolytic gel that

reduces contact impedance and improves signal quality [44]. A 12-lead ECG system provides a comprehensive view of cardiac electrical activity by acquiring signals from different anatomical perspectives. A standard 12-lead ECG records electrical signals from 10 electrode placements, creating a comprehensive representation of heart function [38]. It consists of three bipolar limb leads (I, II, III), three augmented unipolar leads (aVR, aVL, aVF), and six precordial leads (V1-V6). These leads are placed at standardized positions on the body to obtain signals from multiple orientations of the heart's electrical activity [45].

In addition to the standard 12-lead system, ECGs can be recorded using fewer leads depending on clinical requirements. Single-lead ECGs are systems that use only one pair of electrodes to measure cardiac activity, commonly used in wearable monitors and personal ECG devices for rhythm monitoring rather than full cardiac diagnosis [46]. Three-lead ECGs are typically used for basic cardiac monitoring in hospital settings; three-lead ECGs record leads I, II, and III, providing a limited but essential view of cardiac rhythm and rate. They are widely used in ambulatory monitoring and emergency settings [47]. Five-lead ECGs are frequently used in intensive care and telemetry units; five-lead systems provide continuous cardiac monitoring, adding a precordial lead (V1 or V5) for additional waveform detail [48]. Six-lead ECGs are an advancement seen in portable devices; six-lead ECGs add more precordial lead perspectives to enhance rhythm analysis while maintaining a compact form factor [49].

### **2.2.2 Instrumentation Amplifier**

Once the signals are acquired through the electrodes, they pass through an amplifier since cardiac electrical activity is relatively weak (typically 0.5–5 mv). The amplifier enhances these signals while reducing common-mode noise, ensuring a clear representation of heart activity [50]. Common-mode noise refers to unwanted electrical interference that appears simultaneously on



both input terminals of a differential measurement system, often originating from power line coupling or electromagnetic interference [51]. This is accomplished using an instrumentation amplifier (IA), which consists of three operational amplifiers in a high common-mode rejection ratio (CMRR) configuration [52]. The IA consists of two buffer amplifiers that provide high input impedance and a differential amplifier that subtracts the signals while rejecting noise. A commonly used circuit is the three-op-amp IA, as shown in Figure 16, where the two input buffers ensure minimal loading on the electrodes, while the differential stage provides gain [53].

The IA's resistors set the gain, typically determined by equation 1.

$$G = 1 + \frac{2R1}{R_{gain}} \quad \text{equation 1}$$

Where R1 and R<sub>gain</sub> define the amplification factor [54].

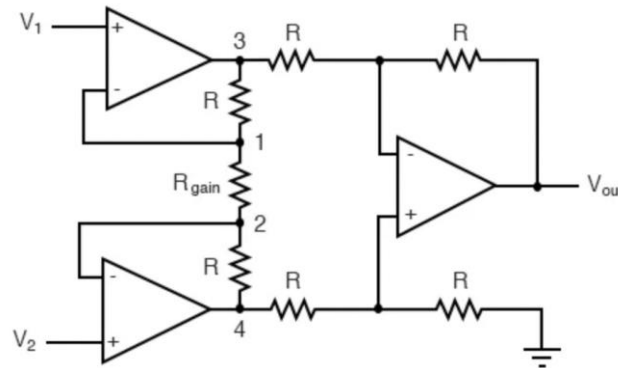


Figure 16. Instrumentation Amplifier circuit [55].

### 2.2.3 Right Leg Drive Circuit

A Right Leg Drive (RLD) Circuit is integrated for Common-Mode Noise Cancellation. The RLD circuit actively drives the patient's right leg with an inverted common-mode signal to cancel out power line interference [56]. It consists of an operational amplifier configured as an inverting integrator, which senses the common-mode voltage and injects an out-of-phase signal back into the body to minimize interference [57].

### 2.2.4 Filters

After acquisition and amplification, the signals are processed through filters to remove unwanted noise from muscle activity (EMG artifacts), electrical interference, and motion artifacts. Low-pass, high-pass, and notch filters help refine the signals, ensuring accuracy [58]. The key filtering stages include:

- High-Pass Filter (HPF): Typically set at 0.5 Hz to eliminate baseline wander caused by respiration and electrode drift [59]. The Filter circuit is seen in Figure 17.

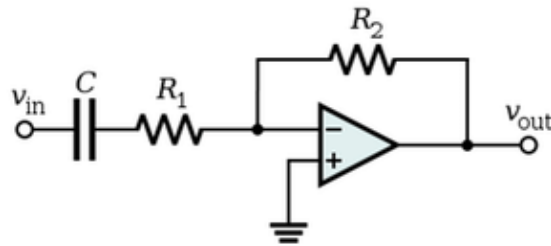


Figure 17. An active high pass filter circuit [60].

- Low-Pass Filter (LPF): Set at 150 Hz to remove high-frequency noise and EMG artifacts [61]. The Filter circuit is seen in Figure 18.

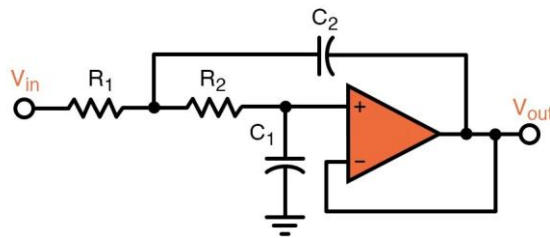


Figure 18. A second order low pass filter circuit [62].

- Notch Filter: Designed at 50/60 Hz to suppress power line interference using a twin-T topology or active filter implementation [63]. The Filter circuit is seen in Figure 19.

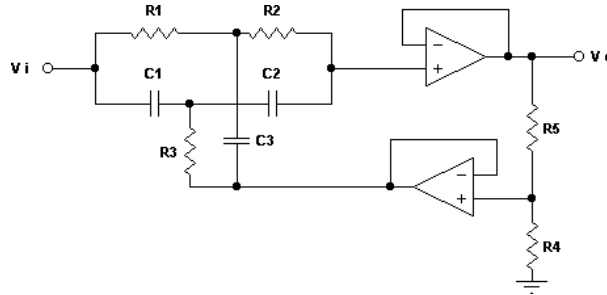


Figure 19. A noninverting Twin-T second order notch filter [64].

### 2.2.5 Isolation

For safety, electrical isolation is crucial to prevent hazardous currents from reaching the patient. An optocoupler (optoisolator) is used for galvanic isolation. It consists of an infrared LED and a phototransistor within a sealed package, as shown in Figure 20. When the LED is energized, the phototransistor conducts, allowing signal transmission without direct electrical connection [65]. This protects against leakage currents exceeding safety thresholds defined by IEC 60601-1 standards [66].

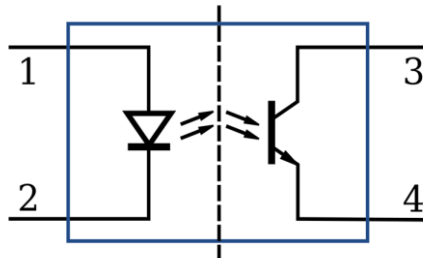


Figure 20. An optocoupler [67].

## **2.3 Signal Processing of an ECG signal**

### **2.3.1 Analog-to-digital Conversion**

Following signal conditioning, the electrical impulses undergo analog-to-digital conversion (ADC). In modern ECG machines, ADCs use high-resolution sampling rates (often above 500 samples per second) to preserve temporal precision. The digitized signals are processed to extract critical features such as heart rate, QRS duration, and ST-segment deviations, which are crucial for diagnosing various heart conditions [68]. The signal is digitized using an ADC with at least 12-bit resolution to ensure adequate signal fidelity. The sampling rate must be at least 500 Hz per lead to comply with medical standards [69].

### **2.3.2 Microcontroller Unit**

The digitized ECG signals are then processed by a microcontroller unit (MCU), which performs several key functions, including Digital Filtering to further noise reduction using software-based finite impulse response (FIR) or infinite impulse response (IIR) filters [70]. Lead Calculation for Generation of augmented leads from limb leads [71], which is the mathematical derivation of additional leads from recorded signals to provide a comprehensive view of cardiac electrical activity. For example, the augmented limb leads (aVR, aVL, aVF) are derived from the standard limb leads (I, II, III) using Einthoven's triangle, as shown in Figure 21, and Kirchhoff's voltage law, while precordial leads (V1-V6) provide direct measurements from the chest [72].

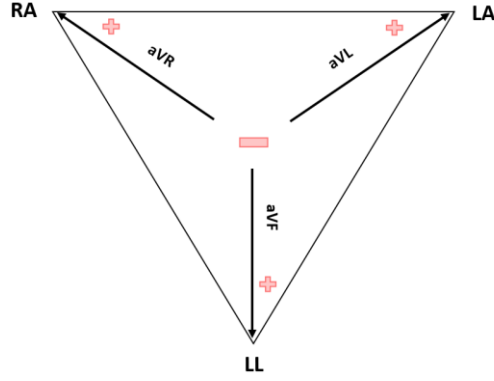


Figure 21. Einthoven's Triangle [73].

The augmented leads are defined as:

$$aVR = RA - \frac{(LA + LL)}{2}$$

$$aVL = LA - \frac{(RA + LL)}{2}$$

$$aVF = LL - \frac{(RA + LA)}{2}$$

The augmented leads are formed by referencing the potential of one limb electrode to the average of the remaining two electrodes, with amplitude scaling applied to enhance signal magnitude [74].

This computational approach expands the diagnostic information available from a limited number of physical electrodes and is commonly implemented in digital ECG systems to support multi-lead visualization and analysis.

## 2.4 Data Transmission Mechanisms

Once the ECG signal is processed, it must be transmitted to an external interface for display, storage, or further analysis. Common communication technologies employed include USB,

Bluetooth, and Wi-Fi, each suited to specific applications depending on latency, power, and accessibility requirements [75].

#### **2.4.1 USB Communication**

USB (Universal Serial Bus) provides a stable, high bandwidth wired connection commonly used during development or clinical scenarios requiring low latency and reliable data transfer. ECG data, once processed, is formatted as a serial data stream and sent through a USB-to-serial converter. On the receiving end, the signal may be decoded and visualized in real-time using custom desktop applications, Python scripts, or MATLAB interfaces [76].

#### **2.4.2 Wi-Fi Transmission**

Wi-Fi communication provides high data throughput and extended operating range, making it well suited for Internet-of-Things (IoT)–based biomedical monitoring systems [77]. In ECG monitoring applications, Wi-Fi can support continuous wireless transmission of digitized physiological signals using standard communication protocols such as HTTP or WebSocket-based streaming, enabling reliable real-time data transfer and interoperability with web technologies.

Although Wi-Fi generally consumes more power than short-range wireless alternatives, it offers higher bandwidth, multi-client access, and stable long-duration communication. These characteristics make it particularly suitable for centralized monitoring environments and IoT-enabled medical systems where continuous data availability and reliable connectivity are required [78].

### **2.5 IoT Integration**

The Internet of Things (IoT) paradigm enables conventional ECG hardware to operate as networked medical monitoring devices, where sensing, processing, and communication

capabilities are integrated within embedded or web-based systems [79]. In IoT-enabled ECG architectures, each device can function as an independent node within a distributed monitoring network, enabling scalable and flexible deployment models.

IoT-based healthcare systems commonly support remote accessibility, real-time data visualization, and retrospective analysis through network-connected interfaces [80]. By abstracting data acquisition from data presentation layers, such architectures allow physiological data to be accessed by multiple authorized clients while maintaining interoperability with standard web technologies.

Through the integration of sensing, embedded processing, wireless communication, and web-based visualization, IoT-enabled ECG systems enhance accessibility and continuity of cardiac monitoring across diverse deployment environments [81], [82].

## **2.6 Display Mechanisms**

After transmission, the ECG data must be visualized for interpretation and clinical decision-making. This can be achieved through various output methods, some of which are:

- **Real-Time Graphical Interface:** Waveforms are displayed through custom desktop GUIs, mobile apps, or medical software, showing heart rate, waveform shape, and time markers [83].
- **Web-Based Visualization with Storage:** Web-based ECG systems may integrate real-time waveform visualization with centralized storage, allowing recorded sessions to be archived and accessed later for review or offline analysis [84]
- **Printed Output:** ECG recordings can be saved as PDF reports or printed on thermal paper, simulating traditional hospital-grade ECG outputs [85].

Together, these visualization and output methods enable ECG data to be observed in real time or reviewed retrospectively, supporting clinical interpretation and documentation across home-based, outpatient, and clinical monitoring environments.

## 2.7 Analog Front-End (AD8232)

The AD8232 is an integrated signal conditioning block specifically designed for ECG and other biopotential measurements. Manufactured by Analog Devices, it acts as an analog front-end (AFE) to amplify, filter, and stabilize low-level physiological signals before digitization. Internally, the AD8232 includes an instrumentation amplifier, operational amplifiers configured as active filters, and a right-leg drive (RLD) amplifier, as seen in Figure 22. [86]

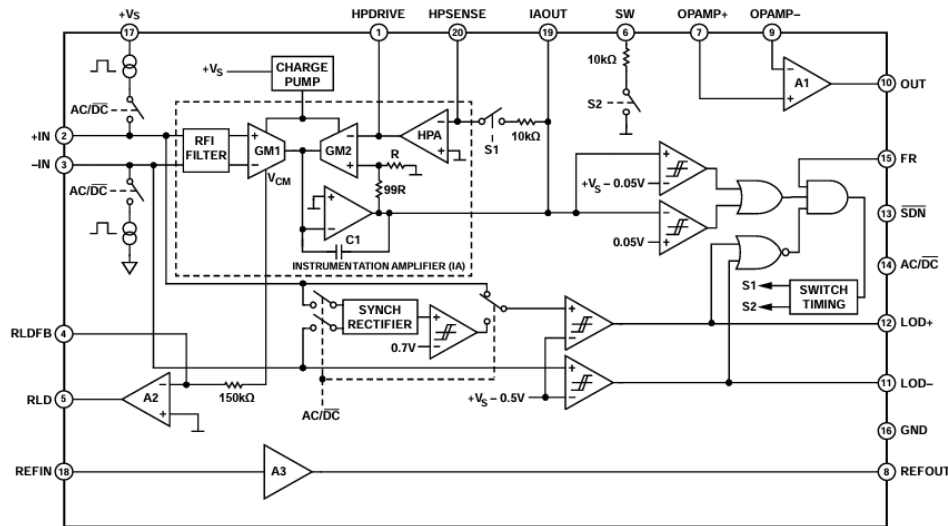


Figure 22. Schematic diagram [86].

The chip outputs a single-ended analog signal representing the ECG waveform, which can be directly interfaced with the ADC of a microcontroller. The instrumentation amplifier offers a high common-mode rejection ratio (CMRR), essential for minimizing interference from external sources such as power lines or motion artifacts [86].



The device supports a typical supply voltage of 3.3V and is capable of operating at ultra-low current consumption (typically  $\sim 170 \mu\text{A}$ ), making it highly suitable for portable battery-operated systems.

One of the unique features of the AD8232 is its built-in lead-off detection and right-leg drive circuitry, which stabilizes the patient's reference potential and reduces common-mode noise [86].

### **2.7.1 AD8232 Instrumentation Amplifier**

After the ECG signal is collected from the electrodes and safely limited using series protection resistors, it is fed into the instrumentation amplifier stage of the AD8232. The purpose of this stage is to amplify the very small differential ECG signal while rejecting unwanted signals that appear equally on both electrodes, such as power-line interference and environmental noise. This makes the IA a critical stage in ensuring that the recorded ECG waveform is accurate and suitable for further processing.

Internally, the instrumentation amplifier shown in Figure 23 is composed of two closely matched transconductance amplifiers (GM1 and GM2), a dc-blocking amplifier (HPA), and an integrator formed by capacitor C1 and an operational amplifier. The first transconductance amplifier, GM1, senses the voltage difference between the two input electrodes and converts this differential voltage into a proportional current. Through the internal feedback network, the circuit forces the same voltage to appear across the inputs of the second transconductance amplifier, GM2. When both amplifiers are perfectly balanced, the generated currents match. Any small mismatch between these currents produces an error current, which is then integrated across capacitor C1. This integration process generates a voltage at the output of the instrumentation amplifier (IAOUT), resulting in a stable and amplified version of the original ECG signal.

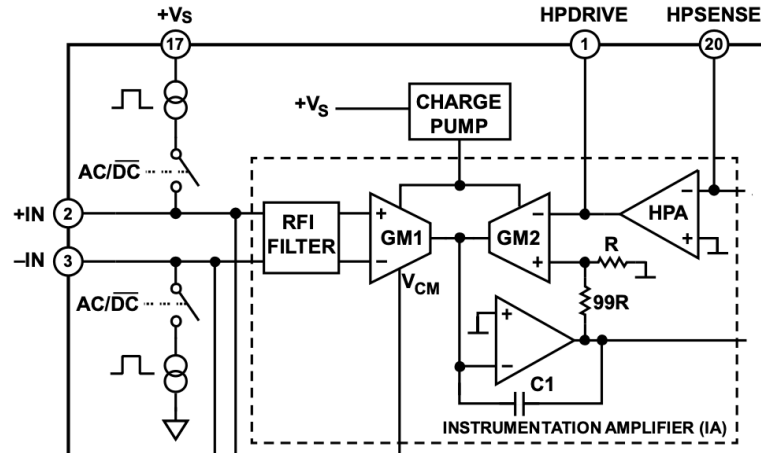


Figure 23. Internal instrumentation amplifier of the AD8232 [86].

The overall gain of the instrumentation amplifier is internally set to 100 using a resistive feedback network that feeds a scaled version of the output signal back through GM2. At the same time, the dc-blocking amplifier continuously integrates slow changes and deviations from the reference voltage.

This design allows the instrumentation amplifier to handle large dc offsets at electrode inputs (up to  $\pm 300$  mV) without saturating. These offsets are effectively shifted to the GM2 inputs with equal magnitude and opposite polarity, ensuring that the small-amplitude ECG signal of interest remains within the linear operating range of the amplifier.

To further improve performance in real-world conditions, the AD8232 includes an internal charge pump that increases the effective supply voltage available to the transconductance amplifiers. This additional headroom expands the common-mode voltage range of the instrumentation amplifier and reduces the risk of saturation in the presence of large common-mode signals, such as 50/60 Hz mains interference. The charge pump operates using an internal oscillator with a frequency of approximately 500 kHz, which is well above the ECG frequency band and therefore does not interfere with the recorded signal.

In addition to amplification, the instrumentation amplifier input stage also incorporates protection against radio frequency interference (RFI). Each input includes an internal  $10\text{ k}\Omega$  resistor and a small gate capacitance of approximately  $15\text{ pF}$  as seen in Figure 24. Together, these elements form a low-pass filter at each input, which attenuates high-frequency RF signals before they can be rectified and appear as unwanted dc offsets at the output. This internal filtering reduces susceptibility to electromagnetic interference without requiring additional external components.

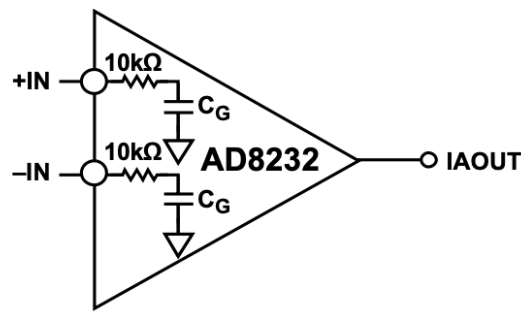


Figure 24. RFI filter without external capacitors [86].

While the instrumentation amplifier provides high differential gain and excellent common-mode rejection, its output may still contain low-frequency components arising from electrode dc offsets, baseline wander due to respiration, and slow motion artifacts. Therefore, the IA stage is followed by a high pass filtering stage. This next stage removes residual dc and very low-frequency components from the amplified signal, stabilizing the baseline and preparing the ECG waveform for subsequent low-pass filtering and digitization. The combination of the IA and the high-pass filter ensures that only the clinically relevant ECG frequency band is preserved for further processing. [86]

### **2.7.2 AD8232 Operational Amplifiers as Active Filters**

Following the instrumentation amplifier stage, the ECG signal is passed through a high-pass filter (HPF) to remove unwanted low-frequency components and residual dc offsets caused by slow motion artifacts and respiration, and baseline wander caused by changes in electrode–skin impedance. Although the instrumentation amplifier can tolerate large dc electrode offsets without saturation, additional filtering is required to stabilize the ECG baseline and improve signal clarity. To achieve this offset rejection, the AD8232 implements an RC network connected between the instrumentation amplifier output (IAOUT), HPSENSE, and HPDRIVE pins, forming a dc-blocking high-pass filter stage. The AD8232 also supports the implementation of extending the high-pass filter using a dc-blocking topology to form a second pole for improved low-frequency rejection.

After baseline stabilization through high-pass filtering, the signal is subjected to low-pass filtering to suppress unwanted high-frequency components arising from electromyographic noise, electromagnetic sources, and switching activity from nearby electronic circuits. If not attenuated, these components can obscure clinically relevant waveform features and degrade signal interpretability. Low-pass filtering is implemented in the AD8232 using an internal operational amplifier in conjunction with external RC components, forming an active low-pass filter stage. This filtering limits the upper bandwidth of the signal, ensuring that the preserved frequency content corresponds to the physiological ECG range while reducing noise prior to analog-to-digital conversion. The AD8232 supports a two or three-pole low-pass filter as well, with cutoff frequencies that can be externally configured via resistor-capacitor (RC) networks. These filters are crucial for isolating the ECG signal from noise components like baseline wander and muscle noise.

In both the high-pass and low-pass filtering stages, the AD8232 reference output (REFOUT) is used to establish a stable mid-supply DC operating point for the ECG signal. In the high-pass filter, REFOUT re-biases the signal after DC blocking, ensuring that the ECG waveform remains centered within the linear operating range of subsequent stages. Similarly, in the low-pass filter, REFOUT functions as a virtual ground that biases the internal operational amplifier and its feedback network, allowing linear filtering of the bipolar ECG signal in a single-supply system. Referencing either filter stage to ground instead of REFOUT would shift the signal bias, drive the amplifiers toward saturation, and result in waveform distortion or clipping. [86]

### **2.7.3 Right Leg Drive (RLD)**

The right leg drive circuitry is an essential component in ECG acquisition systems and is primarily used to improve common-mode noise rejection, particularly against power-line interference. In practical environments, the human body behaves like an antenna and can pick up electromagnetic interference from surrounding electrical sources. This interference appears as a common-mode voltage at the ECG electrode inputs and can significantly degrade signal quality if not properly suppressed.

The RLD amplifier operates by sensing the common-mode voltage present at the inputs of the instrumentation amplifier and actively feeding back an inverted version of this signal into the subject through the driven electrode, typically connected to the right leg. By injecting this compensating signal, the RLD circuit counteracts common-mode voltage variations at the electrode–skin interface, thereby reducing the amplitude of interference seen at the differential inputs and improving the overall common-mode rejection ratio (CMRR) of the ECG system.

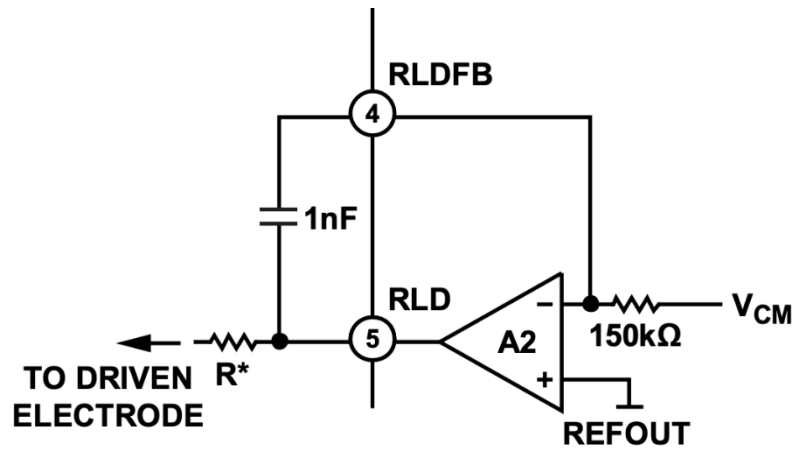


Figure 25. RLD circuit [86].

In the AD8232, the common-mode signal is derived internally from the transconductance amplifier GM1 and routed to the inverting input of the internal RLD amplifier (A2) through an internal 150 kΩ resistor, as shown in Figure 25.

The non-inverting input of the RLD amplifier is biased to the reference voltage (REFOUT), ensuring stable operation in a single-supply configuration. This internal architecture allows the RLD loop to function as an active feedback system that stabilizes the patient’s common-mode potential relative to the analog front end.

Proper operation of the RLD circuit relies on the availability of a stable reference voltage rather than true ground. In single-supply ECG systems, using 0 V ground as the reference would cause

the instrumentation amplifier, filtering stages, and RLD loop to lose their correct dc operating point. This would lead to amplifier saturation, waveform clipping, unstable common-mode feedback, and severe signal distortion. The REFOUT pin of the AD8232 provides a buffered mid-supply virtual ground that allows all analog stages, including the RLD amplifier, to process bipolar ECG signals while operating from a unipolar supply.

Overall, the RLD circuitry plays a critical role in stabilizing the ECG signal, suppressing common-mode interference, and maintaining measurement accuracy. By actively controlling the subject's common-mode potential and integrating seamlessly with the reference buffer and instrumentation amplifier stages, the RLD circuit significantly enhances signal quality without compromising patient safety. [86]

#### **2.7.4 Reference Buffer and Network Biasing**

The AD8232 operates from a single-supply voltage, which requires all internal analog signal processing to be referenced around a stable bias level rather than true ground. To support this mode of operation, the device includes an internal reference buffer that generates a virtual ground at a defined dc voltage between the supply and ground. All amplified and filtered ECG signals within the device are centered around this reference voltage, ensuring proper handling of bipolar ECG waveforms under single-supply conditions. The reference voltage is established at the REFIN pin using an external resistive divider connected between the positive supply and ground shown in Figure 26.

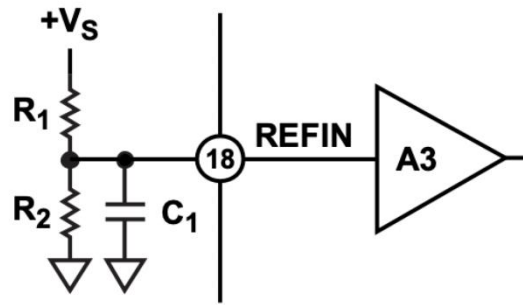


Figure 26. Setting the internal reference [86].

The voltage present at the REFIN pin is internally buffered by the AD8232 and made available at the REFOUT pin, providing a low-impedance and stable bias voltage. This reference level defines the dc operating point of the instrumentation amplifier and subsequent filtering stages. As a result, when the differential ECG input is zero, the output of the analog front end is centered around the reference voltage rather than ground.

By biasing the signal chain at REFOUT, the system ensures linear operation of the amplifiers and filters while preserving the full dynamic range of the ECG waveform within the available single-supply voltage limits. The REFOUT node may also be used externally as a stable bias reference for downstream analog circuitry or for interfacing with an analog-to-digital converter.

Overall, the reference buffer and biasing network establish a stable mid-supply operating point for the entire ECG signal chain, enabling low-power, single-supply operation while maintaining signal integrity and noise immunity. [86]

### 2.7.5 Leads Off Detection Mode

The AD8232 analog front end includes built-in lead-off detection circuitry to identify disconnected or poorly attached electrodes during ECG acquisition. The device supports both dc and ac lead-off detection modes, which are optimized for three-electrode and two-electrode ECG configurations,



respectively. The selected detection mode determines how electrode disconnection is detected and how lead-off status is reported.

DC lead-off detection is intended for three-electrode ECG systems that include a driven right-leg electrode. In this mode, each instrumentation amplifier input is weakly biased toward the positive supply rail through high-value pull-up resistors. During normal operation, the subject's body potential, which is stabilized by the right-leg drive (RLD) amplifier, maintains both inputs within the allowable common-mode range of the instrumentation amplifier.

If an electrode becomes disconnected, the affected input is no longer constrained by the subject's potential and drifts toward the positive supply rail due to the pull-up resistor. When the input voltage approaches within approximately 0.5 V of the positive supply, the AD8232 detects a dc lead-off condition. Because each input is monitored independently, dc lead-off detection allows identification of which specific electrode has been disconnected.

Lead-off status in dc mode is reported through two digital comparator outputs. The LOD+ pin is driven high when the electrode connected to the non-inverting input (+IN) is disconnected, while the LOD− pin is driven high when the electrode connected to the inverting input (−IN) is disconnected. DC lead-off detection mode is selected by driving the AC/DC control pin low.

AC lead-off detection is designed for two-electrode ECG configurations and does not require a driven right-leg electrode. In this mode, the AD8232 injects a small high-frequency current into the input terminals and detects changes in the resulting differential voltage caused by variations in electrode impedance. If an electrode becomes disconnected, the increased impedance triggers a lead-off condition.

Unlike dc lead-off detection, ac lead-off detection cannot distinguish which electrode has been disconnected. In this mode, the LOD+ output is driven high during a lead-off event, while LOD− remains low. AC lead-off detection is selected by driving the AC/DC control pin high. [86]

## 2.8 Microcontroller (ESP32)

The ESP32, shown in Figure 27, was used as the microcontroller unit in this project. It is a low-cost, low-power system-on-chip (SoC) microcontroller developed by Espressif Systems. It features a dual-core processor with integrated Wi-Fi and Bluetooth capabilities, making it ideal for wireless communication and Internet of Things applications.



Figure 27. The ESP32 microcontroller.

The ESP32 operates at a core voltage of 3.3V and supports a wide input voltage range via external regulators. It includes a variety of peripheral interfaces such as Serial Peripheral Interface (SPI), Inter-Integrated Circuit (I<sup>2</sup>C), Universal Asynchronous Receiver-Transmitter (UART), Pulse-Width Modulation (PWM), and a 12-bit analog-to-digital converter (ADC) that can digitize analog input signals at rates suitable for real-time biomedical monitoring [87].

I<sup>2</sup>C is a synchronous serial communication interface that uses two shared lines: a serial clock line (SCL) and a serial data line (SDA). The clock line synchronizes data transfer, while the data line

carries bidirectional data between the master and the peripheral device (slave). The ESP32 includes hardware I<sup>2</sup>C controllers that can be routed to general-purpose I/O pins through the internal GPIO matrix.

The ESP32 also includes internal hardware timers, Real-Time Clock (RTC) memory, and ultra-low power co-processors, allowing for efficient power management. Firmware development is supported through the Arduino IDE or the native ESP-IDF environment, with a rich set of libraries available for wireless protocols and sensor interfacing [88].

## **2.9 ADS1115 Analog-to-Digital Converter**

The ADS1115 is a precision, low-power, 16-bit ADC designed for accurate measurement of low-frequency, low-amplitude signals. It employs a delta-sigma ( $\Delta\Sigma$ ) conversion architecture, where the input signal is oversampled and processed through a noise-shaping modulator, followed by a digital low-pass filter and decimation stage. This process shifts quantization noise to higher frequencies and suppresses it digitally, resulting in improved effective resolution within the signal band of interest.

A key feature of the ADS1115 is its support for differential input measurements. Differential conversion allows the ADC to measure the voltage difference between two input terminals while rejecting common-mode voltages that appear equally on both inputs.

To support low-power and portable systems, the ADS1115 includes an internal voltage reference and multiple operating modes, including single-shot and continuous conversion. Communication with a host microcontroller is achieved through an I<sup>2</sup>C interface, which provides a compact and efficient means of configuring the converter and retrieving digitized samples. [89]

## **2.10 Web-Based System Architecture and Communication Technologies**

Modern biomedical monitoring systems increasingly rely on web-based architectures to enable remote access, real-time visualization, and scalable data management. Such systems are commonly structured according to a client–server model, where system functionality is divided between a backend server and one or more frontend clients. This architectural separation improves modularity, security, and maintainability, particularly in systems that continuously acquire physiological signals such as ECGs [90].

The backend forms the central control layer of the system. It is responsible for receiving physiological data from acquisition devices, managing system state, performing signal handling and processing, enforcing access control, and storing data for later retrieval. In continuous monitoring applications, the backend must be capable of handling time-stamped data streams while organizing them into well-defined recording sessions. This approach allows data to be validated, processed, and archived in a controlled and reproducible manner before being made available to user interfaces [91].

Interaction between system components is typically achieved through RESTful web services, which are standard web-based communication methods that allow system components to communicate by sending and receiving data through simple web requests based on widely used internet standards including HTTPs. In REST-based systems, system resources such as sessions, users, and recorded data are accessed through clearly defined specific server addresses designed to perform particular actions, called endpoints, using simple request–response interactions [92].

While RESTful communication is effective for control operations and data retrieval, real-time biomedical signal visualization requires continuous, data transfer with little delay. For this purpose, many modern systems employ WebSocket-based communication, which establishes a persistent,

bidirectional connection between the backend and connected user interfaces (such as web browsers). It enables near-real-time streaming of physiological signals and system events, allowing ECG waveforms and alerts to be delivered to user interfaces with minimal delay [93].

The front end represents the client-side component of the system and is typically implemented as a browser-based interface. Its primary role is to present physiological data in an interpretable form, provide controls for session management, and allow users to review recordings. By performing data processing and storage on the backend server, the frontend remains focused on displaying information and interacting with the user. [90], [91].

### **Chapter 3: Literature Review**

Wearable ECG devices are central to offering real-time monitoring through lightweight and user-friendly form factors, helping reduce hospital visits and improving patient compliance [94][95]. These systems frequently utilize cloud services not just for data storage, but also for analysis and visualization through mobile or web-based interfaces [96][97]. This remote accessibility enables physicians to make timely decisions based on continuous and historical ECG data, while also allowing patients to engage more actively with their own cardiovascular health.

The integration of AI and ML into these platforms enables early detection of arrhythmias, heart rate anomalies, and ischemic patterns, contributing to faster and more reliable diagnoses [98][99]. These algorithms are often trained on large, annotated datasets and embedded into the processing pipeline to operate autonomously on either the device or in the cloud. This end-to-end automation ensures that potentially life-threatening conditions are flagged in near real-time, even in the absence of clinical supervision.

All these innovations are made possible by improvements in analog-to-digital conversion, low-power microcontrollers, and advanced signal processing algorithms [100][101]. Enhanced ADC resolution allows for more detailed waveform capture, which improves diagnostic granularity. Meanwhile, energy-efficient MCUs and optimized signal processing reduce power consumption and allow for continuous operation without frequent recharging, a critical aspect in long-term patient monitoring and ambulatory care.

### **3.1 Portable ECG Monitoring Systems**

Recent advancements in portable ECG monitoring systems reflect a broader movement toward compact, energy-efficient designs that support real-time and remote health diagnostics. One recurring approach centers on low-power microcontrollers and readily available analog front-end chips, which together enable minimal hardware configurations without compromising signal quality. Bravo-Zanoguera et al. [102] and Iskandar et al. [103], for instance, implemented systems using the AD8232 module coupled with Arduino platforms, illustrating how readily accessible components can deliver clinically viable single-lead ECG readings. Their reliance on open-source development environments, paired with Bluetooth-based transmission and custom PCB implementations, underscores how portability is increasingly linked to modular design and wireless capability.

This foundation expands in complexity and functionality with the work of Guan [104], who employed the STM32F429 microcontroller to integrate signal acquisition, preprocessing, and cloud-based transmission into a unified device. By embedding all processing onto a compact board, this system addresses the limitations of bulkier setups and reduces user dependency on external infrastructure, thereby enhancing both physical portability and functional autonomy. X. Li et al. [105] and J. Li et al. [106] extended this vision by integrating wireless modules such as

nRF905 and nRF24L01+ respectively, showcasing how flexibility in communication standards allows tailoring of devices to varied environmental and use-case constraints.

Energy efficiency remains a persistent theme, as shown by Vo et al. [107] who developed a solar-powered ECG device capable of operating autonomously in off-grid locations. The use of high-resolution ADCs and minimal power draw makes such devices especially suited for deployment in rural or emergency settings where power infrastructure is lacking. These enhancements to energy management translate directly into improved device portability and sustained usage without frequent recharging.

Portability, however, is not solely a function of circuit design. Ergonomics and user comfort also play a decisive role in long-term monitoring adoption. Wang et al. [108] tackled this aspect through non-contact, flexible electrodes fabricated via printed circuit technologies. These electrodes, capable of maintaining signal fidelity during physical activity, introduce a new dimension of wearability that aligns with the needs of continuous patient monitoring, especially in ambulatory or athletic scenarios. The importance of this is reflected in the handheld miniECG system proposed by Zepeda-Echavarria et al. [109], which emerged from iterative, user-centered design. Their four-electrode prototype integrates feedback from clinical environments to optimize size, shape, and ease of use, demonstrating how attention to form factors can elevate a device from prototype to real-world utility.

Such attention to usability continues in the system by Ehresh et al. [110], where the use of ZigBee wireless technology and a user-friendly LabVIEW GUI allows seamless real-time visualization. By minimizing circuit complexity through INA128 amplifiers and pairing with portable computing platforms, the device becomes feasible for non-specialist users in home-care scenarios. Meanwhile, Li et al. [111] advanced telemetry reliability by embedding Bluetooth communication

directly within the ECG circuit, thus ensuring continuous signal streaming while reducing software overhead and hardware bulk. This approach further reinforces the notion that true portability is not achieved through downsizing alone, but through intelligent integration of transmission, acquisition, and interface design.

Cumulatively, these studies illustrate how portable ECG technology has progressed through iterative improvements in both design and functionality. From the use of simple microcontroller-based systems to more advanced embedded platforms with integrated communication and power solutions whether through wireless adaptability, ergonomic enhancements, or self-sustaining power options; these efforts point toward an evolving understanding of how portability can be achieved without compromising diagnostic utility.

### **3.2 IoT/Cloud-Enabled ECG Devices**

Modern developments in IoT-integrated ECG systems have transformed how cardiac data is collected, transmitted, and analyzed, shifting diagnostics from clinical environments to remote, real-time platforms. Across several studies, a recurring methodology is the integration of ECG acquisition modules such as AD8232 with wireless microcontrollers like NodeMCU or ESP8266. Islam et al. [112] exemplified this through a real-time ECG monitoring system that employed AD8232 for signal acquisition and NodeMCU for wireless transmission, with data sent to ThingSpeak for cloud visualization. Their system demonstrated high signal fidelity and stable cloud synchronization, illustrating the feasibility of low-cost remote cardiac monitoring solutions.

The importance of cloud data storage and analytics was further emphasized by the system presented by Uddin et al. [113], which also used AD8232 and NodeMCU but expanded its design with mobile interfacing and alert functionalities. The system reliably sent ECG data to cloud servers, where threshold-based notifications could be triggered, underscoring how simple IoT



systems can enhance proactive healthcare. Likewise, the work by Rehman et al. [114] featured a multi-layered architecture consisting of an ECG sensor node, microcontroller, and cloud-based dashboard, achieving consistent performance in both live streaming and storage applications. The ability to integrate cloud dashboards with physiological data expands the diagnostic horizon while maintaining portability.

Several implementations explored extended communication protocols and real-time monitoring enhancements. For example, in the study by Jan et al. [115], MQTT protocols were used alongside ESP32 to ensure lightweight yet reliable data transfer to cloud storage. The system achieved low latency and minimal packet loss in remote transmission, marking a key advancement in robustness and scalability. Complementary to this, the system by Arunkumar et al. [116] integrated Bluetooth and Wi-Fi modules to support both short- and long-range ECG monitoring. Their platform featured adaptive data compression to manage bandwidth, which significantly extended the usability of the device in low-resource settings.

A notable emphasis on rural and underserved healthcare is evident in the work by Rao et al. [117], who designed an IoT-based ECG system optimized for remote communities. Utilizing GSM and cloud support, the device ensured uninterrupted transmission of critical cardiac signals despite unstable local networks. Results from this implementation validated its diagnostic capability in real-world settings. This priority is echoed by the system of Aghamohammadi et al. [118], which introduced redundancy in data transfer by supporting both local and cloud-based storage. Their testing confirmed continuous functionality even when internet access was interrupted, offering a valuable model for robust IoT ECG design.

Sensor optimization and energy efficiency were tackled in the study by Gao et al. [119], where low-power electronics and dynamic sampling techniques were employed. Their system leveraged

IoT integration to batch-transfer ECG data at optimal intervals, reducing energy drain while preserving signal quality. These findings reinforce that cloud connectivity does not necessarily require constant streaming, allowing for smarter, resource-conscious deployments. Similarly, the approach by Jafari et al. [120] emphasized scalable design through modular sensor arrays and mobile-cloud interfacing, enabling expansion across multiple patients and locations.

Lastly, the work by Sadat et al. [121] highlighted the importance of user interface and real-time feedback. Their mobile-integrated ECG system pushed data to a cloud server while simultaneously displaying results locally, merging IoT flexibility with immediate usability. By demonstrating accurate real-time signal visualization and cloud archiving, this dual-mode setup bridged convenience and reliability, a recurring theme across all successful implementations.

Collectively, these studies demonstrate how IoT-enabled ECG systems evolve through consistent refinement in cloud architecture, transmission stability, and user adaptability. Whether through efficient protocols, mobile integration, or redundancy planning, the common trajectory is toward decentralizing healthcare diagnostics without compromising performance. These approaches illuminate viable frameworks for achieving reliability, accessibility, and autonomy in future ECG system designs.

### **3.3 Diagnosing ECGs**

The diagnostic capabilities of modern ECG systems have seen marked advancement, propelled by both hardware and algorithmic innovation. One recurring pattern among recent works is the utilization of embedded platforms such as Raspberry Pi, which allow the integration of real-time classification algorithms directly within the acquisition device. A notable example is the system by Thiagaraja et al. [122], which paired a Raspberry Pi with an analog front-end to acquire ECG signals and apply arrhythmia detection algorithms on-board. This design achieved live

classification using pre-trained models and emphasized the potential for real-time feedback in a compact, portable form, which is essential for remote diagnostics and on-site decision making.

Cloud-connected classification models further extend diagnostic capacity. In a study by Ryu et al. [123], ECG signals were acquired using wearable sensors and transmitted to a cloud-based diagnostic engine that used deep learning to identify abnormal patterns. This approach enabled scalable deployment of automated diagnosis across different care settings and highlighted the potential for integrating wearable biosensors with hospital infrastructure. Meanwhile, Zhu et al. [124] implemented their system using MATLAB and signal processing tools to classify arrhythmias, achieving high classification accuracy on benchmark datasets like MIT-BIH. Their focus on preprocessing methods—bandpass filtering and segmentation—along with feature extraction techniques such as wavelet transforms, showed how algorithmic decisions directly impact diagnostic robustness.

Bridging hardware and software domains, Paviglianiti et al. [125] leveraged wearable ECG acquisition modules and coupled them with predictive models aimed at detecting atrial fibrillation. Their methodology involved time-domain and frequency-domain analysis, demonstrating how even limited leads can be repurposed for detecting specific conditions. Similarly, the study by Roy et al. [126] proposed a machine learning-based ECG analyzer, using CNN and LSTM architectures to distinguish between different cardiac events. Their model's ability to generalize across datasets indicated promise for deployment in clinical triage systems.

Large-scale studies have confirmed the scalability of such techniques. Raghunath et al. [127], published in *The Lancet Digital Health*, trained deep neural networks on extensive ECG datasets, and validated their models against cardiologist consensus. The system reached diagnostic accuracy on par with human experts, reinforcing the clinical viability of ML-based diagnosis. Similarly,

Mazumder et al. [128] focused on lightweight diagnostic algorithms capable of running on resource-constrained devices. Their work demonstrated that even edge devices can handle accurate arrhythmia classification, lowering barriers for widespread adoption.

Real-time feedback and live visualization remain crucial for actionable diagnostics. Pajooyan et al. [129] used discrete wavelet transforms and deep learning to enable multi-class arrhythmia detection, achieving strong performance across noisy datasets. Their methodology was particularly centered around adaptive filtering and dynamic windowing, useful for diagnosing patients in non-clinical environments. Complementing this, Yousefi et al. [130] addressed ECG denoising and signal integrity, critical prerequisites for any diagnostic algorithm. Their results showed improved classification when preprocessing stages were fine-tuned to real-world noise.

The study by Roshini et al. [131] combined ECG acquisition with Bluetooth-enabled data transfer and MATLAB-based arrhythmia classification. The choice of Bluetooth highlighted a balance between portability and connectivity, while the software approach allowed post-acquisition diagnostic analysis. Finally, Di Noto et al. [132] proposed a deep learning system focused specifically on ECG waveform classification. Their results reached high F1-scores across multiple rhythm types, illustrating the growing maturity of end-to-end diagnostic systems that require minimal user intervention.

Collectively, these efforts reveal how diagnostics can now be decentralized from hospital settings without compromising accuracy. Whether through embedded platforms like Raspberry Pi, adaptive filtering and signal enhancement, or cloud-synced learning models, these systems reflect a trend toward empowering frontline and remote users with reliable tools for immediate cardiac insight.

### 3.4 Integrated ECG Systems: Combining Portability, IoT, and Diagnostic Capability

Numerous studies have investigated the integration of IoT and Cloud communications and diagnostic intelligence into portable ECG systems, with recurring architectures and components shaping the standard design approach.

Qtaish et al. [133] developed a portable ECG monitoring system that transmits three-lead ECG data via Wi-Fi using HTTP and MQTT protocols. The system utilizes the AD620 amplifier, followed by cascaded filtering (high-pass, low-pass, notch) to clean the signal before digitization. A microcontroller handles data preprocessing and uploads the results to a cloud database. Their design emphasizes signal integrity and leverages machine learning models (SVM, CNN, decision trees) for automated arrhythmia detection. The device uses a 3-lead setup based on 3 electrodes, with no explicit mention of physical dimensions, but it is designed for portability.

In a similar approach, Zagan et al. [134] proposed a multi-lead IoT-enabled ECG system using AD620 amplification, multi-stage analog filtering, and wireless transmission through Wi-Fi, Bluetooth, or LoRa. Real-time data processing is carried out on a microcontroller before being sent to a cloud platform for further analysis. The authors also implemented ML algorithms to support classification of cardiac rhythms, mirroring the diagnostic intent found in [133]. This system uses a multi-lead setup, likely 3 or more leads, with typical 3 to 5 electrodes. Physical size is not provided, though portability is emphasized.

Jeon et al. [135] contributed a lightweight, wearable ECG device focused on real-time signal analysis. Following the now-standard filtering pipeline (0.5 Hz high-pass, 100 Hz low-pass, and notch), the device performs onboard feature extraction, detecting QRS complexes and R peaks. It supports immediate data display and wireless transmission, and integrates adaptive digital filtering and ML models, including CNNs and random forests, for abnormality detection. The paper

mentions the system is lightweight and wearable and employs a multi-lead configuration (likely 3-lead) using surface electrodes.

Ehresh et al. [136] presented a cost-effective and power-efficient ECG system based on the INA128 amplifier and ATmega328P microcontroller. Though simpler in architecture, the system achieves high diagnostic sensitivity and specificity through robust signal conditioning and transmission via ZigBee. A custom GUI developed in MATLAB and LabVIEW allows real-time visualization, demonstrating how effective signal filtering alone can maintain clinical accuracy without high-end processing. It uses a 3-lead configuration with 3 electrodes. The device is wearable, with no specific dimensions reported.

Rahaman et al. [137] emphasized the role of machine learning in ECG classification. Their system incorporates AD620 amplification, standard filtering stages, and wavelet-based digital noise reduction. After extracting features like R peaks and QRS intervals, the data is transmitted wirelessly to a cloud server where it is analyzed using supervised ML techniques (SVM, k-NN, CNN), enabling real-time identification of abnormal cardiac patterns. This system also operates with a 3-lead configuration and 3 electrodes, and is described as suitable for long-term, real-time monitoring.

Yi et al. [138] explored wearable ECG monitoring with a focus on atrial fibrillation detection. Using high-fidelity electrodes and a low-noise analog front end, their system applies lossy compression before transmission to conserve bandwidth. On the cloud, a ResNet-based deep neural network classifies ECG signals with accuracy above 98%, demonstrating the potential of deep learning in personalized, scalable healthcare. The system uses a multi-lead configuration and wearable form factor; while specific lead count or device dimensions are not listed, its design supports ambulatory use.

Alimbayeva et al. [139] proposed a portable ECG system designed for real-time cardiac assessment, employing AD620 amplification, standard analog filtering, and wireless data transmission via Bluetooth or Wi-Fi. Their diagnostic framework includes ML classifiers (SVM, k-NN, ANN) trained on ECG datasets, enabling automated detection of arrhythmias and supporting continuous remote monitoring. It supports standard bipolar and unipolar lead configurations, likely 3 to 6 leads, with 3+ electrodes. The system emphasizes portability and wireless access, with no explicit size dimensions.

A clear trend emerges across these studies. Nearly all systems rely on surface electrodes with AD620 amplification [133][134][135][137][139] and universally apply a three-stage filtering approach: a high-pass filter ( $\sim 0.5$  Hz) to remove baseline drift, a low-pass filter ( $\sim 100$  Hz) to reduce high-frequency noise, and a notch filter (50/60 Hz) for powerline interference. Digital enhancement through adaptive filters and wavelet transforms is also common [135][136][137][139]. Wireless data transmission (via Wi-Fi, Bluetooth, ZigBee, or GSM) and cloud-based platforms are consistently used to facilitate remote diagnostics. On the software side, machine learning plays a central role in five of the seven studies, with SVM, CNN, and k-NN being the most frequently employed algorithms for ECG signal classification, [133][134][135][137][139] while one uniquely implements a deep ResNet model [138]. Together, these recurring strategies form a foundational blueprint for our proposed system, validating our choice of components, architecture, and diagnostic goals.

### **3.5 Commercially Available Devices**

To better understand the landscape of portable ECG technology, a selection of commercial devices was analyzed based on their technical features, connectivity, power options, and diagnostic capabilities. This comparative review provides context for the development of our system and

highlights critical design considerations such as number of leads, data accuracy, IoT integration, and power management.

TeleCardio is a single-lead ECG device that integrates AI and IoT for real-time arrhythmia detection with an accuracy of 94% [140]. It uses edge devices to acquire ECG data, which is filtered, analyzed, and transmitted to a web dashboard for remote access by healthcare providers. The device supports cloud upload, local storage exceeding 20GB, and over 10 hours of battery life.

PocketECG offers a more clinical-grade monitoring solution, with versions supporting 3, 6, or 12 leads [141]. It captures continuous ECG data and classifies every heartbeat in real time. Full disclosure monitoring, IoT streaming to medical portals, and integration with hospital systems make it suitable for both home and hospital use. It features an onboard accelerometer and long battery life, lasting up to 30 days.

Withings BPM Core combines ECG, blood pressure measurement, and a digital stethoscope [142]. It records a single-lead ECG and transmits data over Wi-Fi and Bluetooth to a companion app. Data is automatically synchronized to the cloud and is accessible for multiple users. Its design emphasizes personal and family heart health monitoring rather than clinical diagnostics.

CardioNS is a 12-lead ECG device with plug-and-play functionality using a USB OTG connection to Android devices [143]. It allows real-time streaming and report generation in PDF format, enabling seamless communication between patients and healthcare professionals.

FL10 [144][145][146], Wellue Pulsebit EX Personal EKG Monitor [147] and VC6O EO1 [148], and are compact, single-lead ECG monitors optimized for portability and ease of use. They capture short ECG segments and use Bluetooth to transmit data to mobile apps for review and storage.



Some models incorporate built-in displays, external electrodes, and AI-driven analysis, but they generally lack continuous monitoring capability.

The KardiaMobile 6L offers six-lead ECG recording, striking a balance between clinical utility and portability [149]. It supports Bluetooth transmission, cloud storage, and optional subscription-based diagnostic features. Its affordability and multi-lead capability make it a competitive personal ECG solution.

Finally, the Humeds ECG Monitor uses a 3-lead configuration and advanced AI to detect over 99% of arrhythmias in under 30 seconds [150]. It integrates Bluetooth connectivity and cloud synchronization via a dedicated app, although continuous monitoring is not supported. Some features require a paid subscription, and the device is still in pre-order, indicating limited availability.

A summary of all analyzed devices is presented in Table 1, which compares their ECG configurations, connectivity platforms, price range, and notable features. These insights informed design decisions in our own ECG system development, especially regarding lead configuration, user interface simplicity, and cloud compatibility.

TABLE 1 Summary of the Market analysis Devices

| Device     | Operation Principle                                                                                       | Number of Leads | IoT Platforms                                                                                 | Price      | Notes/ Additional Features                                                                                  |
|------------|-----------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------|
| TeleCardio | Single-lead (Lead II) ECG with real-time acquisition, noise filtering, and AI-driven arrhythmia detection | 1 (Lead II)     | IoT-enabled edge devices; data transmitted to a cloud-based web portal via mobile app/website | Not stated | 94% arrhythmia detection accuracy; local storage capacity >20GB; rechargeable battery lasting over 10 hours |

| Device                    | Operation Principle                                                                                                                                         | Number of Leads                                                  | IoT Platforms                                                                                             | Price                       | Notes/ Additional Features                                                                                                                            |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| PocketECG                 | Continuous, full-disclosure ECG monitoring that captures every heartbeat and classifies arrhythmias using statistical analysis                              | Basic Version: 3 electrodes; Advanced: 6/12 leads; Pro: 12 leads | Real-time streaming to mobile app and website; integration with hospital systems and cloud connectivity   | Not stated                  | Offers different versions for personal and clinical use; built-in accelerometer; long-lasting rechargeable battery (up to 30 days on a single charge) |
| Withings BPM Core         | Multifunctional device combining blood pressure measurement, single-lead ECG (via 3 electrodes), and digital stethoscope for comprehensive heart assessment | 1 (via 3 electrodes integrated in the cuff)                      | Wi-Fi and Bluetooth; automatically synchronizes data with the Withings Health Mate app and cloud platform | Not stated (premium device) | Supports multiple user profiles (up to 8); free cloud storage with automatic data sync; sequential measurement process                                |
| CardioNS                  | 12-lead resting ECG device that records comprehensive cardiac signals and supports real-time telemetry                                                      | 12-lead (using 10 electrodes)                                    | Wired USB OTG connection to Android devices for plug-and-play functionality                               | Not stated                  | Generates customizable, printable PDF reports; emphasizes ease-of-use and rapid data sharing                                                          |
| FL10 Portable ECG Monitor | Handheld device with built-in sensors that record single-lead ECG without external wires or electrodes                                                      | 1 (Single lead)                                                  | Bluetooth connectivity to Android and iOS devices; data uploaded to cloud via dedicated mobile app        | Not stated                  | 1.77-inch color TFT-LCD display; rechargeable lithium battery supports up to 500 recordings; can be used on chest, wrist, ankle, or hand              |

| Device                                  | Operation Principle                                                                                                  | Number of Leads                     | IoT Platforms                                                                                                                | Price               | Notes/ Additional Features                                                                                                                              |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wellue Pulsebit EX Personal EKG Monitor | Personal EKG monitor offering both cable-free and external electrode modes for capturing ECG readings                | Likely 1 (Single lead)              | Bluetooth connectivity; syncs with a mobile app offering AI-driven analysis                                                  | Not stated          | 2.4-inch touchscreen interface; supports dual operation modes (built-in sensors vs. external electrodes); real-time AI analysis                         |
| VC6O EO1 Portable ECG Monitor           | Handheld single-lead ECG device with real-time display and local storage of readings                                 | 1 (Single lead)                     | Bluetooth connectivity to a smartphone app with cloud integration                                                            | Not stated          | 1.77-inch color TFT-LCD display; stores up to 500 ECG recordings; relatively shorter battery life requiring frequent recharging                         |
| KardiaMobile 6L                         | Six-lead personal ECG monitor that provides a comprehensive view of heart activity for detailed arrhythmia detection | 6 (Leads I, II, III, aVL, aVR, aVF) | Bluetooth connectivity; works with the Kardia app; data stored in the cloud with optional subscription for expanded analysis | Approximately \$129 | Quick measurement (results within 30 seconds); not suitable for pacemakers or ICD users; subscription-based features enhance diagnostic capabilities    |
| Humeds ECG Monitor                      | AI-powered device with a 3-lead ECG system designed for rapid arrhythmia detection and real-time data sharing        | 3 (3-lead)                          | Bluetooth connectivity; dedicated smartphone app with cloud integration for automatic data sharing                           | Not stated          | Detects over 99% of arrhythmias in approximately 30 seconds; some advanced features require a subscription; does not support continuous 24/7 monitoring |

## **Chapter 4: Methodology**

### **4.1 Introduction to Methodology**

This section presents the methodology followed in the design and implementation of the portable ECG monitoring system. The adopted methodology follows a structured engineering approach encompassing hardware design of the Analog Front end, digitization, transmission and embedded system integration, backend software development, and web-based frontend implementation.

The system was developed as a set of independent modules that together implement the complete ECG monitoring workflow, from signal acquisition to browser-based visualization. Each subsystem was designed and implemented separately and then integrated to ensure reliable operation across the full acquisition and monitoring pipeline.

The following subsections describe the hardware architecture, embedded firmware, backend processing system, frontend interface, and validation methodology employed in this project.

### **4.2 System Overview**

The ECG monitoring system implements a sequential acquisition and data-handling pipeline that spans analog signal conditioning, digital transmission, backend processing, and web-based visualization, as illustrated in the system block diagram in Figure 28.

At the hardware level, ECG biopotential signals are acquired using surface electrodes arranged in three limb-lead configurations (RA–LA, RA–LL, and LA–LL), each referenced to a driven right-leg (RLD) electrode. Each electrode pair is connected to a dedicated AD8232 analog front-end module, which outputs a conditioned analog ECG waveform. All analog front-end modules are powered from a common 3.3 V supply.

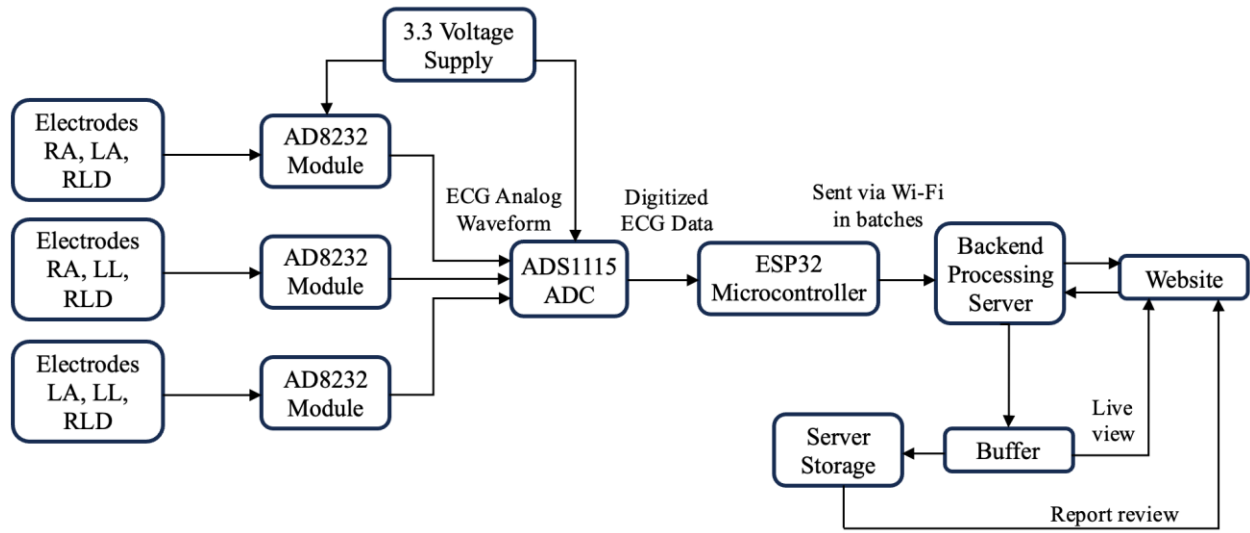


Figure 28. The system block diagram.

The conditioned analog ECG signals are digitized by an external ADS1115 analog-to-digital converter, with each signal routed to a separate analog input channel. The digitized samples are transferred to an ESP32 microcontroller. The ESP32 acts as the system's embedded communication unit rather than a signal-processing platform; it coordinates data acquisition and transmits the ECG data wirelessly over Wi-Fi in batch form to the backend processing server.

The backend handles data validation, session management, lead derivation, and formatting. Incoming ECG data are received and temporarily stored in an in-memory buffer. This buffer supports real-time streaming of ECG waveforms to the web interface during active recording sessions, while buffered data are also periodically written to persistent server storage for session archiving. Stored ECG sessions can later be retrieved for post-recording review.

Visualization and user interaction are provided through a browser-based web interface that communicates bidirectionally with the backend server. Live ECG data are streamed from the buffer

for real-time display, while archived data are accessed from server storage for report review. This separation enables responsive real-time monitoring alongside reliable long-term data storage within the deployed local network environment. Although deployed locally in this project, the system architecture allows the interface to be extended to wider network access without requiring dedicated display hardware.

Overall, this approach simplifies development, supports scalability and future expansion (such as cloud integration or advanced analytics), and provides a practical foundation for remote ECG.

### 4.3 Analog Front-End Design

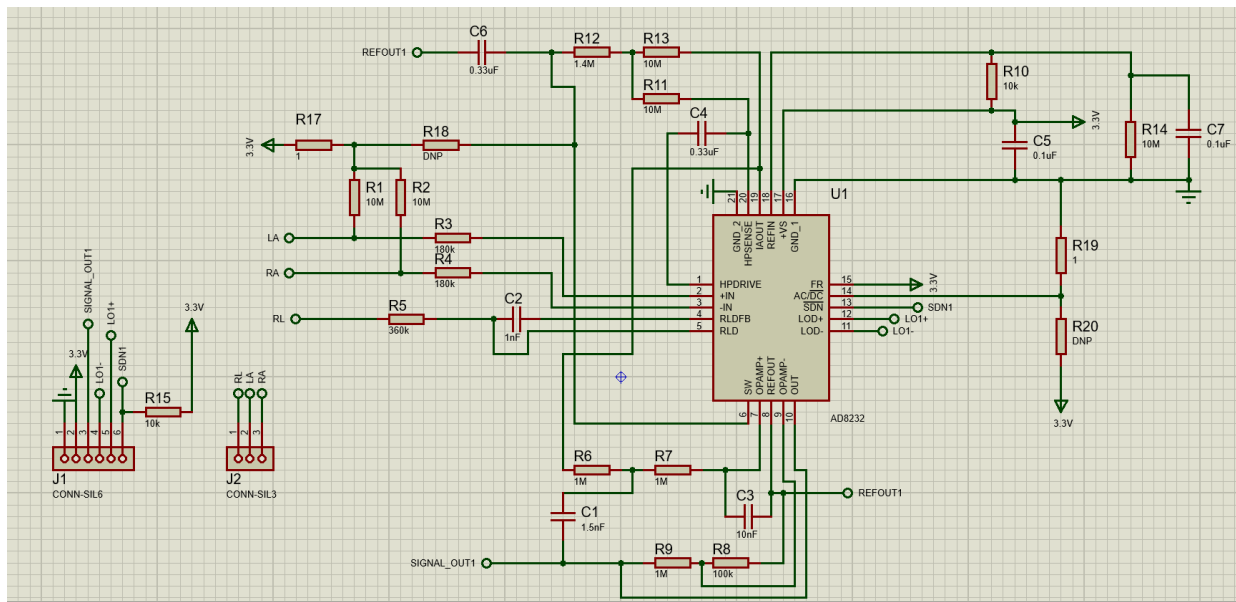


Figure 29. Circuit topology of the AD8232 on proteus

The analog front-end (AFE) of the proposed ECG system was designed to acquire biopotential signals from surface electrodes and condition them for digitization and transmission. The complete AFE implementation was done on the Proteus application, shown in Figure 29, based on the AD8232 ECG front-end and includes the required supporting circuitry for stable and safe ECG

signal acquisition. This section describes the stages of the AFE and their implementation in the system.

#### 4.3.1 Signal Acquisition

The ECG signal acquisition stage is designed to interface with standard surface biopotential electrodes, which serve as the primary sensing elements for capturing cardiac electrical activity. In the intended system configuration, electrodes are placed at the Right Arm (RA), Left Arm (LA), Left Leg (LL), and Right Leg (RL). The RA and LA electrodes provide a differential ECG signal measured by the instrumentation amplifier, corresponding to Lead I of the ECG. The LL electrode when paired with RA provides Lead II and when paired with LA provides Lead III, each measured by a dedicated instrumentation amplifier channel, giving an output as differential ECG signals. The RL electrode functions as a reference electrode driven by the right-leg drive circuit to stabilize the common-mode voltage and improve noise rejection.

Each electrode input includes a  $180\text{ k}\Omega$  series resistor, while the reference/drive electrode path (RL) includes a  $360\text{ k}\Omega$  series resistor, as shown in Figure 30. These series resistors are placed between the electrodes and the analog front-end to limit fault currents and provide protection against voltages beyond the supply rails. Therefore, the electrode inputs interface with the

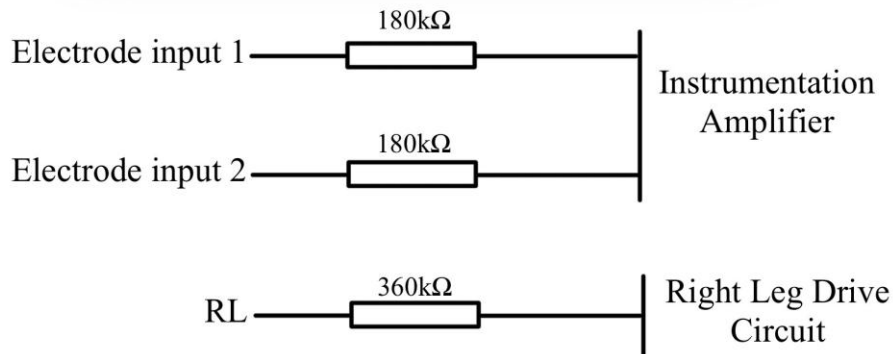


Figure 30. Electrode input resistors.

AD8232's instrumentation amplifier through a high-impedance front-end, which minimizes loading on the electrodes and preserves the low-amplitude ECG signal.

#### **4.3.2 Instrumentation Amplifier (IA) Stage**

In this design, the instrumentation amplifier stage is implemented using the internal IA of the AD8232, which provides fixed gain and high common-mode rejection suitable for ECG acquisition. Because the internal architecture and gain of the instrumentation amplifier are predefined by the device, the design focus at this stage was primarily on input interfacing, protection, and noise mitigation, rather than gain configuration.

The differential ECG inputs are connected to the AD8232 through the series input resistors seen in Figure 30. Not only do these resistors provide patient protection and overload current limiting, ensuring that input currents remain within safe limits under fault conditions, they also contribute to radio-frequency interference (RFI) suppression by increasing the effectiveness of the internal low-pass filtering formed by the AD8232 input resistance and parasitic input capacitance. Placing these resistors close to the instrumentation amplifier inputs helps attenuate high-frequency interference before it can be rectified and appear as dc offsets at the amplifier output.

No additional external RFI filtering components were added beyond these series resistors, as the combination of internal input filtering and external protection resistors was sufficient for the intended operating environment. This approach reduces component count while maintaining adequate immunity to electromagnetic interference.

The output of the instrumentation amplifier is routed directly to the subsequent high-pass filter stage.



### 4.3.3 High-Pass Filter (HPF)

To reduce residual DC offsets and low-frequency baseline drift present after instrumentation amplification, a high-pass filtering stage follows the IA stage. Since the AD8232 provides a configurable DC-blocking high-pass filter architecture that can be extended to improve low-frequency attenuation while preserving the ECG signal of interest, this capability was utilized to implement a two-pole high-pass filter configuration. The use of a two-pole topology provides a steeper attenuation of low-frequency artifacts compared to a single-pole filter, improving baseline stability without requiring a higher cutoff frequency that could distort ECG morphology.

The filter consists of two capacitors ( $C1$  and  $C2$ ), two large resistors ( $R1$  and  $R2$ ), and an additional compensation resistor,  $R_{COMP}$ , as shown in Figure 31. Together, these components form a two-pole high-pass filter.

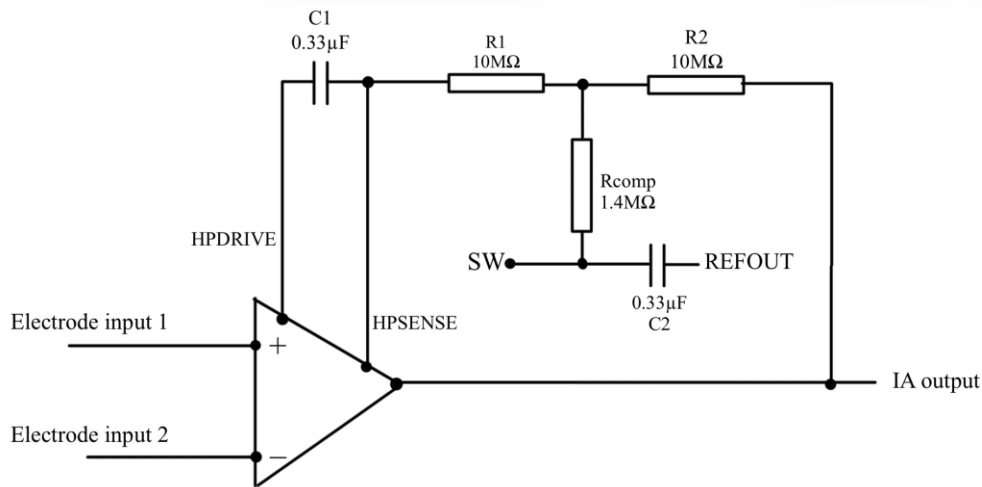


Figure 31. Two-pole HPF circuit.

In the implemented circuit, the selected component values are  $C1 = C2 = 0.33\ \mu\text{F}$ ,  $R1 = R2 = 10\ \text{M}\Omega$ , and  $R_{\text{COMP}} = 1.4\ \text{M}\Omega$ . The large resistor values allow the cutoff frequency to be placed very low without requiring excessively large capacitors, which is particularly important for ECG

applications where frequencies below 1 Hz must be preserved while still rejecting dc and slow drift. The cutoff frequency of this high-pass filter is given by equation 2.

$$f_c = \frac{10}{2\pi\sqrt{R_1C_1R_2C_2}} \quad \text{equation 2}$$

Using equal resistor and capacitor values simplifies the design and ensures symmetrical filter behavior. The cutoff frequency of the two-pole high-pass filter was calculated to be approximately 0.482 Hz, which lies within the commonly accepted range for ECG front-end design and is well suited for cardiac monitoring applications. A cutoff frequency close to 0.5 Hz effectively suppresses dc offsets introduced by the electrode–skin interface as well as very low-frequency baseline wander caused by respiration and slow body movements, while preserving the clinically relevant low-frequency components of the ECG waveform.

The inclusion of the resistor RCOMP plays an important role in shaping the frequency response of the filter. This resistor controls the quality factor (Q) of the HPF and therefore determines whether the passband is maximally flat or exhibits peaking. Lowering the value of RCOMP would increase the Q of the filter and introduce peaking near the cutoff frequency, which may be useful for heart-rate-focused applications, but can distort the ECG waveform if excessive. So, selecting RCOMP to be approximately 0.14 times the value of R1 and R2 optimizes the filter for a maximally flat passband, which is desirable for cardiac monitoring applications. In this design, RCOMP = 1.4 MΩ satisfies this condition relative to R1 = R2 = 10 MΩ. Extremely small values of RCOMP may also lead to instability and are therefore avoided.

After high-pass filtering, the ECG signal has a stabilized baseline and is largely free of dc offsets and slow drift. However, high frequency may still be present. Therefore, the output of the high-pass filter is fed into a subsequent low-pass filter stage

#### 4.3.4 Low-Pass Filter (LPF)

After the ECG signal has been amplified by the instrumentation amplifier and stabilized using the high-pass filter, it is passed to the low-pass filter (LPF) stage. This ensures that only the clinically relevant frequency content of the ECG signal is preserved before digitization.

Compared to a single-pole low-pass filter, a two-pole filter provides a steeper attenuation beyond the cutoff frequency, improving noise suppression while maintaining good signal fidelity within the passband. This is particularly important for ECG monitoring, where the useful signal content typically lies below 40 Hz. In this design, a two-pole Sallen-Key low-pass filter topology, as seen in Figure 32, was used.

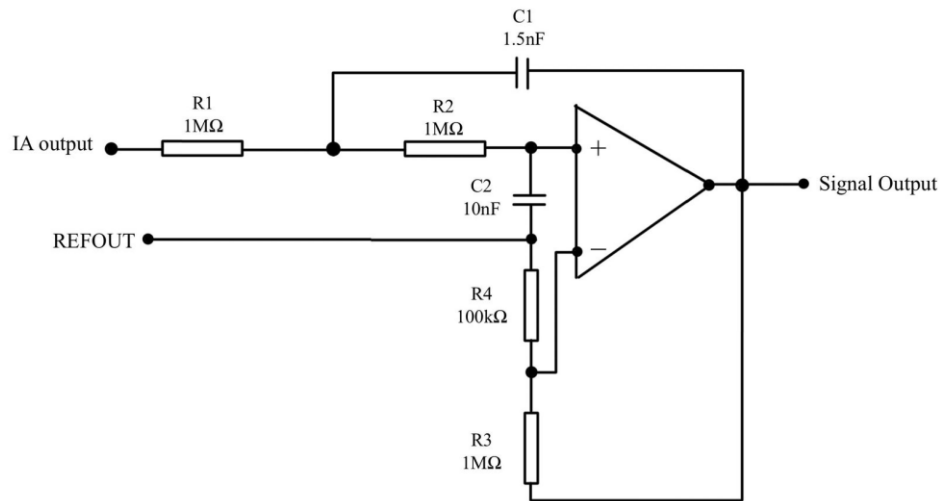


Figure 32. Two-pole Sallen-Key LPF circuit.

The Sallen-Key configuration uses an operational amplifier (A1) together with two resistors (R1 and R2) and two capacitors (C1 and C2) to realize the filtering function. The op-amp is referenced to the REFOUT pin of the AD8232, which biases the signal around a stable mid-supply voltage.

This allows the filter to operate correctly from a single supply while maintaining symmetrical headroom for the ECG waveform.

The cutoff frequency of the low-pass filter is determined by the values of R1, R2, C1, and C2 and is given by equation 3.

$$f_c = \frac{1}{2\pi\sqrt{R1C1R2C2}} \quad \text{equation 3}$$

In the implemented design, the selected component values are R1 = R2 = 1 MΩ, C1 = 1.5 nF and C2 = 10 nF. Using relatively large resistor values allows small capacitor values to be used, which helps reduce component size while still achieving the desired cutoff frequency for ECG monitoring.

In addition to filtering, the Sallen-Key topology provides gain, which is set by the resistor pair R3 and R4 according to equation 4.

$$Gain = 1 + \frac{R3}{R4} \quad \text{equation 4}$$

With R3 = 1 MΩ and R4 = 100 kΩ, the resulting gain of this stage is approximately 11. This additional gain further amplifies the ECG signal after band-limiting, improving the signal-to-noise ratio before it is fed into the analog-to-digital converter.

Using the selected component values, the cutoff frequency of the two-pole Sallen-Key low-pass filter was calculated to be approximately 41 Hz. This cutoff frequency is well suited for ECG monitoring applications, as it preserves the essential spectral content of the ECG signal while effectively attenuating higher-frequency noise sources such as muscle activity (EMG), electromagnetic interference, and switching noise.

The resulting quality factor of the filter, which was calculated by equation 5, was found to be approximately 0.77, which is close to the value required for a maximally flat passband response. This ensures a sharp cutoff without excessive peaking near the cutoff frequency, thereby maintaining ECG waveform integrity and preventing distortion.

$$Q = \frac{\sqrt{R1C1R2C2}}{R1C2 + R2C2 + R1C1(1 - Gain)} \quad \text{equation 5}$$

An important characteristic of the Sallen-Key low-pass filter is that the filter quality factor (Q) depends on both the component values and the gain of the amplifier. Increasing the gain increases Q, but excessive gain can lead to instability. To maintain stability at higher gain values, the capacitor C2 is chosen to be significantly larger than C1. In this design, C2 is more than six times larger than C1, which helps stabilize the circuit and prevents oscillations.

Overall, this two-pole Sallen-Key low-pass filter combined with the preceding two-pole high-pass filter makes the analog front end effectively behave as a band-pass filter shown in Figure 33, passing only the frequency range of interest for ECG monitoring. The overall analog gain of the signal conditioning chain is determined by the cascaded gain stages. The AD8232 instrumentation

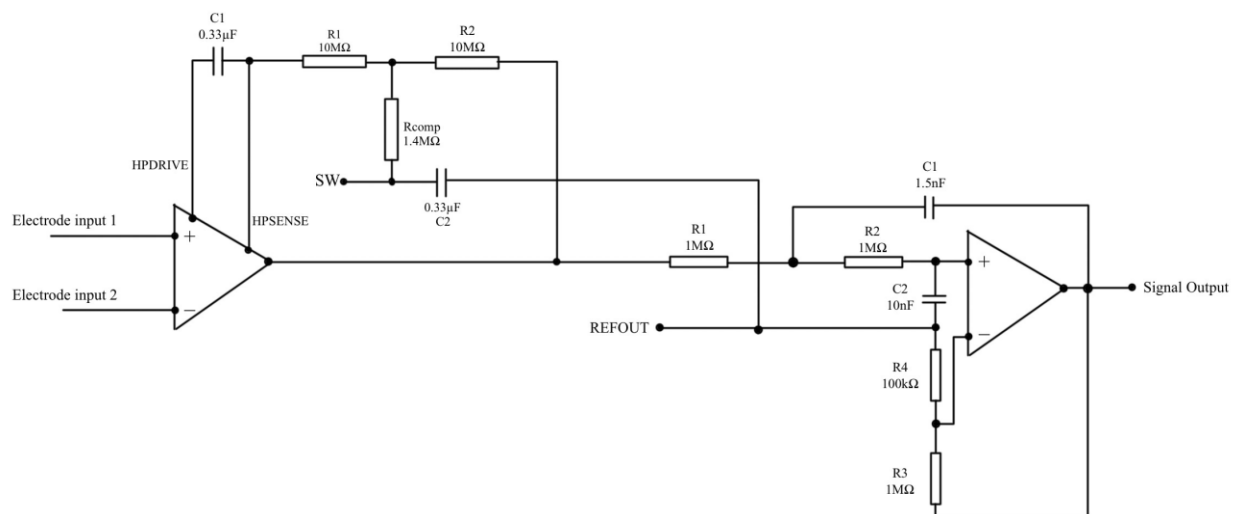


Figure 33. The cascaded HPF and LPF circuits.

amplifier provides an internal gain of 100, while the high-pass filter stage is designed for unity gain (x1) in the passband. Additional amplification is provided by the second-order Sallen-Key low-pass filter, which is configured with a non-inverting gain of 11. As a result, the total midband analog gain of the ECG acquisition chain is approximately 1100. This gain applies within the passband defined by the high-pass and low-pass cutoff frequencies, while attenuation occurs outside this range.

After low-pass filtering, the ECG signal is fully conditioned in the analog domain. Baseline drift and dc offsets have been removed by the high-pass filter, and high-frequency noise has been suppressed by the low-pass filter. The resulting band-limited signal is therefore well suited for sampling by the ADC, enabling accurate digital processing, visualization, and analysis in the subsequent stages of the system.

#### 4.3.5 Right Leg Drive (RLD) Circuit

The right leg drive (RLD) circuit is implemented using the internal RLD amplifier of the AD8232. Since the core RLD architecture is fixed internally by the device, the design focus was placed on

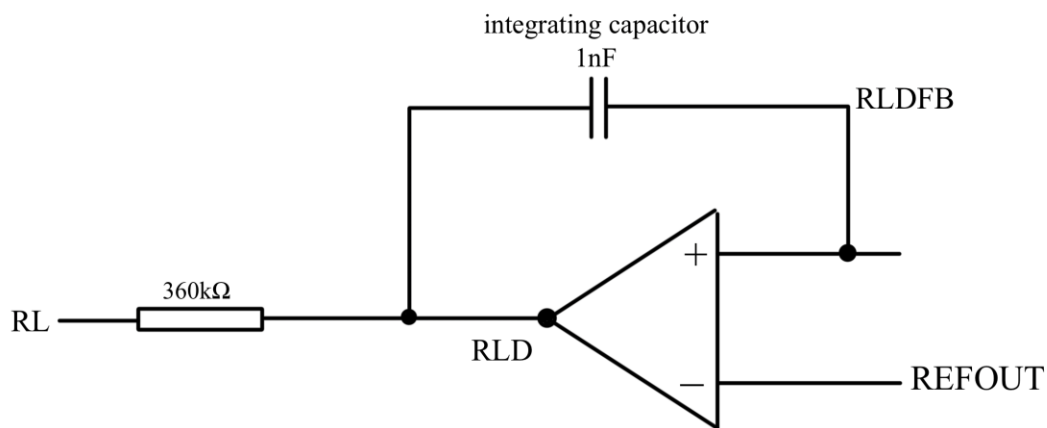


Figure 34. The RLD circuit.

loop stabilization and patient safety, which are controlled through external component selection, as seen in Figure 34.

An integrator is formed by connecting an external capacitor between the RLD feedback (RLDFB) and RLD pins. In this implementation, a 1 nF capacitor is used, which sets the RLD loop crossover frequency to approximately 1 kHz. This selection provides sufficient loop gain in the 50–60 Hz frequency range, where power-line interference is most prominent, thereby improving common-mode noise rejection without compromising system stability.

The value of the integration capacitor represents a tradeoff between interference suppression and loop stability. Increasing the capacitance lowers the crossover frequency, reducing the available loop gain at power-line frequencies and allowing more line noise to appear at the inputs. Conversely, reducing the capacitance increases loop gain but may lead to instability or saturation of the RLD amplifier. The selected value of 1 nF represents a well-established compromise that provides effective common-mode noise suppression while maintaining stable operation.

For patient safety, the output of the RLD amplifier is current limited using a 360 k $\Omega$  series resistor connected to the driven electrode. This resistor ensures that the injected current remains well below the recommended safety limit of 10  $\mu$ A, even under fault conditions. At the same time, the chosen value allows the RLD loop to operate effectively without significantly degrading common-mode feedback performance.

#### **4.3.6 Reference Buffer and Biasing Network**

The reference buffer and biasing network is designed to establish a stable dc operating point for the analog ECG signal chain under single-supply operation while minimizing power consumption

and susceptibility to noise. The system operates from a 3.3 V supply, requiring the ECG signal to be biased around a mid-supply reference rather than ground.

A high-impedance resistive divider formed by R10 and R14 was used to define the reference voltage at the REFIN pin shown in Figure 35. High resistor values were intentionally selected to

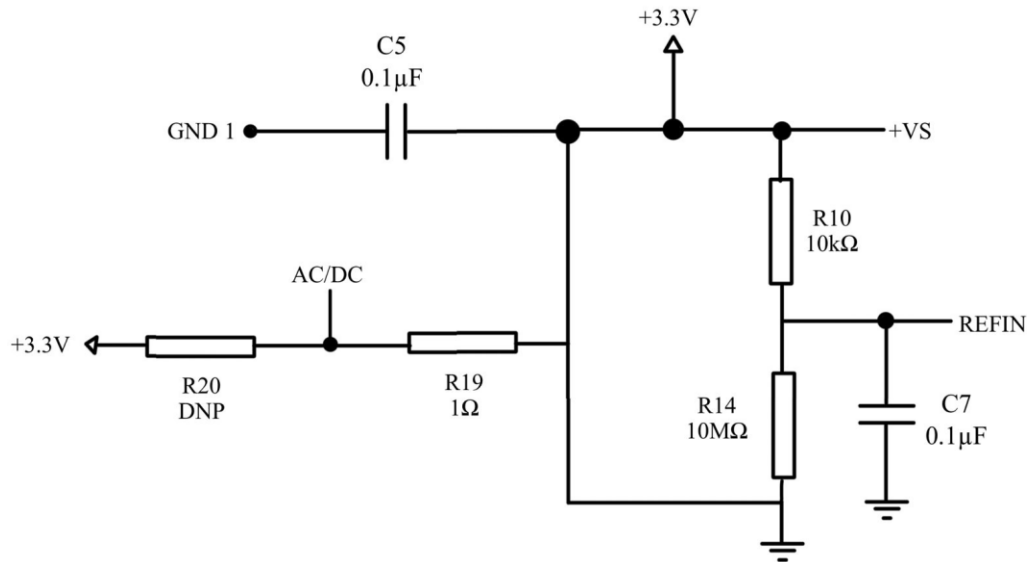


Figure 35. Reference buffer and biasing network circuitry.

reduce static current draw to the nanoamp range, making the design suitable for low-power, portable ECG applications. The reference voltage applied at REFIN is internally buffered by the analog front-end device and used to bias the instrumentation amplifier and subsequent filtering stages.

Due to the high impedance of the divider, the REFIN node is sensitive to noise coupling. To improve reference stability, a capacitor (C7) was connected between REFIN and ground, forming a low-pass RC filter. This filtering suppresses high-frequency interference and stabilizes the virtual ground used by the analog circuitry, thereby reducing baseline fluctuations and preserving ECG signal integrity. The capacitor value was selected as a compromise between noise suppression and reference settling time following power-up.



Additionally, local supply decoupling was implemented using capacitor C5 to reduce high-frequency noise on the 3.3 V rail before it reaches the reference network. Overall, the implemented reference buffer and biasing network provide a low-noise, low-power mid-supply bias that enables reliable single-supply ECG signal acquisition while maintaining signal fidelity.

#### 4.3.7 Lead-Off Detection Mode

DC lead-off detection mode was selected to enable detection of disconnected or improperly attached electrodes in the three-electrode ECG configuration used in this system. DC lead-off detection was chosen because it is optimized for ECG systems that include a driven reference electrode and provides lead-off monitoring without introducing additional signal interference, unlike ac lead-off detection. This mode was chosen due to its compatibility with static electrode monitoring and its minimal impact on normal ECG signal acquisition.

To implement DC lead-off detection, each electrode input was connected to the positive supply through a high-value pull-up resistor of 10 M $\Omega$ , as shown in Figure 36. These resistors establish a defined bias path for the input nodes when an electrode becomes disconnected, while presenting negligible loading during normal operation.

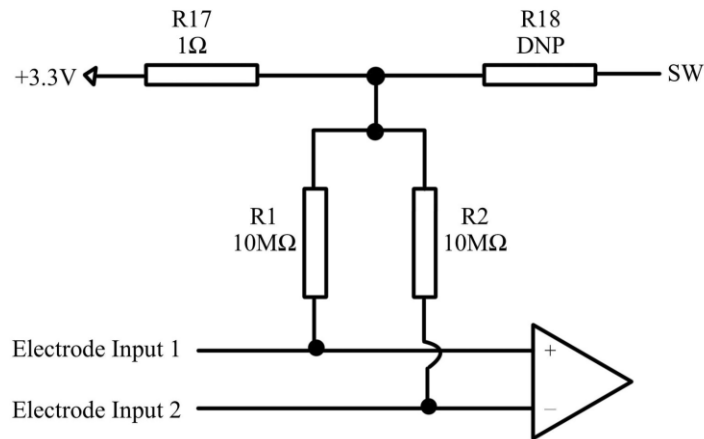


Figure 36. Lead-off detection circuit.

The DC lead-off detection mode was selected using the AC/DC control pin of the analog front-end device. The optional pull-up resistor connecting the AC/DC pin to the supply was intentionally not populated (DNP), as seen in Figure 36 leaving the pin pulled low by default through a small series protection resistor. This configuration ensures the selection of DC lead-off detection mode, consistent with the three-electrode ECG operation. The series resistor serves primarily as a current-limiting and protection element and does not significantly affect the logic level at the control pin. Although DC lead-off detection was enabled at the hardware level, the comparator output pins (LOD+ and LOD–) were not routed to the microcontroller in the current PCB revision and were therefore not utilized for software-level lead status monitoring.

#### **4.3.8 Power Supply and Decoupling**

Reliable ECG acquisition requires a clean and stable power supply, as excessive noise on the supply pins can directly couple into the sensitive analog front-end circuitry and degrade signal quality. The AD8232, like other precision analog devices, is therefore decoupled using local bypass capacitors to suppress supply noise across a wide frequency range.

In this design, the AD8232 is powered from a 3.3 V supply, and a 0.1  $\mu\text{F}$  capacitor is placed as close as possible to the supply pin, as seen in Figure 35. This capacitor provides low impedance decoupling at high frequencies and attenuates fast transients and switching noise that may originate from digital circuitry or external power sources. Close placement is critical, as long trace lengths can introduce parasitic inductance that reduces the effectiveness of high-frequency decoupling.

While adequate decoupling is essential for stable operation, excessive decoupling capacitance can increase inrush current and power dissipation during power cycling. Therefore, the selected capacitor values represent a balance between noise suppression and power efficiency. Together,

the local 0.1  $\mu\text{F}$  bypass capacitor and the bulk decoupling capacitor ensure that the AD8232 operates from a low-noise supply, minimizing interference with the analog ECG signal and supporting reliable performance of the instrumentation amplifier, filtering stages, and reference circuitry.

#### **4.4 ADC Interface and Signal Conditioning**

After analog conditioning by the instrumentation amplifier, high-pass filter, and low-pass filter stages, the ECG signal is routed directly to the analog-to-digital converter for sampling and digital processing. In this design, the output of each AD8232 low-pass filter stage is connected directly to the AIN0, AIN1, and AIN2 inputs of the ADS1115 ADC.

The AD8232 analog front-end is designed to interface directly with data acquisition systems and provides an output stage suitable for driving ADC inputs without the need for an additional buffer amplifier. While ADC inputs may draw transient currents during sampling, the ECG signal in this design is bandwidth-limited and sourced from an actively filtered output stage, allowing stable operation when interfaced directly with the ADS1115.

The voltage presented to the ADC input corresponds to the filtered ECG signal biased around the internal reference voltage generated by the AD8232. This biasing ensures that the signal remains within the allowable input range of the ADS1115 while preserving ECG waveform integrity. The direct ADC interface simplifies the signal path, reduces component count, and enables reliable digitization of the conditioned ECG signal.

## 4.5 ESP32 Microcontroller and Embedded Software

The ESP32 microcontroller acts as the central processing and communication unit of the designed ECG monitoring system. It receives digitized ECG samples from the external ADS1115 analog-to-digital converter, performs acquisition control and data handling, and transmits the acquired data wirelessly to the backend system. The ESP32 integrates both digital peripheral interfaces and a Wi-Fi subsystem on a single chip, enabling compact PCB integration and real-time data transmission without external communication hardware.

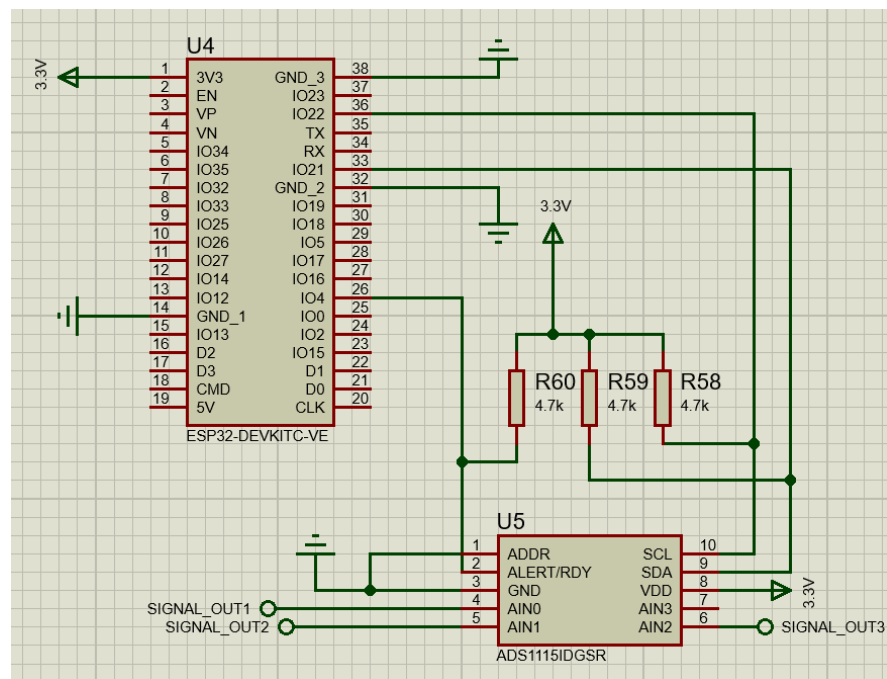


Figure 37. The ESP32 and ADS1115 pin connections on Proteus.

Communication between the ESP32 and the ADS1115 is implemented using the Inter-Integrated Circuit (I<sup>2</sup>C) protocol. In this design, GPIO22 is configured as SCL and GPIO21 as SDA, connecting the ESP32 to the corresponding pins of the ADS1115. Because the I<sup>2</sup>C bus uses open-drain signaling, both lines are externally pulled up to the 3.3 V supply using 4.7 k $\Omega$  resistors as seen in Figure 37, ensuring correct logic-high levels and reliable communication. Through this

interface, the ESP32 initiates ADC conversions, reads the resulting digital samples, and transfers them into its internal memory.

To support precise sampling control, the ALERT/RDY output of the ADS1115 is connected to GPIO4 on the ESP32. This pin is used to signal conversion completion, allowing the ESP32 to read samples immediately when they are ready rather than relying solely on software timing. This hardware-assisted signaling improves sampling consistency while the system is operating continuously.

Once ECG samples are received over the I<sup>2</sup>C bus, they are handled by the ESP32's internal CPU subsystem. The embedded firmware running on the ESP32 enforces a fixed sampling rate of 250 samples per second to ensure consistent temporal spacing between samples. The firmware sequentially acquires digitized ECG data from multiple ADC channels, converts the raw readings to voltage values, and assigns a timestamp to each sample at the time of acquisition. Acquired samples are temporarily stored in memory and grouped into fixed-size batches to reduce communication overhead communication. Once a batch is complete, the data is formatted into network-ready packets and transmitted wirelessly to the backend server, enabling continuous real-time ECG data streaming for visualization, processing, and storage.

Because the radio subsystem is implemented entirely on-chip, no external RF transceiver is required. The PCB design therefore focuses on stable power delivery and grounding rather than discrete RF circuitry. The ESP32 is powered from the regulated 3.3 V rail, and multiple ground pins are tied to the PCB ground plane to provide low-impedance return paths and stable operation during wireless activity.

At this point, the ECG signal transitions from an embedded data stream into a network-based data flow, marking the endpoint of the embedded signal chain and the handoff to the backend system for visualization, processing, and storage.

This processing stage represents the final step of the signal chain: raw analog ECG signals are converted by the analog front end, digitized by the ADC, transferred to the ESP32 over I<sup>2</sup>C, and prepared for wireless delivery.

#### **4.6 PCB Design and Hardware Integration**

Using Proteus, the printed circuit board was implemented as a two-layer PCB consisting of a top copper layer and a bottom copper layer. The top layer is primarily used for component placement and signal routing, while the bottom layer is predominantly utilized as a ground reference plane with additional routing where required. This configuration provides a practical balance between electrical performance, manufacturability, and cost, and is appropriate for low-frequency biomedical signal acquisition such as ECG, where signal bandwidth is limited and trace lengths are relatively short.

To preserve signal integrity, functional separation between analog and digital circuitry was maintained during PCB layout. The ECG analog front-end was placed in a dedicated region of the board, spatially separated from digital components such as the microcontroller and communication interface, as shown in Figure 38. This layout strategy reduces the likelihood of digital switching noise coupling into sensitive analog signal paths.

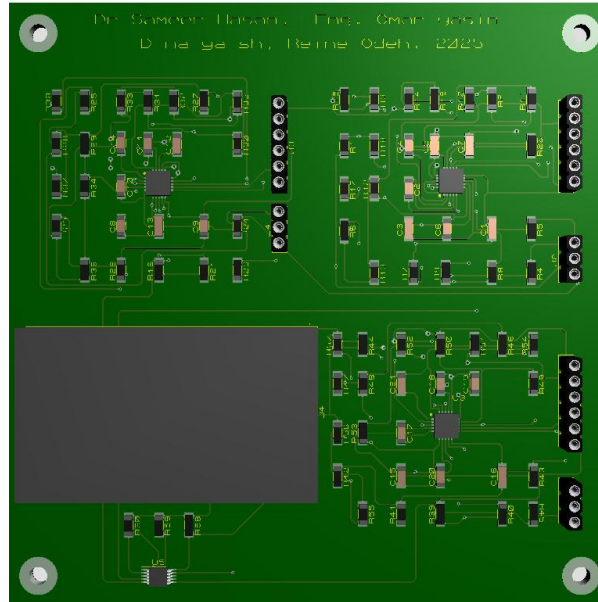


Figure 38. The PCB visualized on Proteus.

A structured grounding approach was employed to improve measurement stability and reduce noise susceptibility. The bottom copper layer serves primarily as a continuous ground reference, providing low-impedance return paths for both analog and digital signals and helping to minimize ground voltage fluctuations.

Signal routing was performed with emphasis on minimizing electromagnetic interference. Critical low-level analog traces were routed away from digital clock and communication lines, which contain fast voltage transitions capable of coupling into nearby ECG traces through capacitive and inductive mechanisms. Long parallel routing between sensitive analog traces and noisy digital signals was avoided to reduce unwanted coupling and preserve signal integrity.

Additional layout-level noise mitigation practices included placing passive components close to their associated integrated circuit pins to minimize parasitic effects and reduce loop areas. Collectively, these layout and routing measures contribute to stable and reliable ECG signal acquisition.

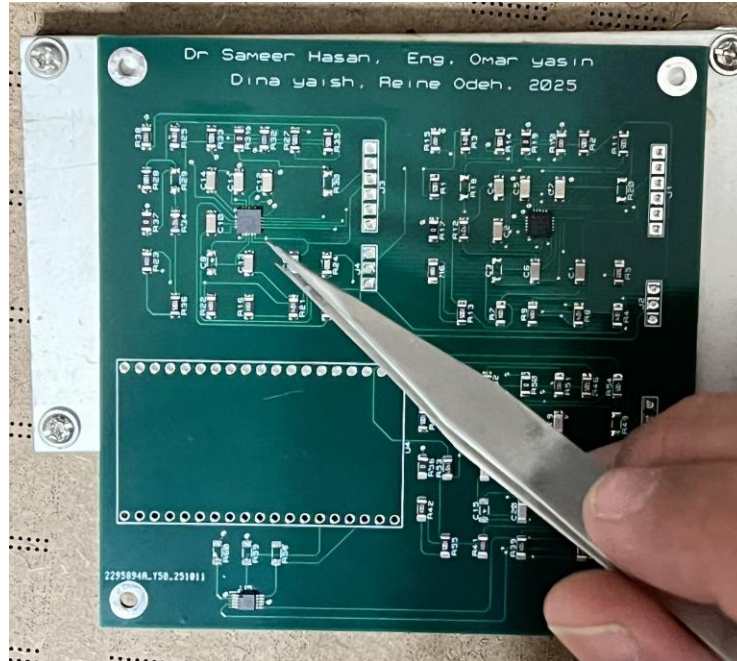


Figure 39. Soldering the PCB components on a hot plate.

For PCB fabrication and assembly, solder paste and a hot-plate reflow process were used, as shown in Figure 39. The ESP32 microcontroller was subsequently soldered using a soldering pen. The completed hardware integrates the analog ECG front-end, digital processing unit, and external interfaces into a compact and modular platform. The final assembled system supports reliable ECG signal acquisition, processing, and wireless transmission, forming a robust hardware foundation for the proposed monitoring system. The PCB being ordered, along with all its components and the ESP32 microcontroller, all fell under 30 USD, which is under 21.27 JOD.

#### 4.7 System Backend

The backend of the ECG system was implemented as a centralized control and processing server responsible for coordinating data acquisition, session management, real-time visualization, and long-term storage. The backend was developed using Node.js, a server-side JavaScript runtime that enables asynchronous, event-driven handling of networked applications, together with the Express.js framework, which provides structured routing and middleware support for



implementing RESTful web services. In this context, Express.js serves as the practical implementation framework, while REST defines the architectural principles governing the design of stateless endpoints and data exchanges. The server was deployed on a local machine within the same Wi-Fi network as the acquisition hardware and client interfaces.

A single-server architecture was adopted, where all backend functionality is contained within one main server application (server.js). This approach was selected to simplify deployment, debugging, and system integration during prototyping while still maintaining clear logical separation between subsystems such as authentication, data ingestion, signal processing, and reporting.

The backend acts as an intermediary between three primary system components:

1. The acquisition and transmission device (ESP32), which transmits digitized ECG samples to the server.
2. Web-based frontend clients, including patient and doctor dashboards which are expanded in the frontend design, are used for real-time visualization and session control.
3. A local SQLite database, which stores patient information, recording sessions, ECG samples, and alert metadata.

The backend was designed to be session-centric, meaning that all incoming ECG samples are associated with an explicitly started recording session. This design ensures that ECG data are temporally organized, reproducible, and traceable to a specific patient and recording interval. Session lifecycle control (start, stop, stop with doctor notification) is handled exclusively by backend logic to prevent inconsistent or incomplete recordings.

To support both control operations and live signal visualization, the backend combines HTTP-based REST endpoints with WebSocket communication. REST endpoints are used for

authentication, session control, profile management, and historical data access, while WebSocket is used for low-latency broadcast of real-time events and alerts to connected clients. WebSocket communication is used exclusively for real-time visualization and event notification, while persistent ECG data storage is handled through HTTP-based ingestion endpoints

All digital ECG signal processing performed beyond the analog front-end is handled on the backend server. This includes digital filtering of digitized channels, derivation of standard limb leads, heart-rate estimation, and preparation of data for visualization and reporting.

#### **4.7.1 ECG Data Ingestion and Session Handling**

ECG data ingestion in the proposed system is initiated only after an explicit recording session has been started by the patient through the web interface. This design enforces a controlled acquisition workflow in which all incoming ECG samples are associated with a known patient and a clearly defined recording interval. This approach improves system reliability and protects data integrity during network interruptions or user-side errors.

Once a session is active, the ESP32 transmits digitized ECG samples to the backend server using HTTP POST requests. Data are sent in batch form, where each request contains a short sequence of consecutive samples along with their corresponding timestamps. Batch-based transmission was selected to reduce network overhead while maintaining sufficient temporal resolution for real-time monitoring.

Upon reception, the backend associates incoming ECG samples with a valid active recording session, ensuring that all acquired data are correctly organized within a defined acquisition interval and preventing loss or misassignment during continuous monitoring.

### **4.7.2 Data Filtering and Processing**

Each ECG channel is sampled at 250 Hz on the hardware, timestamped, and transmitted to the backend in short batches. Prior to storage and visualization, the digitized signals are digitally filtered using a high-pass filter with a cutoff frequency of 0.5 Hz to suppress baseline wander and a low-pass filter with a cutoff frequency of 40 Hz to attenuate high-frequency noise. This passband preserves the primary morphological features of the ECG waveform required for visualization and heart-rate estimation. Following filtering, the standard limb leads (Lead I, Lead II, Lead III) and augmented leads (aVR, aVL, aVF) are derived mathematically from the filtered signals using standard Einthoven and Goldberger relationships.

Processed samples are not written on the database immediately on arrival. Instead, samples are temporarily stored in an in-memory buffer and periodically committed to the database in grouped transactions. This buffering strategy reduces disk write overhead and ensures stable system performance during continuous high-rate data acquisition.

### **4.7.3 Heart Rate Extraction and Processing**

Heart rate estimation was implemented on the backend server using a real-time, Pan–Tompkins–inspired signal processing algorithm applied to the derived Lead II ECG signal. Incoming ECG samples are processed sequentially, where the signal first undergoes differentiation to emphasize rapid slope changes associated with QRS complexes, followed by squaring to enhance peak prominence. A moving window integration stage of approximately 150 ms is then applied to capture the energy of the QRS complex while suppressing high-frequency noise.

An adaptive thresholding mechanism is used to identify candidate R-peaks based on the integrated signal amplitude. To prevent false detections, a refractory period of 350 ms is enforced between

successive detections, and physiologically implausible RR intervals are rejected. Detected RR intervals are stored in a rolling buffer, and heart rate is computed using the median RR value to reduce sensitivity to outliers and transient artifacts.

To improve display stability, heart rate updates are rate-limited such that changes occur only at fixed time intervals and are constrained to gradual transitions. The resulting heart rate values are stored alongside ECG samples and made available for real-time visualization and post-session analysis.

#### **4.7.4 Authentication**

To protect sensitive patient data and ensure controlled system operation, the backend enforces authentication and role-based access control. User roles, including patient and clinician accounts, are managed by backend logic, restricting access to system functions such as session initiation, historical data retrieval, and alert review. This design ensures that ECG data and session controls are accessible only to authorized users, which is particularly important in medical monitoring applications.

#### **4.7.5 Session Termination, Data Storage, and Report Generation**

Recording sessions are explicitly terminated by the user through the web interface, signaling the end of ECG acquisition. Upon session termination, the backend finalizes data handling by closing the active recording interval and preventing any further samples from being associated with the session. This controlled termination ensures that each ECG recording represents a well-defined and complete acquisition period.

All ECG samples are associated with timestamps to preserve temporal accuracy across acquisition, processing, and storage. This temporal organization ensures that ECG waveforms, derived

parameters, and session reports can be accurately reconstructed and traced to specific recording intervals during post-acquisition reviews. The data acquired during the session are stored persistently and organized under the corresponding patient and session identifiers. This structure enables reliable retrieval of historical recordings and ensures that recorded data remain traceable to a specific patient and acquisition event, which is essential for post-recording review and comparison.

Following session termination, the backend automatically generates a structured ECG session report. The report includes session timing information and multi-lead ECG waveform plots derived from the stored data. Reports are archived locally and made accessible through the web interface for both patients and clinicians, allowing recorded ECG data to be reviewed independently of real-time acquisition. The backend also supports clinician review by flagging selected recording sessions and associating them with alerts that can be accessed by authorized clinical users.

While the current implementation adopts a single-server architecture suitable for prototyping and controlled testing, the modular design of the backend allows future extension toward multi-device support, distributed deployment, or cloud-based storage. This flexibility enables the system to scale beyond the prototype stage without fundamental changes to the underlying backend architecture.

## **4.8 Frontend**

The front end of the ECG monitoring system was implemented as a browser-based web interface that supports user authentication, ECG session control, real-time visualization, and post-recording review. The interface communicates with the backend server using HTTP and WebSocket connections and can be accessed through standard web browsers without requiring dedicated software installation.

To improve accessibility and usability for local users, the patient interface includes language support for both English and Arabic. Users can toggle the interface language at any time, with all labels, buttons, and informational fields translated. This design consideration aims to improve usability and reduce user error, particularly in non-clinical home-monitoring scenarios.

Frontend functionality is distributed across multiple pages, each corresponding to a specific stage of the user interaction workflow.

#### **4.8.1 User Authentication and Profile Management Pages**

The login, signup, and account settings pages collectively manage user access and profile information. New users can create patient accounts by providing personal and basic clinical information, while existing users log in using stored credentials, as seen in Figure 40. Upon successful authentication, users are redirected to role-appropriate interfaces based on whether they are registered as patients or clinicians.

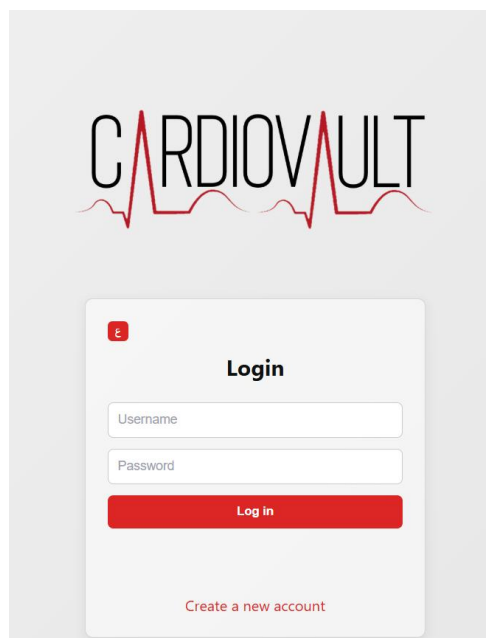


Figure 40. The login interface.

During signup and within the account settings page, patients can enter and update personal attributes such as sex, height, weight, and date of birth, as seen in Figure 41. The system automatically calculates the patient's age based on the stored birth date. Patients can also specify existing medical conditions using a structured multi-selection interface, with support for predefined conditions as well as custom entries. This information is stored centrally and associated with future ECG sessions, providing contextual background during data review.

Figure 41. The sign-up and account settings interfaces.

#### 4.8.2 Patient Dashboard and Live ECG Monitoring Interface

The patient dashboard serves as the primary interface for ECG acquisition and live monitoring, as shown in Figure 42. From this page, patients can initiate and terminate ECG recording sessions. During an active session, six ECG leads (Lead I, Lead II, Lead III, aVR, aVL, aVF) are displayed simultaneously as waveforms, enabling multi-lead observation consistent with standard ECG presentation. The ability to zoom in to each lead provides an interactive feature for the users.

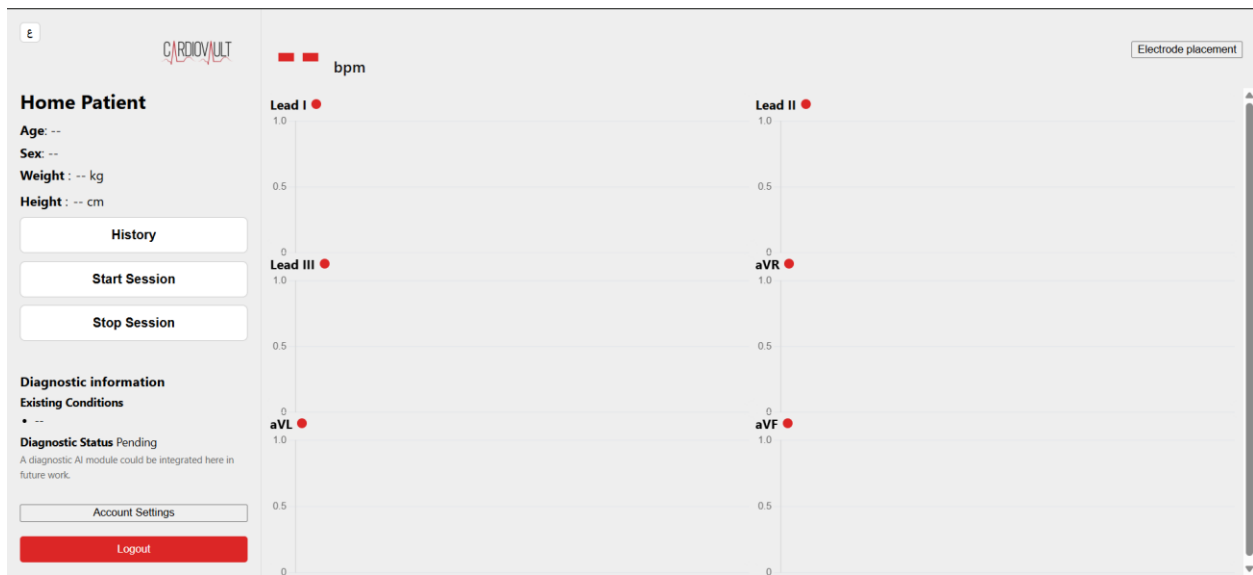


Figure 42. The patient dashboard.

The dashboard also provides electrode placement guidance, allowing users to verify correct electrode placement before or during recording, as seen in Figure 43. Real-time heart rate is displayed alongside waveform data, providing physiological feedback during acquisition. Basic patient information is shown on the dashboard to maintain context throughout the session.

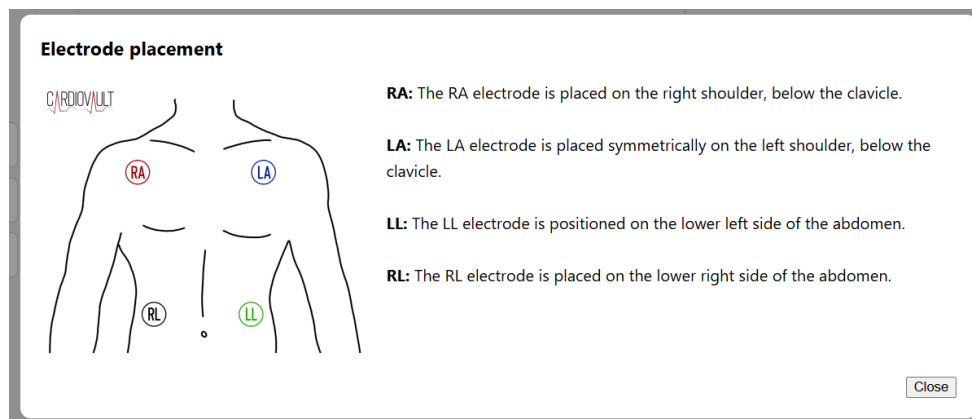


Figure 43. Electrode placement guide.

Visual indicators are displayed alongside each ECG lead to provide real-time feedback on signal activity. These indicators reflect the presence and continuity of incoming data rather than true

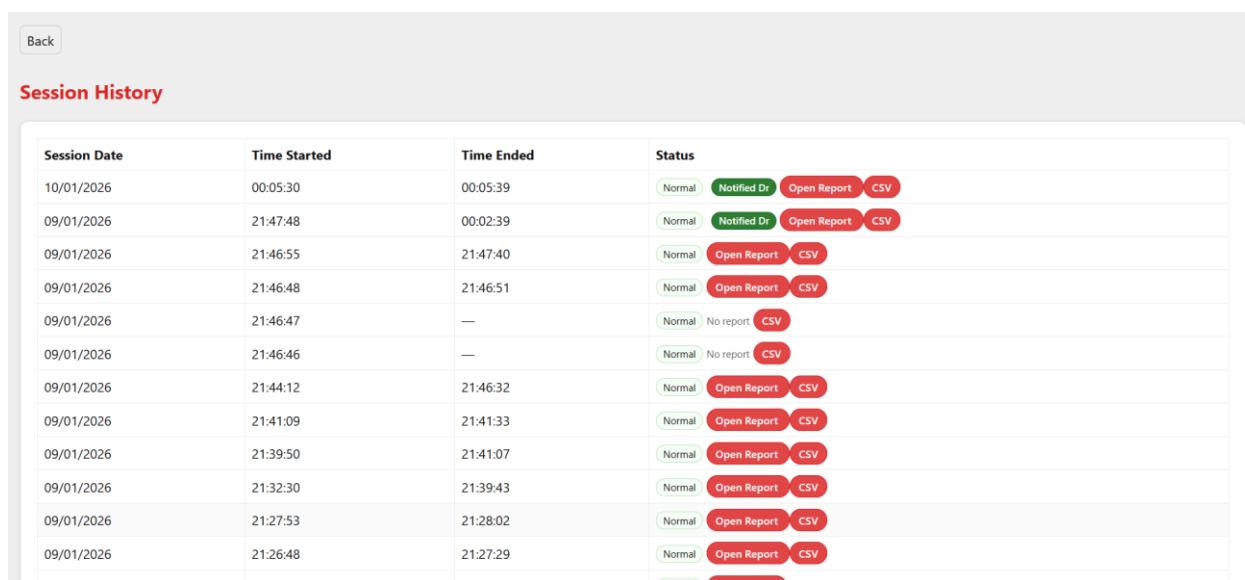


electrode impedance-based lead-off detection. Their purpose is to help users verify that signals are being received during acquisition. This visual framework also establishes a foundation for future integration of true lead-off detection mechanisms, should sensing be added at the hardware level.

Session termination is handled through an explicit control that allows the patient to either stop the session normally or stop the session and notify a clinician. This option supports realistic monitoring scenarios in which a patient records an ECG during symptoms and requests professional review. Interactive features such as waveform zooming are available to improve visibility during live monitoring.

### 4.8.3 Session History and Report Review Pages

The history page allows patients to view a list of previously recorded ECG sessions, as shown in Figure 44. Each session entry includes status indicators, such as whether the session has been notified for clinician review. From this interface, users can open archived ECG reports or download session data in standardized formats, including CSV files for offline analysis.



| Session Date | Time Started | Time Ended | Status                                                                     |
|--------------|--------------|------------|----------------------------------------------------------------------------|
| 10/01/2026   | 00:05:30     | 00:05:39   | Normal <span>Notified Dr.</span> <span>Open Report</span> <span>CSV</span> |
| 09/01/2026   | 21:47:48     | 00:02:39   | Normal <span>Notified Dr.</span> <span>Open Report</span> <span>CSV</span> |
| 09/01/2026   | 21:46:55     | 21:47:40   | Normal <span>Open Report</span> <span>CSV</span>                           |
| 09/01/2026   | 21:46:48     | 21:46:51   | Normal <span>Open Report</span> <span>CSV</span>                           |
| 09/01/2026   | 21:46:47     | —          | Normal <span>No report</span> <span>CSV</span>                             |
| 09/01/2026   | 21:46:46     | —          | Normal <span>No report</span> <span>CSV</span>                             |
| 09/01/2026   | 21:44:12     | 21:46:32   | Normal <span>Open Report</span> <span>CSV</span>                           |
| 09/01/2026   | 21:41:09     | 21:41:33   | Normal <span>Open Report</span> <span>CSV</span>                           |
| 09/01/2026   | 21:39:50     | 21:41:07   | Normal <span>Open Report</span> <span>CSV</span>                           |
| 09/01/2026   | 21:32:30     | 21:39:43   | Normal <span>Open Report</span> <span>CSV</span>                           |
| 09/01/2026   | 21:27:53     | 21:28:02   | Normal <span>Open Report</span> <span>CSV</span>                           |
| 09/01/2026   | 21:26:48     | 21:27:29   | Normal <span>Open Report</span> <span>CSV</span>                           |

Figure 44. History interface (patient view).

The “Report review” page presents the session information, including the recording date, time, duration, and session number, as well as the recorded ECG data, as demonstrated in Figure 45. The data is structured in a multi-lead layout similar to the live monitoring view. Waveforms are labeled according to standard ECG lead conventions, and an interactive zooming and scrolling feature are supported to allow detailed inspection of waveform morphology across the recording duration. These features enable post-session review independently of real-time acquisition.



Figure 45. The report review interface.

#### 4.8.4 Clinician Dashboard and Review Interface

A separate dashboard for the clinician provides authorized healthcare professionals with access to patient ECG data for review purposes. A notification indicator displays the number of active alerts, allowing clinicians to quickly identify sessions that require attention. The interface includes a list of patients with recorded sessions, as well as a view of recent ECG recordings across patients as seen in Figure 46.

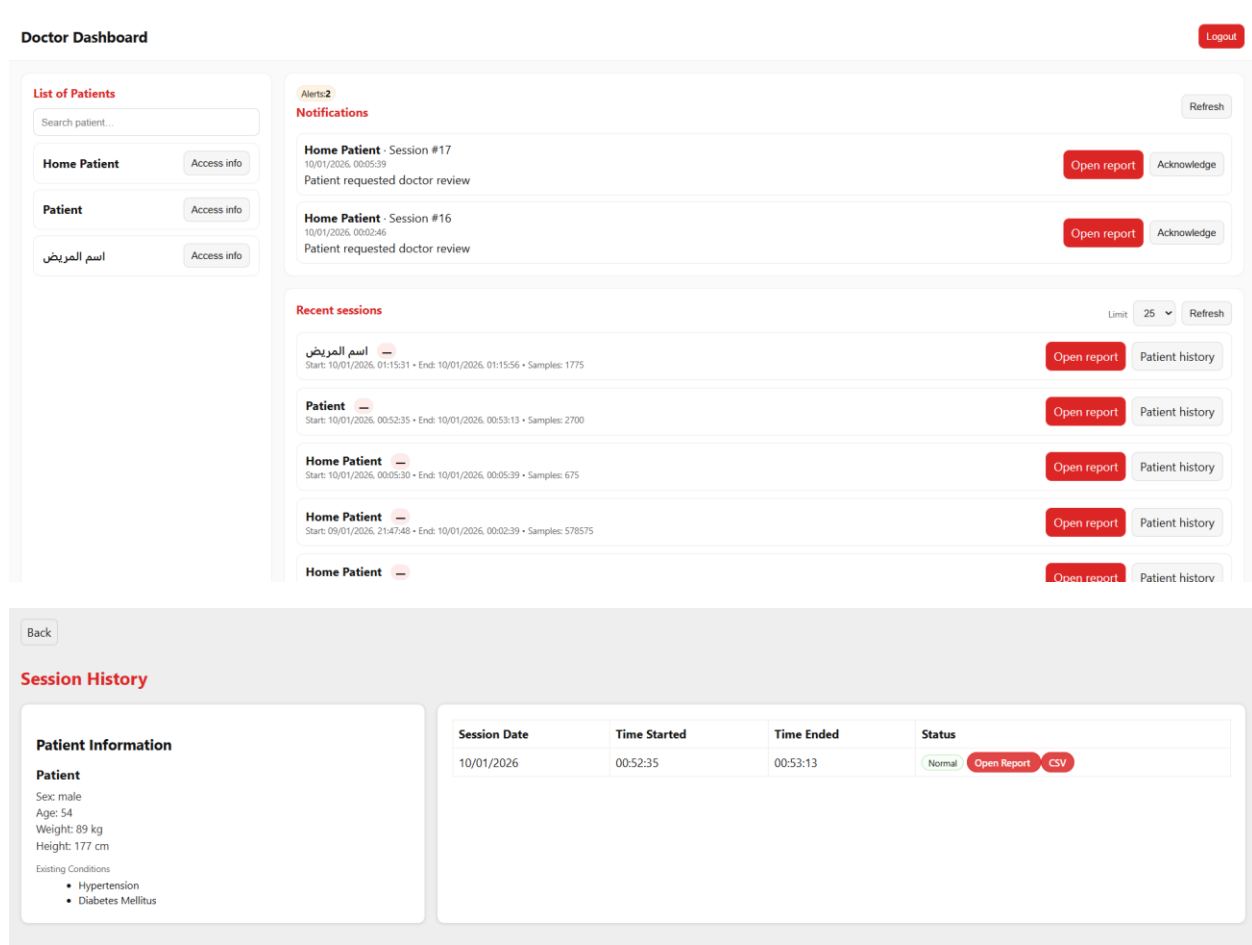


Figure 46. (Top) Clinician dashboard. (Bottom) Patient history - clinician view.

From the clinician interface, clinicians can open patient history and archive ECG reports for detailed waveform inspection. The dashboard is designed as read-only with respect to ECG acquisition, ensuring that recording control remains exclusively within the patient interface. This separation supports safe operation and reflects realistic clinical review workflows in remote monitoring systems.

#### 4.9 External Design Concept and Visual Identity

The proposed external design and visual identity of the ECG system were developed to support clarity, usability, and future product readiness, while remaining consistent with the functional scope of the current prototype. An AI-generated image was used to visualize the anticipated

physical form of the device as seen in Figure 47. The envisioned external form emphasizes portability, simplicity, and safe user interaction. A compact enclosure is proposed to protect the internal circuitry, organize electrode connections, and improve electrical isolation, while providing a visually clean and approachable medical appearance. The inclusion of a screen or interface element reflects a future direction in which real-time status indicators, session control, or summarized metrics may be displayed locally, complementing the existing web-based visualization platform. This concept image serves as a design reference rather than a finalized industrial design and is intended to guide future enclosure development and user-interface integration.



Figure 47. The proposed final product design.

The system named CardioVault was selected to reflect the core objectives of the project. “Cardio” refers to the system’s focus on cardiac monitoring, while “Vault” symbolizes secure storage, protection, and long-term preservation of ECG data. The CardioVault logo incorporates a stylized ECG waveform integrated into the text.

While the current implementation focuses on functional validation using a prototype hardware platform, the external design concept and visual identity provide a structured foundation for future development. These elements support a transition toward a more clinically presentable and product-oriented system, including the integration of advanced processing, AI-assisted analysis, secure cloud connectivity, and battery-powered operation, as outlined in future work considerations.

#### **4.10 Optimization Considerations**

In the implemented analog front-end design, using relatively large resistor values allows for the use of smaller capacitors. Filter cutoff frequencies and gain settings were selected to balance noise suppression and signal fidelity while minimizing external component count. These approaches reduce overall component footprint and support compact PCB layout without compromising filter performance in the ECG bandwidth.

The use of the ESP32 microcontroller consolidates multiple system functions into a single device, including digital processing, peripheral interfacing, and wireless communication. By integrating these subsystems directly on the microcontroller, the need for external communication modules and associated supporting circuitry is eliminated. This reduces PCB area, component count, power routing complexity, and potential points of failure.

The system was optimized for single-supply operations across both analog and digital domains. The AD8232 analog front-end operates from the same low-voltage supply as the ESP32, eliminating the need for dual-rail power generation or level-shifting circuitry. Biasing the ECG signal around an internal mid-supply reference enables safe and stable signal acquisition while minimizing power consumption and simplifying power management.

To improve digitization accuracy without increasing system complexity, an external high-resolution ADC (ADS1115) was used instead of relying on the ESP32's internal ADC. This choice enables precise capture of low-amplitude ECG signals while allowing the ESP32 to focus on communication and control tasks. Separating signal acquisition and wireless transmission improves overall system efficiency and scalability.

The architecture was optimized for scalability by using identical analog front-end channels and a shared digital acquisition and transmission pipeline. This modular approach allows additional ECG channels to be added with minimal changes to firmware and backend infrastructure, supporting future expansion toward higher-lead configurations without redesigning the core system.

## **Chapter 5. Testing and Validation**

This chapter presents the testing and validation procedures carried out to verify the correct operation and performance of the ECG monitoring system. Testing was conducted progressively, starting from basic functional verification and individual subsystem validation, and ending in full system-level evaluation. Both laboratory instruments and an ECG signal simulator were used to ensure that the system performs as intended under controlled conditions.

During development, individual firmware sketches were used to validate specific subsystems, including Wi-Fi connectivity, ADC communication, and data streaming, before integration into the final system firmware.

### **5.1 Power Supply and Continuity**

Following PCB assembly and soldering, initial testing focused on verifying correct power distribution across the printed circuit board. The ESP32, ADS1115, and all AD8232 modules were supplied with the intended 3.3 V operating voltage. Voltage levels were measured using a digital

multimeter to confirm that power reached each subsystem and that no unintended shorts or voltage drops were present.

Continuity testing was performed across power, ground, and signal paths to verify correct electrical connectivity and to identify any unintended open or short circuits on the PCB.

## 5.2 Analog Front-End Circuitry

Signal testing was performed to verify the correct operation of each AD8232 analog front-end module prior to full system integration. For this test, an ECG signal simulator configured to output a recorded analog ECG waveform was used as the input stimulus. The simulator provided electrode-level analog outputs corresponding to a physiological ECG signal, which were connected directly to the differential input junctions of an AD8232 module, replicating the intended electrode interface used during normal system operation.

The conditioned analog output of the AD8232 module was taken from the SIGNAL OUT junction and monitored using a NI ELVIS–based oscilloscope interface. Each AD8232 channel was tested

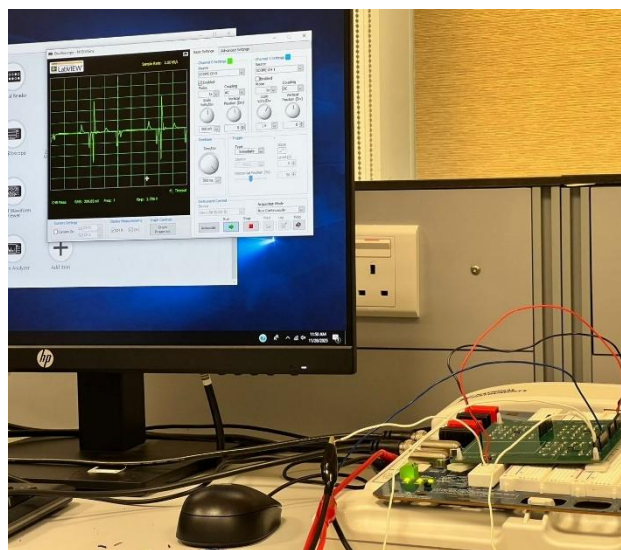


Figure 48. AFE test set up and oscilloscope result.

individually to verify correct signal routing, proper operation of the instrumentation amplifier, and correct functioning of the biasing and filtering stages.

Clear and stable ECG waveforms were observed at the output of each module, as shown in Figure 48, confirming that the analog front-end circuitry correctly amplified and conditioned the recorded ECG signal. Low-frequency baseline drift components were visibly attenuated, consistent with the operation of the high-pass filtering stage, while high-frequency noise components were suppressed, indicating effective low-pass filtering within the intended ECG bandwidth. The resulting waveforms exhibited stable baselines and well-defined cardiac features across all tested channels.

This test provided confidence that each AD8232 analog front-end channel was functioning as intended prior to ADC integration and full system-level testing.

### **5.3 Backend–Frontend Streaming Verification**

Prior to full hardware integration, the real-time data streaming pipeline between the backend server and the web-based frontend was validated using a controlled test signal. A continuous waveform was processed by the backend and transmitted to the frontend via WebSocket, where it was rendered in real time on the patient dashboard.

Figure 49 demonstrates the successful live visualization of a continuous test waveform on the web interface, confirming correct operation of the backend–frontend communication, data handling, and real-time plotting mechanisms independent of the physical acquisition hardware.



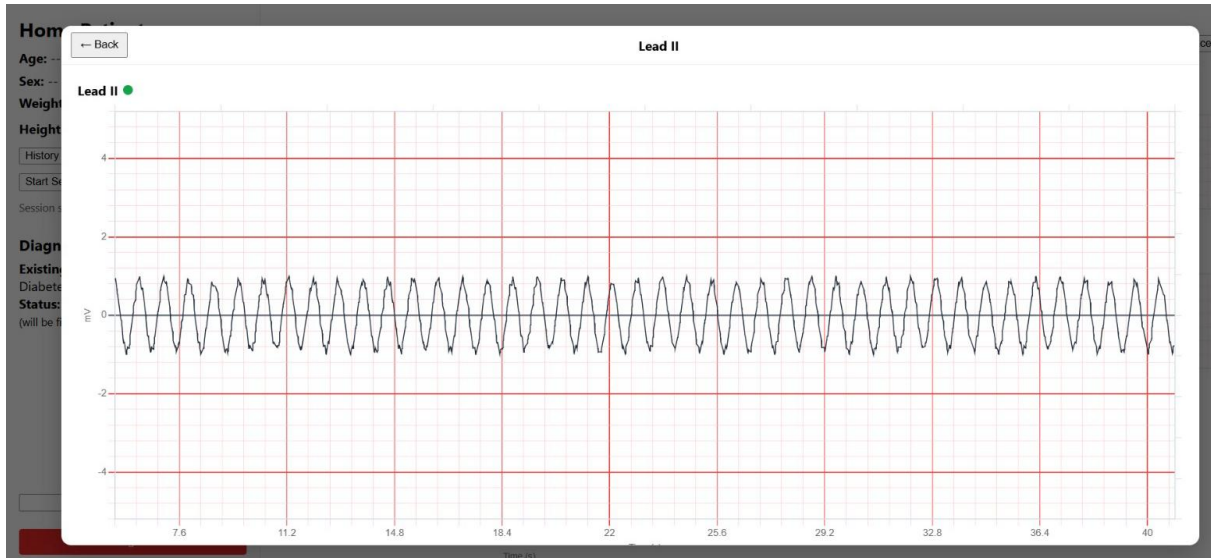


Figure 49. Result of the Backend–Frontend Streaming test.

#### 5.4 Reference and Bias Voltage Test

This test was performed to verify the correct operation of the reference buffer, biasing network, and polarity response of the AD8232 analog front-end under single-supply operation, while observing the resulting behavior through the complete acquisition chain. Each AD8232 channel was evaluated individually.

A controlled DC differential voltage was applied across the input terminals of the AD8232. When the negative input ( $-IN$ ) was held at ground and the positive input ( $+IN$ ) was driven, the instrumentation amplifier output shifted toward the upper end of its allowable output range. Conversely, when the positive input ( $+IN$ ) was held at ground and the negative input ( $-IN$ ) was driven, the output shifted toward the lower end of the range. This was demonstrated in Figure 50. The resulting output behavior was observed on the web-based frontend after digitization by the ADS1115 and transmission via the ESP32 and backend server.

The observed output level of approximately 2.7 V on one of the AD8232 modules, as shown in Figure 50, reflects the single-supply operation of the analog front-end, in which the ECG signal is biased around an internal reference voltage rather than ground. In this configuration, the REFOUT pin provides a mid-supply reference of approximately 1.65 V (half of the 3.3 V supply), which acts as a virtual ground for the instrumentation amplifier and subsequent filtering stages.

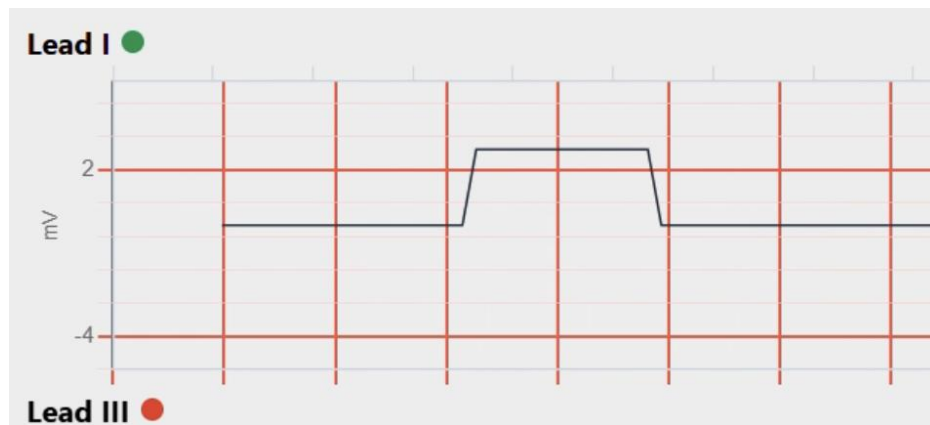


Figure 50 Results of the reference and bias voltage test.

When a DC differential input is applied, the output does not swing around 0 V but instead moves relative to the REFOUT bias level. As the output attempts to increase above this reference, it eventually reaches the positive swing limit of the internal amplifier, resulting in saturation at approximately 2.7 V rather than the full 3.3 V supply. This reduced output headroom is a consequence of the internal amplifier architecture and single-supply operation of the AD8232.

The limited output swing and polarity-dependent response observed on the front end are therefore consistent with the biasing scheme and saturation characteristics of the AD8232 analog front-end. Although the signal was visualized through the complete digital acquisition and transmission pipeline, this test was primarily intended to validate the biasing, reference generation, and polarity behavior of the analog front-end prior to full system-level testing.

### 5.5 Single lead ECG Visualization (Arduino IDE)

Full system testing was performed by injecting recorded analog ECG signals from a simulator into the analog front-end and observing the reconstructed waveforms first on a serial plotter on Arduino IDE, prior to visualizing on the web interface. The first lead was tested alone to verify connectivity and signal clarity shown in Figure 51.

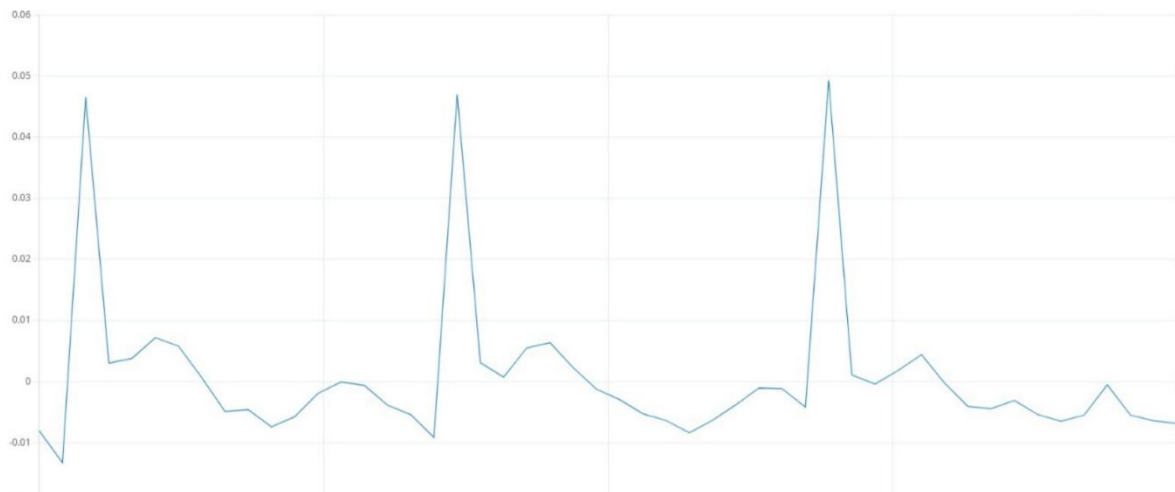


Figure 51. Lead visualization (Arduino).

### 5.6 Multi-Lead Reconstruction and Visualization (Arduino)

Multi Lead ECG acquisition was tested starting with the three standard limb leads. Due to the use of a simulator of a recorded ECG providing only RA and LA outputs, these signals were distributed using a breadboard-based fan-out configuration, where the same RA and LA nodes were branched and connected to the electrode input junctions of the 3 AD8232 analog front-end modules on the custom PCB. Each module employed a different electrode mapping configuration to emulate distinct limb-lead orientations.

The system successfully acquired and transmitted signals from each analog front-end channel, enabling correct operation of the acquisition, digitization, and transmission pipeline. The resulting waveforms were cleanly displayed on the serial plotter as shown in Figure 52.

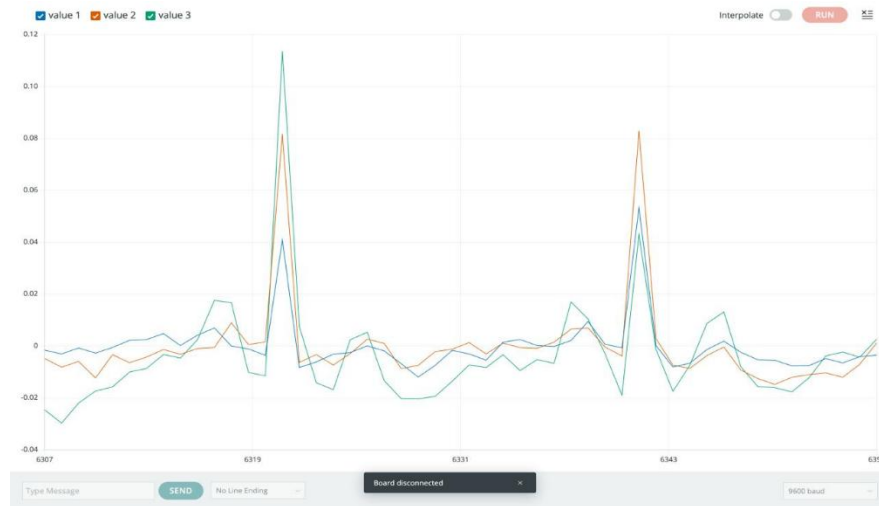


Figure 52. 3 Lead visualization (Arduino).

Despite originating from the same RA–LA source, distinct waveform characteristics were observed across channels due to differences in electrode mapping and analog front-end behavior. Subsequently, the augmented limb leads aVR, aVL, and aVF were mathematically derived within the Arduino IDE using standard Einthoven and Goldberger lead relationships. All six leads were displayed simultaneously on the serial plotter to assess signal smoothness, stability, and real-time rendering performance, as illustrated in Figure 53. This validation confirmed correct lead derivation logic and multi-channel visualization functionality under the constraints of a simplified two-electrode signal source.



Figure 53. 6 Lead visualization (Arduino).

## 5.7 Web-Based Visualization

After successful signal visualization using the Arduino IDE, ECG processing and visualization were migrated to the backend and web-based frontend. An initial streaming test was conducted to evaluate real-time visualization performance by directly streaming ECG data to the web interface and continuously updating six ECG charts.

In the initial configuration, each incoming ECG sample was added directly to the displayed waveform and plotted immediately on the web interface. As streaming continued, the number of displayed data points increased continuously without being limited, forcing the browser to repeatedly redraw the entire ECG waveform.

After approximately 10–15 seconds of continuous operation, the web interface became unresponsive, and the ECG waveform stopped updating. This issue occurred because the browser was required to process and redraw an increasingly large amount of data at a high update rate, exceeding its rendering capacity.

Based on this limitation, the streaming architecture was revised. Instead of directly pushing all samples to the frontend, ECG data is now transmitted in batches to the backend server for buffering and storage, while the web interface retrieves and renders data at controlled update intervals. This modification significantly reduced frontend processing load and enabled stable, continuous real-time visualization.

### 5.8 Signal Digitization and Amplitude Consistency Verification

Signal digitization and amplitude consistency were verified to ensure that the analog ECG signal was correctly scaled, digitized, transmitted, and stored without saturation or clipping. An ECG signal simulator providing a recorded analog ECG waveform was used as the input stimulus. The

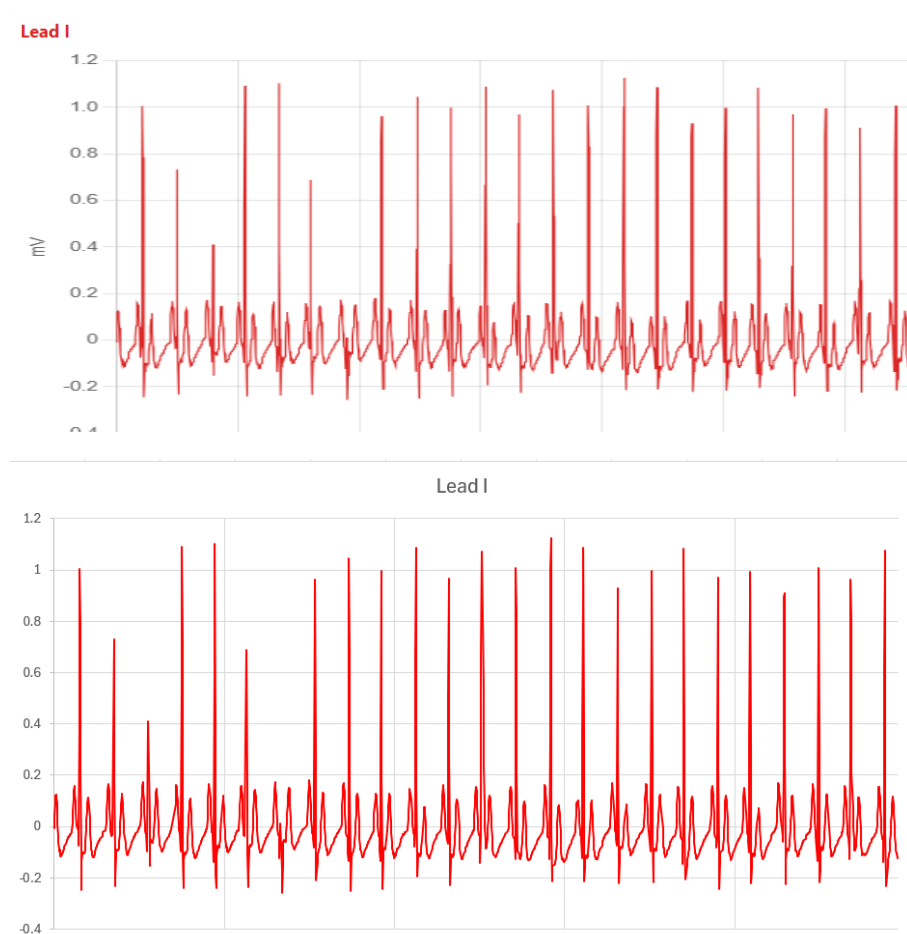


Figure 54. ECG view on the front end (top) v.s. Plotted data from csv file (bottom).

conditioned analog output of the AD8232 front-end was digitized by the ADS1115 ADC and transmitted via the ESP32 to the backend server.

During live operation, the ECG waveform was visualized in real time on the web-based frontend. In parallel, the same digitized samples were stored by the backend and later exported as CSV files. These exported datasets were plotted offline and compared qualitatively with the live frontend visualization, as seen in figure 54.

The observed waveform amplitudes remained well within the ADS1115 input range, with no evidence of saturation or clipping in either the live display or the exported data. The consistent waveform shape and relative amplitude observed across both representations confirmed appropriate gain selection in the analog front-end and correct signal scaling throughout the acquisition pipeline. This test provided qualitative verification that the ECG signal amplitude was preserved from analog input through digitization, transmission, and storage.

### **5.9 Sampling Timing and Multi-Channel Consistency**

The stability of the sampling rate and consistency of multi-channel acquisition were evaluated during continuous system operation. Sampling timing was verified by examining timestamped ECG samples transmitted by the ESP32, which demonstrated uniform temporal spacing over extended recording intervals.

Multi-channel sampling behavior was assessed by reconstructing ECG leads derived from multiple ADC channels and observing their relative alignment. No observable timing skew or distortion affecting waveform morphology was detected across channels, as seen in Figure 55, indicating that

sequential ADC sampling and firmware timing control were sufficiently consistent for multi-lead ECG reconstruction within the intended application.



Figure 55. 6 Lead graphs correctly aligned.

## 5.10 Wireless Transmission, Backend Data Handling, and Session Duration Validation

Wireless data transmission between the ESP32 and the backend server was evaluated under continuous streaming conditions. During extended acquisition sessions, ECG data was streamed over Wi-Fi without observable interruptions, packet loss, or discontinuities in the received data stream.

Backend data handling was validated by monitoring incoming ECG samples and confirming correct session segmentation, buffering, storage, and retrieval. Exported ECG datasets were complete and temporally consistent with the received data, indicating reliable backend ingestion and data management.

System reliability over time was further evaluated through extended recording sessions lasting up to over 2 hours. During these sessions, ECG data was continuously received, stored, and retrieved



without loss of data integrity, system crashes, or degradation in waveform continuity, confirming stable operation for prolonged monitoring scenarios.

### **5.11 Lead Connection Functional Response**

Basic functional behavior of lead connections was observed during testing using the ECG simulator. Changes in signal behavior and the frontend were monitored when leads were disconnected or reattached, confirming that the system responded appropriately to variations in input connectivity. This test served as a functional verification of signal presence rather than a quantitative lead-off detection evaluation.

### **5.12 Data Storage and Retrieval Verification**

Recorded ECG sessions were stored locally by the system and later retrieved through the web interface. Retrieved data matched the originally acquired waveforms, confirming correct session archiving and retrieval functionality.

## **Chapter 6 – Results and Discussion**

It should be noted that most validation was conducted using an ECG signal simulator rather than human subjects. The ECG simulator was not clinical-grade, but rather a recorded analog ECG signal. While this approach ensures repeatability and practicality, real-world physiological variability, such as electrode–skin impedance changes, motion artifacts, and subject-specific noise, was not assessed within the scope of this project. The results presented therefore focus on system-level performance and signal integrity under controlled test conditions.

### **6.1 ECG Signal Quality**

The acquired ECG signals demonstrate clearly recognizable cardiac waveforms when observed through the web-based visualization interface. Across multiple test sessions, the characteristic P

wave, QRS complex, and T wave morphology were visible, indicating correct operation of the analog front-end, digitization stage, and data transmission pipeline. The waveform shape remained relatively stable throughout each recording session, confirming reliable end-to-end signal acquisition from the ECG simulator to the browser display, as shown in Figure 56.

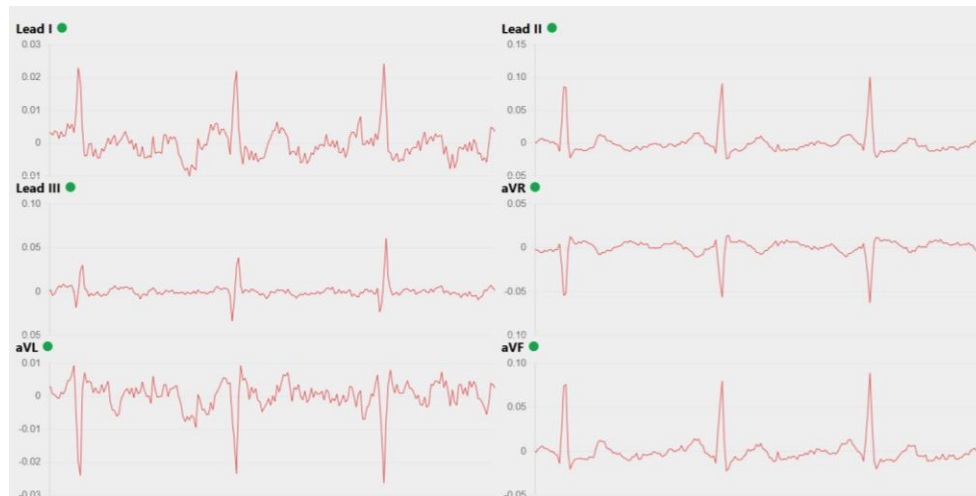


Figure 56. The 6 Lead ECG.

During system testing, low-level noise was observed in the acquired ECG signals. Increased noise activity occurred when the simulator lead wires were physically moved, indicating susceptibility to motion-induced interference. This behavior is consistent with motion artifacts commonly associated with electrode wiring and simulated ECG leads. While the implemented grounding strategy, PCB layout, and integrated analog filtering stages contributed to reduced interference during steady-state operation, complete noise performance validation remains limited by available test resources. The ECG simulator itself may introduce baseline noise, and comprehensive validation would require either human subject testing or a higher-fidelity clinical ECG simulator.

In contrast, the presence or absence of nearby radio-frequency devices, such as mobile phones, produced no observable change in signal quality. This suggests effective operation of the integrated radio-frequency interference (RFI) suppression circuitry within the AD8232 analog front-end modules.

Additionally, the structured PCB layout shown in Figure 57 and the implemented grounding strategy contributed to reduced electromagnetic interference from nearby digital components. No observable digital switching artifacts were present in the acquired ECG signal during simultaneous Wi-Fi transmission and real-time data streaming. This indicates sufficient practical isolation between the analog signal acquisition circuitry and the digital communication subsystem under the tested operating conditions.

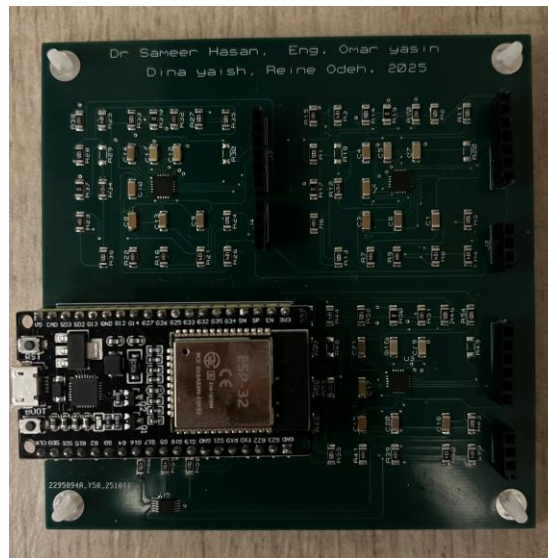


Figure 57. The final PCB

Baseline stability was assessed through long-duration ECG recordings without changes in simulator input; the longest test being just over 2 hours. The signal baseline remained largely centered, with only minor low-frequency drift observed over time, as seen in Figure 56. This behavior is consistent with the expected performance of the implemented high-pass filtering stages designed to attenuate DC offsets and slow baseline wander. Any residual drift was gradual and did not obscure waveform interpretation, confirming acceptable baseline stability for longer monitoring sessions.

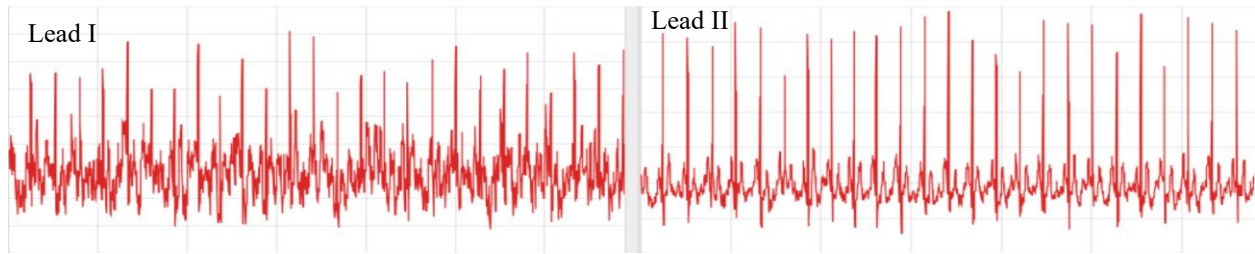


Figure 58. Baseline stability across Lead I and Lead II.

## 6.2 Performance Metrics

The system operated at a fixed sampling frequency of approximately 250 Hz, which falls within the recommended range for diagnostic ECG acquisition. This sampling rate was sufficient to capture rapid transitions within the QRS complex while maintaining manageable data throughput for wireless transmission and browser-based visualization.

The use of the ADS1115 external ADC contributed to improved effective resolution relative to typical on-chip microcontroller ADCs, which is reflected in the smooth appearance of the reconstructed ECG waveform without visible quantization artifacts.

End-to-end system latency was evaluated by observing the delay between signal generation at the simulator and waveform appearance on the web interface. The observed latency was minimal and subjectively perceived as near real-time during operation. This responsiveness is attributed to lightweight packet batching, efficient backend server and WebSocket-based data handling, and local network transmission. While precise millisecond-level latency measurements were not instrumented, the observed performance was sufficient for real-time visualization and interactive monitoring, meeting the functional requirements of the prototype.

No data loss was observed during controlled testing sessions conducted over a stable Wi-Fi network. The displayed ECG waveform remained continuous, with no visible gaps or

discontinuities that would indicate dropped samples or packet loss as seen in Figure 58. This indicates that the buffering and data ingestion mechanisms implemented on the backend successfully handled incoming data streams without overflow or synchronization issues. However, data loss performance under unstable or congested network conditions was not extensively evaluated and remains a consideration for future work.

### **6.3 Multi-Channel ECG Waveform Behavior Under Identical Input Signals**

During system validation, a single ECG simulator or recorded ECG source providing only right-arm (RA) and left-arm (LA) outputs was used. Despite receiving identical RA and LA inputs, the three AD8232 channels produced visibly different ECG waveforms. This behavior arises from the internal architecture of the AD8232, which extends beyond simple differential amplification. Each device incorporates internal biasing, high-pass and low-pass filtering, reference buffering, and an active right-leg drive (RLD) feedback loop.

Variations in electrode pin assignment resulted in observable differences in amplitude scaling, signal polarity, and baseline characteristics. Although all three front-end modules were driven by the same simulator outputs, each channel produced a distinct conditioned waveform due to differences in electrode mapping and internal signal conditioning behavior. The AD8232 is not a simple subtractor; electrode configuration directly influences common-mode rejection, filter response, and reference stabilization.

It is therefore emphasized that the displayed waveforms do not represent independent physiological limb leads in the clinical sense. Instead, they represent distinct conditioned projections of the same ECG signal processed through separate analog front-end pathways under different electrode configurations. While this experimental setup does not enable true Einthoven triangle reconstruction or clinical multi-lead ECG acquisition, it is sufficient for validating analog

front-end functionality, signal conditioning behavior, digitization accuracy, wireless transmission reliability, and multi-channel visualization. The observed waveform differences thus confirm correct system operation while highlighting the limitations imposed by the simplified signal source.

## **Chapter 7: Future Work**

Although the current system successfully demonstrates multi-lead ECG acquisition, wireless streaming, visualization, and local data archiving, several enhancements can elevate it toward a more clinically presentable and product-oriented platform.

In future iterations, the ESP32 will be limited to signal acquisition, timestamping, and wireless transmission, while advanced signal processing and diagnostic computation will be offloaded to a dedicated edge processor such as a Raspberry Pi. This upgrade would enable more robust digital filtering, reliable QRS/R-peak detection, and stable heart rate and heart rate variability (HRV) computation, improving timing accuracy, buffering, and overall system reliability.

Building on this processing capability, future work can introduce an AI-assisted diagnostic layer to support detection of common rhythm abnormalities and clinically relevant signal patterns. Machine learning or validated algorithmic libraries can be deployed on the Raspberry Pi to generate decision-support indicators, which would be displayed through an existing dedicated diagnostic section of the CardioVault web interface.

To enhance portability and scalability, future versions can incorporate secure cloud connectivity for remote monitoring and long-term storage using encrypted communication protocols, alongside local buffering to ensure reliability during network interruptions.

A dedicated power management subsystem incorporating battery operation, an integrated charging and protection circuit, and seamless switching between battery power and externally supplied DC power would further enhance system robustness and support extended continuous operation. Additionally, a compact, electrically insulating (non-conductive) enclosure could be employed to protect internal circuitry, prevent accidental electrical contact, and further enhance user safety.

Although the hardware layout includes support for lead-off detection through the AD8232 analog front-end, this feature was not fully utilized in the current system implementation. The lead-off detection output was not connected to the ESP32 microcontroller, and corresponding firmware and backend handling were therefore not implemented.

Future work could involve routing the lead-off detection signal to a dedicated ESP32 GPIO and integrating firmware logic to monitor electrode connection status in real time. This would enable automatic detection of disconnected or poorly attached electrodes and allow the system to generate user alerts through the web interface. Incorporating lead-off status into recorded sessions would further enhance data reliability and improve user feedback during home-based ECG acquisition.

Finally, expanded testing and validation, including Clinical Level ECG simulator, testing under unstable or congested network, longer continuous sessions, and human trials under ethical approval, would be essential steps toward improved clinical credibility and potential certification readiness.

## **Chapter 8: Conclusion**

This project successfully demonstrated the design and implementation of a complete end-to-end portable ECG monitoring system that integrates custom analog front-end hardware, digital signal acquisition, wireless data transmission, and a web-based visualization and storage platform. The

developed system illustrates how a compact, modular architecture can support real-time ECG acquisition and remote access.

Experimental validation showed that the system was able to acquire and transmit ECG signals with stable operation under controlled testing conditions. The results demonstrate that the system is capable of acquiring, transmitting, and visualizing ECG signals with sufficient quality for basic rhythm observation and waveform morphology analysis. While the system cannot currently replace certified clinical ECG equipment, it provides meaningful cardiac signal visualization suitable for preliminary assessment, remote monitoring, and educational or research applications. It also lays down the foundation for a scalable, upgradeable, locally available, and affordable ECG device. The ability to stream and archive ECG sessions through a web interface further enhances system usability and aligns with current trends in IoT-based healthcare platforms.

Despite these positive outcomes, the project scope also introduced limitations that define clear directions for future work. Validation relied primarily on an ECG simulator and a recorded signal rather than human subjects, and the multi-channel measurements represent conditioned projections derived from shared inputs rather than fully independent physiological limb leads.

In summary, the system fulfills its primary objective as a functional proof-of-concept for portable, IoT-based ECG monitoring. It establishes a solid technical foundation upon which more clinically robust signal processing, expanded validation, and product-level refinements can be built, supporting future development toward a practical and accessible cardiac monitoring solution.

## **Chapter 9: References**

- [1] World Health Organization, "Cardiovascular diseases (CVDs)," 2021.
- [2] Mayo Clinic, "Electrocardiogram (ECG or EKG)," 2022.



- [3] American Heart Association, "Understanding your EKG results," 2020.
- [4] National Heart, Lung, and Blood Institute, "What is an electrocardiogram?" 2021.
- [5] A. Serhani, H. T. El Kassabi, H. Ismail, and A. Nujum Navaz, "ECG Monitoring Systems: Review, Architecture, Processes, and Technologies," *IEEE Access*, vol. 8, pp. 11545–11569, 2020.
- [6] L. Zepeda-Echavarria, A. A. Aguilar-Mejia, J. R. Rodriguez-Rodriguez, and M. A. Reyna, "Electrocardiogram Devices for Home Use: Technological and Clinical Scoping Review," *Sensors*, vol. 21, no. 16, 2021.
- [7] A. C. Guyton and J. E. Hall, *Textbook of Medical Physiology*, 11th ed. Philadelphia, PA: Saunders, 2011.
- [8] <https://my.clevelandclinic.org/health/body/23074-heart-chambers>
- [9] E. Braunwald, *Heart Disease: A Textbook of Cardiovascular Medicine*, 9th ed. Philadelphia, PA: Elsevier, 2012.
- [10] A. M. Katz, *Physiology of the Heart*, 5th ed. Philadelphia, PA: Lippincott Williams & Wilkins, 2010.
- [11] D. E. Mohrman and L. J. Heller, *Cardiovascular Physiology*, 8th ed. New York, NY: McGraw-Hill, 2014.
- [12] R. H. Anderson, *The Heart: Structure, Function, and Pathology*. London, UK: Greenwich Medical Media, 2005.
- [13] [buzzle.com](https://www.buzzle.com)
- [13] D. M. Bers, "Cardiac excitation–contraction coupling," *Nature*, vol. 415, no. 6868, pp. 198–205, 2002.
- [14] L. H. Opie, *Heart Physiology: From Cell to Circulation*, 4th ed. Philadelphia, PA: Lippincott Williams & Wilkins, 2004.
- [15] W. J. Lederer et al., "Excitation-Contraction Coupling in Heart Cells: Roles of the Sodium-Calcium Exchange, the Calcium Current, and the Sarcoplasmic Reticulum," *Annals of the New York Academy of Sciences*, vol. 588, no. 1, pp. 190–206, 1990.
- [16] D. M. Bers and S. Despa, "Cardiac Excitation-Contraction Coupling," in *Encyclopedia of Biological Chemistry*, 2nd ed., Elsevier, 2013, pp. 379–383.
- [17] T. F. McDonald, "Excitation-Contraction Coupling," in *Physiology and Pathophysiology of the Heart*, N. Sperelakis, Ed., Boston, MA: Springer, 1984, pp. 315–337.
- [18] A. L. Goldberger, *Clinical Electrocardiography: A Simplified Approach*, 7th ed. St. Louis, MO: Mosby, 2006.

- [19] D. P. Zipes and J. Jalife, *Cardiac Electrophysiology: From Cell to Bedside*, 6th ed. Philadelphia, PA: Saunders, 2013.
- [20] E. N. Marieb and K. Hoehn, *Human Anatomy & Physiology*, 9th ed. San Francisco, CA: Pearson, 2013.
- [21] M. Delmar and N. Makita, "Gap junction remodeling in heart disease," *The Journal of Clinical Investigation*, vol. 122, no. 1, pp. 22-25, 2012.
- [22] B. Surawicz, T. Knilans, and A. L. Fisch, *Official Electrocardiographic Standards and Recommendations*, New York, NY: Elsevier, 2009.
- [23] V. Fuster, R. A. Walsh, and R. A. Harrington, *Hurst's The Heart*, 13th ed. New York, NY: McGraw-Hill, 2011.
- [24] Surawicz, B., & Knilans, T. K. (2008). *Chou's Electrocardiography in Clinical Practice: Adult and Pediatric* (6th ed.). Philadelphia, PA: Saunders Elsevier.
- [25] AHA Scientific Statement, "American Heart Association guidelines on cardiac electrophysiology," *Circulation*, vol. 141, no. 6, pp. e1-e10, 2020.
- [26] R. E. Klabunde, *Cardiovascular Physiology Concepts*, 2nd ed. Philadelphia, PA: Lippincott Williams & Wilkins, 2012.
- [27] Strauss, D.G., et al. (2016). "Prediction of sudden cardiac death with automated high-throughput analysis of heterogeneity in standard resting 12-lead electrocardiograms." *Heart Rhythm*, 13(3), 713-720.
- [28] G. Buckberg, J. Mahajan, S. Saleh, I. Hoffman, and M. Coghlan, "Structure and function relationships of the helical ventricular myocardial band," *The Journal of Thoracic and Cardiovascular Surgery*, vol. 136, no. 3, pp. 578-589, 2008.
- [29] A. M. Younis et al., "The burden of Cardiovascular diseases in Jordan: a longitudinal analysis from the global burden of disease study, 1990–2019," *BMC Public Health*, vol. 24, no. 1, 2024.
- [30] S. Nejadghaderi et al., "Burden of ischemic heart disease in the Middle East and North Africa (MENA) and attributable risk factors: An epidemiological analysis from 1990 to 2019," *Journal of Clinical Hypertension*, vol. 23, no. 11, pp. 1965–1973, 2021

- [31] A. Zepeda-Echavarria, "Electrocardiogram Devices for Home Use: Technological and Clinical Overview," *JMIR Cardio*, vol. 7, no. 1, e44003, 2023.
- [32] M. Carrington et al., "Monitoring and diagnosis of intermittent arrhythmias: evidence and clinical considerations," *Frontiers in Cardiovascular Medicine*, 2022.
- [33] Jordan Investment Commission, "Life Sciences Sector in Jordan – 2023 Report," Invest Jordan, 2023.
- [34] <https://arazygroup.com/ivd-medical-device-registration-jordan>
- [35] P. A. Macfarlane et al., "Recommendations for the Standardization and Interpretation of the Electrocardiogram," *Circulation*, vol. 115, no. 10, pp. 1306–1324, Mar. 2007
- [36] W. Einthoven, "The string galvanometer and the human electrocardiogram," *Pflügers Archiv für die gesamte Physiologie des Menschen und der Tiere*, vol. 122, no. 12, pp. 517–548, 1906.
- [37] J. W. Pick and F. R. Keating, *Electrocardiography: Basic and Clinical Applications*, New York, NY, USA: Grune & Stratton, 1972.
- [38] P. Kligfield, L. S. Gettes, J. J. Bailey, et al., "Recommendations for the standardization and interpretation of the electrocardiogram: Part I: The electrocardiogram and its technology," *Circulation*, vol. 115, no. 10, pp. 1306–1324, 2007.
- [39] N. J. Holter, "New method for heart studies: Continuous electrocardiography," *Science*, vol. 134, no. 3486, pp. 1214–1220, 1957.
- [40] P. Smith, M. P. Büchner, and C. M. Clark, "Advances in digital electrocardiography: The role of AI and machine learning," *Journal of Biomedical Engineering*, vol. 32, no. 4, pp. 253–269, 2015.
- [41] A. Hannun, P. Rajpurkar, M. Haghpanahi, et al., "Cardiologist-level arrhythmia detection with convolutional neural networks," *Nature Medicine*, vol. 25, no. 1, pp. 65–69, 2019.
- [42] R. S. Khandpur, *Handbook of Biomedical Instrumentation*, 3rd ed. McGraw Hill, 2014.
- [43] L. A. Geddes and L. E. Baker, *Principles of Applied Biomedical Instrumentation*, 3rd ed. Hoboken, NJ, USA: Wiley, 1989.

- [44] L. A. Geddes and L. E. Baker, "Principles of electrode-electrolyte interface," *IEEE Transactions on Biomedical Engineering*, vol. 19, no. 5, pp. 313-320, 1972.
- [45] J. G. Webster, *Medical Instrumentation: Application and Design*, 4th ed. Wiley, 2009.
- [46] D. Houstis, G. Apostolou, and D. Koutsouris, "ECG monitoring and diagnostic systems: A review," in *Proceedings of the 2012 IEEE Engineering in Medicine and Biology Society Conference*, San Diego, CA, USA, 2012, pp. 3432–3435.
- [47] A. D. J. Albahadily, A. M. Eid, and N. K. B. Y. Ibrahim, "A review of advanced ECG monitoring systems in critical care: Trends and applications," *J. Clin. Monit. Comput.*, vol. 35, no. 4, pp. 829-842, 2021.
- [48] C. S. Li, A. L. Liu, and M. J. Y. Song, "Utilization of multi-lead ECG for early detection of arrhythmias in ICU: A comprehensive review," *J. Clin. Monit. Comput.*, vol. 36, no. 2, pp. 305-317, 2022.
- [49] J. J. Zhang, S. R. Mohanty, and M. E. M. Shankar, "Artificial intelligence applications in ECG analysis for cardiac arrhythmias detection," *IEEE Rev. Biomed. Eng.*, vol. 14, pp. 29-42, 2021.
- [50] R. M. Rangayyan, *Biomedical Signal Analysis: A Case-Study Approach*, Hoboken, NJ, USA: Wiley, 2001.
- [51] J. G. Webster, *Medical Instrumentation: Application and Design*, 5th ed., Hoboken, NJ, USA: John Wiley & Sons, 2018.
- [52] S. Franco, *Design with Operational Amplifiers and Analog Integrated Circuits*, 4th ed. McGraw-Hill, 2015.
- [53] T. L. Wheeler et al., "A high common-mode rejection instrumentation amplifier for biomedical applications," *IEEE Transactions on Circuits and Systems*, vol. 32, no. 4, pp. 445-452, 1985.
- [54] <https://www.allaboutcircuits.com/textbook/semiconductors/chpt-8/the-instrumentation-amplifier/>
- [55] P. Horowitz and W. Hill, *The Art of Electronics*, 3rd ed. Cambridge University Press, 2015.
- [56] A. S. Morris, *Measurement and Instrumentation Principles*, 3rd ed. Butterworth-Heinemann, 2001.
- [57] D. Prutchi and M. Norris, *Design and Development of Medical Electronic Instrumentation*, Wiley, 2005.
- [58] B. B. Winter and J. G. Webster, "Driven-right-leg circuit design," *IEEE Transactions on Biomedical Engineering*, vol. BME-30, no. 1, pp. 62–66, 1983.

- [59] A. C. MettingVanRijn, "High-performance analog signal processing for ECG applications," IEEE Engineering in Medicine and Biology Magazine, vol. 18, no. 3, pp. 50-62, 1999.
- [60] [https://www.researchgate.net/Figure/Circuit-diagram-for-an-active-high-pass-filter-based-on-an-operational-amplifier-where-V\\_fig11\\_296695687](https://www.researchgate.net/Figure/Circuit-diagram-for-an-active-high-pass-filter-based-on-an-operational-amplifier-where-V_fig11_296695687)
- [61] J. Pan and W. J. Tompkins, "A real-time QRS detection algorithm," IEEE Transactions on Biomedical Engineering, vol. 32, no. 3, pp. 230-236, 1985.
- [62] <https://www.allaboutcircuits.com/video-tutorials/op-amps-low-pass-and-high-pass-active-filters/>
- [63] W. T. Miller and B. L. Olson, "Power-line interference filtering in ECG signals," IEEE Transactions on Biomedical Engineering, vol. 47, no. 3, pp. 343-346, 2000.
- [64] [https://circuitsarchive.org/circuits/filters/second\\_order\\_notch/filter-twin-t-second-order-notch-ii-non-inverting/](https://circuitsarchive.org/circuits/filters/second_order_notch/filter-twin-t-second-order-notch-ii-non-inverting/)
- [65] A. Sedra and K. C. Smith, Microelectronic Circuits, 7th ed. Oxford University Press, 2015.
- [66] IEC 60601-1:2012, Medical electrical equipment—Part 1: General requirements for basic safety and essential performance.
- [67] <https://robu.in/the-basics-of-optocoupler-relay/>
- [68] J. G. Webster, Medical Instrumentation: Application and Design, 4th ed. Hoboken, NJ, USA: Wiley, 2010.
- [69] American National Standards Institute (ANSI), ANSI/AAMI EC11-1991: Diagnostic Electrocardiographic Devices.
- [70] M. Akay, Biomedical Signal Processing, Academic Press, 2012.
- [71] L. Sörnmo and P. Laguna, Bioelectrical Signal Processing in Cardiac and Neurological Applications, Elsevier, 2005.
- [72] P. W. Macfarlane and T. D. Van Oosterom, Comprehensive Electrocardiology, 2nd ed., Springer, 2010.
- [73] <https://thephysiologist.org/study-materials/the-ecg-leads-polarity-and-einthovens-triangle/>
- [74] E. Goldberger, "Unipolar lead electrocardiography," American Heart Journal, vol. 23, no. 4, pp. 483–492, 1942.

- [75] S. M. R. Islam, D. Kwak, M. H. Kabir, M. Hossain, and K. S. Kwak, "The Internet of Things for Health Care: A Comprehensive Survey," *IEEE Access*, vol. 3, pp. 678–708, 2015.
- [76] J. G. Webster, *Medical Instrumentation: Application and Design*, 4th ed. Wiley, 2010.
- [77] S. Farahani, *ZigBee Wireless Networks and Transceivers*. Newnes, 2011.
- [78] M. Patel and J. Wang, "Applications, challenges, and prospective in emerging body area networking technologies," *IEEE Wireless Communications*, vol. 17, no. 1, pp. 80–88, 2010.
- [77] R. Want, "An Introduction to RFID Technology," *IEEE Pervasive Computing*, vol. 5, no. 1, pp. 25–33, 2006.
- [78] T.-M. Kuo, F.-T. Liu, and R. H. Liang, "Cloud computing for healthcare: Opportunities and challenges," *IEEE Computer*, vol. 44, no. 12, pp. 22–27, 2011.
- [79] A. Al-Fuqaha et al., "Internet of Things: A Survey on Enabling Technologies, Protocols, and Applications," *IEEE Communications Surveys & Tutorials*, 2015.
- [80] J. Gubbi et al., "Internet of Things (IoT): A Vision, Architectural Elements, and Future Directions," *Future Generation Computer Systems*, 2013.
- [81] P. Bonato, "Wearable sensors/systems and their impact on biomedical engineering," *IEEE Eng. Med. Biol. Mag.*, vol. 22, no. 3, pp. 18–20, May–Jun. 2003.
- [82] H. Alemdar and C. Ersoy, "Wireless sensor networks for healthcare: A survey," *Computer Networks*, vol. 54, no. 15, pp. 2688–2710, Oct. 2010.
- [83] A. Al-Janabi, "Real-Time Electrocardiogram (ECG) Monitoring Using Android-based Mobile Application," *Procedia Computer Science*, vol. 170, pp. 451–458, 2020.
- [84] Y. Khan et al., "A Review of Wearable Sensors and Systems with Application in Rehabilitation," *Sensors*, 2020.
- [85] P. W. Macfarlane et al., *Comprehensive Electrocardiology*, 2nd ed., Springer, 2010.
- [86] Analog Devices, "AD8232: Single-Lead, Heart Rate Monitor Front End," *Datasheet*, Rev. D, 2020.
- [87] Espressif Systems. "ESP32 Datasheet."

- [88] Neil Kolban, "Kolban's Book on ESP32," 2017.
- [89] Texas Instruments, "ADS1115: Ultra-Small, Low-Power, 16-Bit Analog-to-Digital Converter with Internal Reference," Datasheet, 2009. :contentReference[oaicite:0]{index=0}
- [90] R. T. Fielding and R. N. Taylor, "Principled design of the modern Web architecture," ACM Transactions on Internet Technology, 2002.
- [91] S. M. R. Islam et al., "The Internet of Things for health care: A comprehensive survey," IEEE Access, 2015.
- [92] C. Pautasso, O. Zimmermann, and F. Leymann, "RESTful Web Services vs. Big Web Services," in Proc. WWW Conference, 2008.
- [93] I. Fette and A. Melnikov, "The WebSocket Protocol (RFC 6455)," IETF, 2011.
- [94] A. Patel, H. Park, P. Bonato, L. Chan, and M. Rodgers, "A review of wearable sensors and systems with application in rehabilitation," Journal of NeuroEngineering and Rehabilitation, vol. 9, no. 21, pp. 1–17, 2012.
- [95] P. Bonato, "Advances in wearable technology and systems for health monitoring," IEEE Engineering in Medicine and Biology Magazine, vol. 29, no. 3, pp. 25–36, 2010.
- [96] T.-M. Kuo, F.-T. Liu, and R. H. Liang, "Cloud computing for healthcare: Opportunities and challenges," IEEE Computer, vol. 44, no. 12, pp. 22–27, 2011.
- [97] S. M. R. Islam, D. Kwak, M. H. Kabir, M. Hossain, and K. S. Kwak, "The Internet of Things for health care: A comprehensive survey," IEEE Access, vol. 3, pp. 678–708, 2015.
- [98] P. Rajpurkar, A. Y. Hannun, M. Haghpanahi, C. Bourn, and A. Y. Ng, "Cardiologist-level arrhythmia detection with convolutional neural networks," Nature Medicine, vol. 25, pp. 65–69, 2019.
- [99] S. Kiranyaz, T. Ince, and M. Gabbouj, "Real-time patient-specific ECG classification by 1-D convolutional neural networks," IEEE Transactions on Biomedical Engineering, vol. 63, no. 3, pp. 664–675, 2016.
- [100] World Health Organization, Global diffusion of eHealth: making universal health coverage achievable: report of the third global survey on eHealth, WHO, 2017.
- [101] J. Pan and W. J. Tompkins, "A real-time QRS detection algorithm," IEEE Transactions on Biomedical Engineering, vol. BME-32, no. 3, pp. 230–236, Mar. 1985.10.14569/IJACSA.2022.0130408.

- [102] A. Bravo-Zanoguera et al., "Low-cost portable ECG acquisition system for undergraduate bioinstrumentation laboratories," *SN Applied Sciences*, vol. 2, no. 11, 2020.
- [103] R. Chen, "Low-Cost Arduino-Based Wireless ECG Monitoring System," Cornell University ECE4760 Final Projects, 2019.
- [104] X. Guan, "Design and implementation of a portable ECG monitoring system," *Journal of Physics: Conference Series*, vol. 1438, no. 1, 2020.
- [105] X. Li, Y. Zhao, and Y. Liu, "Wireless Electrocardiograph Monitoring System Based on nRF905," *Proceedings*, vol. 42, no. 1, 2019.
- [106] J. Li, X. Yang, and W. Sun, "Design of a low power consumption wireless ECG monitoring system," *Applied Sciences*, vol. 9, no. 10, 2019.
- [107] M. Vo, H. Nguyen, and P. Tran, "Development of a solar-powered wireless ECG monitoring system for remote applications," *Procedia Computer Science*, vol. 172, pp. 221–226, 2020.
- [108] Y. Wang et al., "Flexible Non-contact ECG Monitoring Based on Printed Electronics," *Advanced Biomedical Research*, 2023.
- [109] D. Zepeda-Echavarria et al., "Handheld miniECG Device for Real-Time Monitoring and Diagnosis: Design and Validation," *Frontiers in Digital Health*, vol. 4, 2024.
- [110] M. Ehresh et al., "Portable ECG system with ZigBee and LabVIEW GUI for real-time visualization," *AIP Conference Proceedings*, vol. 2207, 2019.
- [111] T. Zhang, Y. Li, and M. Chen, "Bluetooth-enabled ECG transmission module for portable devices," *Journal of Biomedical Engineering*, vol. 13, no. 2, 2020.
- [112] J. Uddin, M. H. Kabir, and M. Shabnam, "IoT Based Patient Monitoring System Using ECG Sensor," *International Journal of Scientific & Engineering Research*, vol. 10, no. 4, pp. 248–252, Apr. 2019.
- [113] P. Islam, S. Kundu, and T. Bhuiyan, "Design and implementation of a low-cost IoT-based portable ECG monitoring system," *Biosensors*, vol. 11, no. 6, p. 188, 2021.



- [114] S. Rehman, N. U. Rehman, and M. A. Khan, "Remote ECG Monitoring System by using IoT," Academia.edu, 2022.
- [115] A. Jan, M. A. Khan, S. Khan, and I. Ullah, "A Lightweight MQTT Protocol Based IoT ECG Monitoring System Using ESP32," *Intelligent Automation & Soft Computing*, vol. 33, no. 1, pp. 637–649, 2022.
- [116] K. Arunkumar, P. Karthikeyan, and K. Srinivasan, "Development of an IoT based ECG monitoring system using Wi-Fi and Bluetooth technology," *Sensors*, vol. 20, no. 24, p. 7353, 2020.
- [117] M. R. Rao, M. R. Reddy, and M. R. Goud, "Internet of Things (IoT) Based ECG System for Rural Health Care," Academia.edu, 2021.
- [118] A. Aghamohammadi, S. Kazemi, and M. J. Soleimani, "An IoT-Based ECG Monitoring System with Backup Storage for Internet-Disrupted Scenarios," *Sensors*, vol. 22, no. 18, 2022.
- [119] H. Gao, X. Zhao, and L. Wang, "Low-power wearable ECG monitoring system based on IoT and edge computing," *Micromachines*, vol. 13, no. 12, p. 2153, 2022.
- [120] A. Jafari and H. R. Khalilzadeh, "Design of Modular IoT ECG Monitoring Architecture for Multi-Patient Environments," *Journal of Information Systems and Telecommunication (JIST)*, vol. 9, no. 34, pp. 47–56, 2021.
- [121] S. Sadat, M. B. Ahmed, and S. Iqbal, "An IoT Based Healthcare using ECG," Academia.edu, 2021.
- [122] P. Thiyagaraja, R. Kalidoss, and R. Narayanasamy, "A Portable Real-Time ECG Device for Arrhythmia Detection Using Raspberry Pi," ResearchGate, 2017.
- [123] K. Ryu, S. Kim, D. Choi, and H. Lee, "Development of a Deep Learning-Based Real-Time ECG Monitoring System Using Wearable Devices," *Sensors*, vol. 20, no. 24, p. 7353, Dec. 2020.
- [124] Y. Zhu, J. Tan, and H. Zheng, "Arrhythmia Classification Based on ECG Signal Using DWT and CNN-LSTM," *Procedia Computer Science*, vol. 171, pp. 613–620, 2020.
- [125] A. Paviglianiti, M. Brignoli, L. Mainardi, and P. Ravi, "Wearable ECG Devices for Atrial Fibrillation Detection Based on Machine Learning," *IFMBE Proceedings*, vol. 76, pp. 122–129, 2020.
- [126] S. Roy, R. Ghosh, and D. Chakraborty, "Automatic Classification of Arrhythmias Using CNN and LSTM Networks," Semantic Scholar, 2020.

- [127] S. Raghunath et al., "Deep Neural Networks Can Predict New-Onset Atrial Fibrillation from the 12-Lead ECG and Help Identify Those at Risk of AF-Related Stroke," *The Lancet Digital Health*, vol. 2, no. 4, pp. e196–e206, Apr. 2020.
- [128] S. Mazumder, A. Sinha, and D. Nandi, "Design of a Lightweight Real-Time ECG Classification System for Mobile Devices," *Procedia Computer Science*, vol. 196, pp. 396–403, 2022.
- [129] M. Pajooyan, A. Zargari, and A. Ghaffari, "Deep Learning Approach to ECG Signal Classification Using Discrete Wavelet Transform," *IEEE Access*, vol. 9, pp. 90052–90060, 2021.
- [130] M. Yousefi, A. Shamsollahi, and M. Babaie-Zadeh, "Improving ECG Classification Using Preprocessing and Adaptive Filtering," *Optics & Laser Technology*, vol. 117, pp. 153–160, 2019.
- [131] S. Roshini, R. Renuka, and S. Karthik, "Bluetooth-Based ECG Monitoring System and Arrhythmia Detection Using MATLAB," *ScienceDirect*, 2020.
- [132] T. Di Noto, S. Bruno, and G. Gargiulo, "End-to-End ECG Classification with Deep Neural Networks," *MDPI*, vol. 13, no. 12, p. 2153, Dec. 2022.
- [133] Qtaish, Amjad & Al-Shrouf, Anwar. (2022). A Portable IoT-cloud ECG Monitoring System for Healthcare. 22. 269. 10.22937/IJCSNS.2022.22.1.37.
- [134] I. Zagan, V. G. Găitan, A.-I. Petrariu, N. Iuga, and A. Brezilianu, "Design, Fabrication, and Testing of an IoT Healthcare Cardiac Monitoring Device," *Computers*, vol. 9, no. 1, p. 15, Feb. 2020, doi: <https://doi.org/10.3390/computers9010015>.
- [135] T. Jeon, B. Kim, M. Jeon, and B.-G. Lee, "Implementation of a portable device for real-time ECG signal analysis," *BioMedical Engineering OnLine*, vol. 13, no. 1, p. 160, 2014, doi: <https://doi.org/10.1186/1475-925x-13-160>.
- [136] M. Ehresh, P. Abatis, and F. S. Schlindwein, "A portable electrocardiogram for real-time monitoring of cardiac signals," *SN Applied Sciences*, vol. 2, no. 8, Jul. 2020, doi: <https://doi.org/10.1007/s42452-020-3065-9>.
- [137] Md. O. Rahaman et al., "Internet of things based electrocardiogram monitoring system using machine learning algorithm," *International Journal of Electrical and Computer Engineering (IJECE)*, vol. 12, no. 4, p. 3739, Aug. 2022,

- [138] W.-Y. Yi et al., “Towards Personalized Healthcare in Cardiac Population: The Development of a Wearable ECG Monitoring System, an ECG Lossy Compression Schema, and a ResNet-Based AF Detector,” arXiv.org, 2022.
- [139] Alimbayeva, Zhadyra & Alimbayev, Chingiz & Bayanbay, Nurlan & Ozhikenov, Kassymbek & Bodin, Oleg & Mukazhanov, Yerkat. (2022). Portable ECG Monitoring System. International Journal of Advanced Computer Science and Applications. 13. 10.14569/IJACSA.2022.0130408.
- [140] <https://nust.edu.pk/research-excellence/telecardio>
- [141] <https://www.pocketecg.com/>
- [142] [support.withings.com](https://support.withings.com)
- [143] [cardions.com](https://cardions.com)
- [144] [creative-sz.com](https://creative-sz.com)
- [145] [walmart.com](https://walmart.com)
- [146] [facelake.com](https://facelake.com)
- [147] [getwellue.com](https://getwellue.com)
- [148] <https://vc6oecg.com/product-details>
- [149] <https://alivecor.com/kardiamobile6l>
- [150] <https://www.humeds.com/>